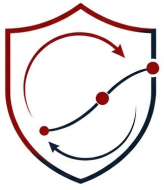
Principles of Robot Autonomy I

Models: Collisions and C-space, Kinematic/dynamics motion models
Planning and Control: A* path planning

Stanford University


SIA Lab
Safe and Intelligent Autonomy Lab

IPRL

Principles of Robot Autonomy I

Announcements:

- Lab section 1 this week
- HW 1 due next Thurs, Oct 8
- Slides uploaded to canvas before lecture
- Total enrolled: 160, total waitlist: 80
- Lab/OH schedule/Zoom links on the course website:
  https://stanfordasl.github.io/PoRA-I/aa274a_aut2627/

Lecture 3:

- Collision checking and C-space
- Robot kinematic/dynamic motion models
- Planning-control stack intro
- A* planning

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SIA Lab
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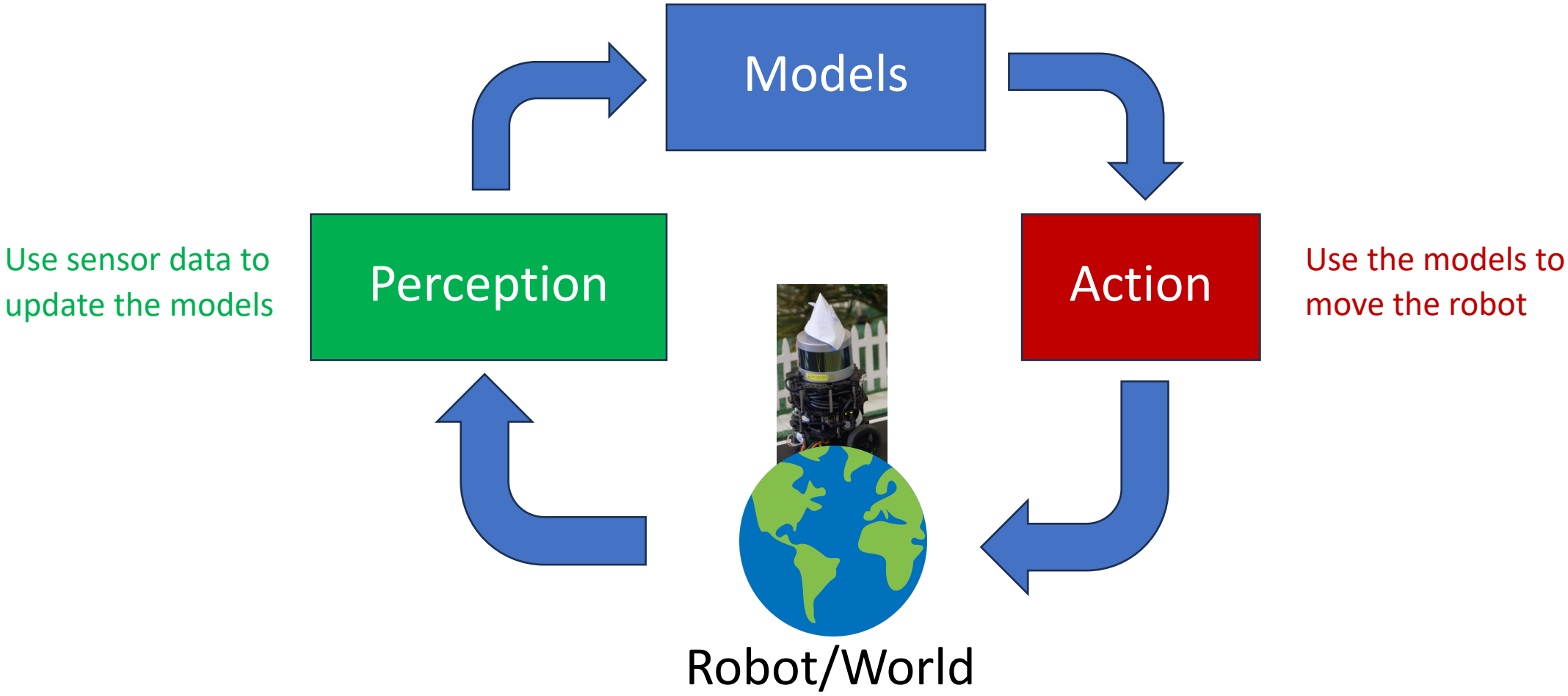
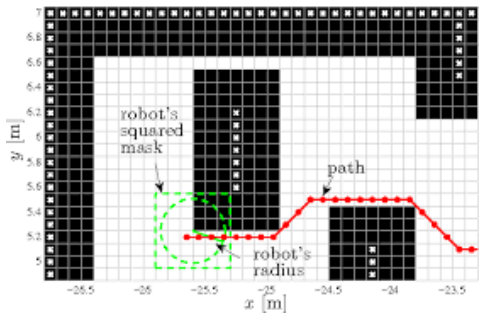
ASL
Autonomous Systems Lab

Full Stack Autonomy: the Perception-Action Loop

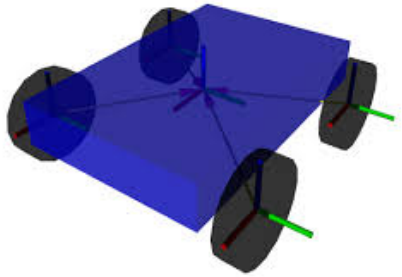


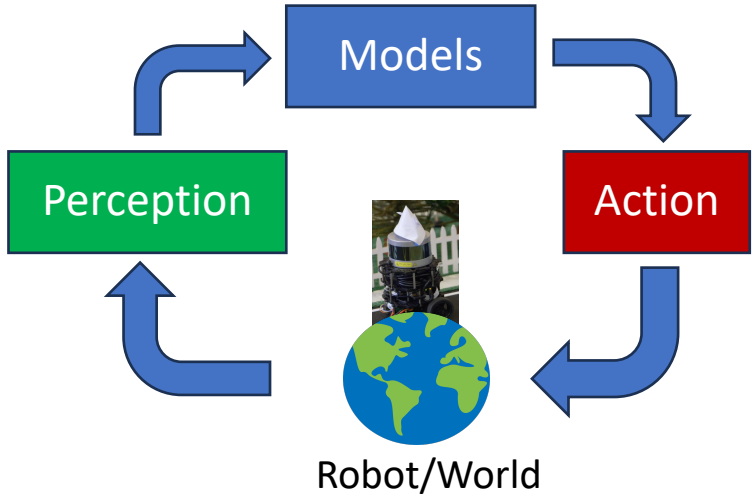
Models: Environment and Robot Representations

Maps:



Robot models
or simulators:



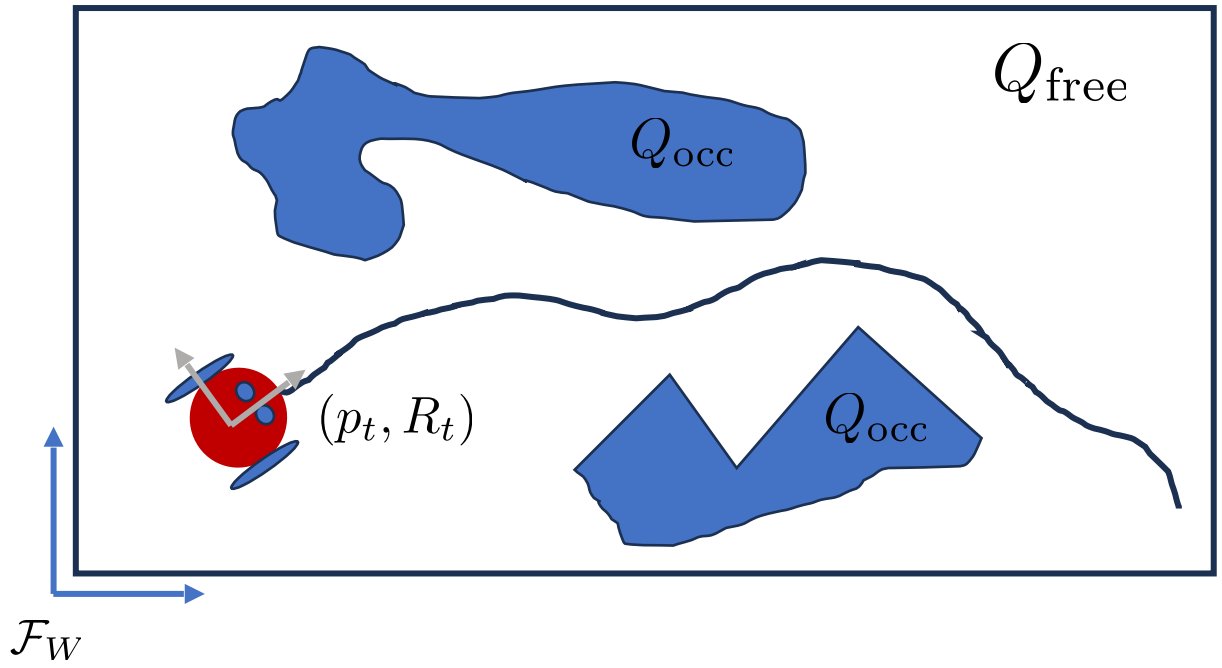


Robot-Map Collisions

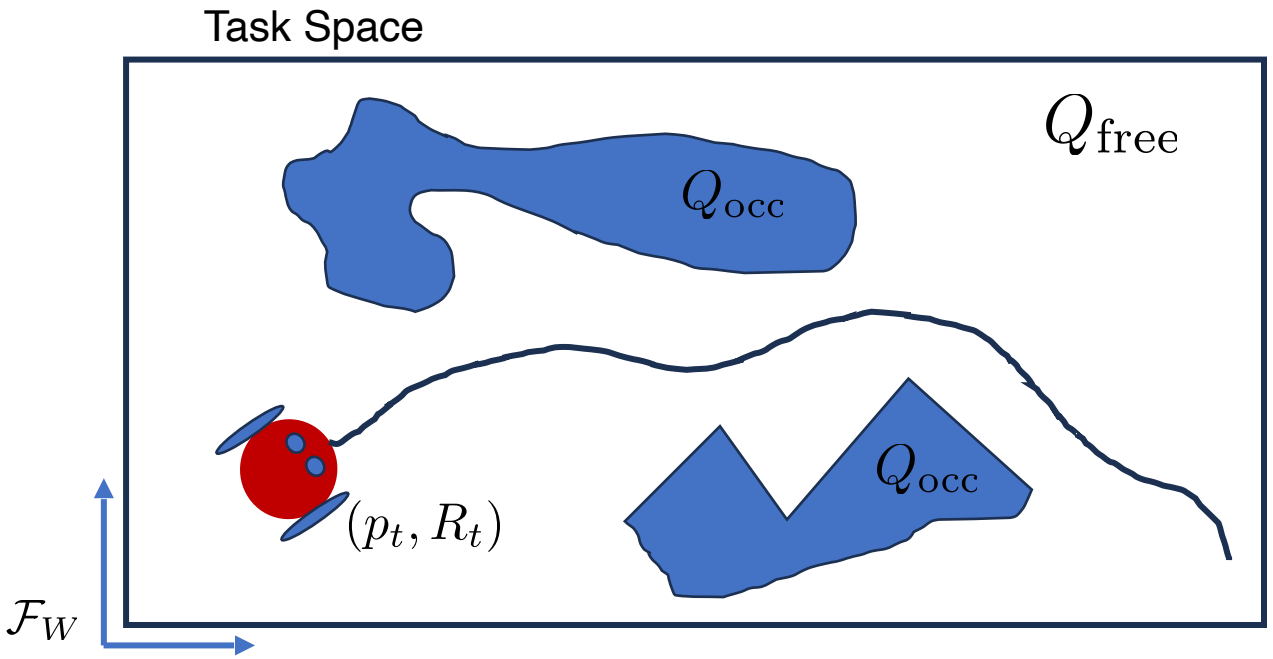
Main point: Use robot pose to represent robot body in world frame to check for **collisions** and to plan **collision free motion.**

Robot body in world frame:

$$B(p_t, R_t) = \{R_t b + p_t \mid b \in B\}$$



Collisions and C-Space



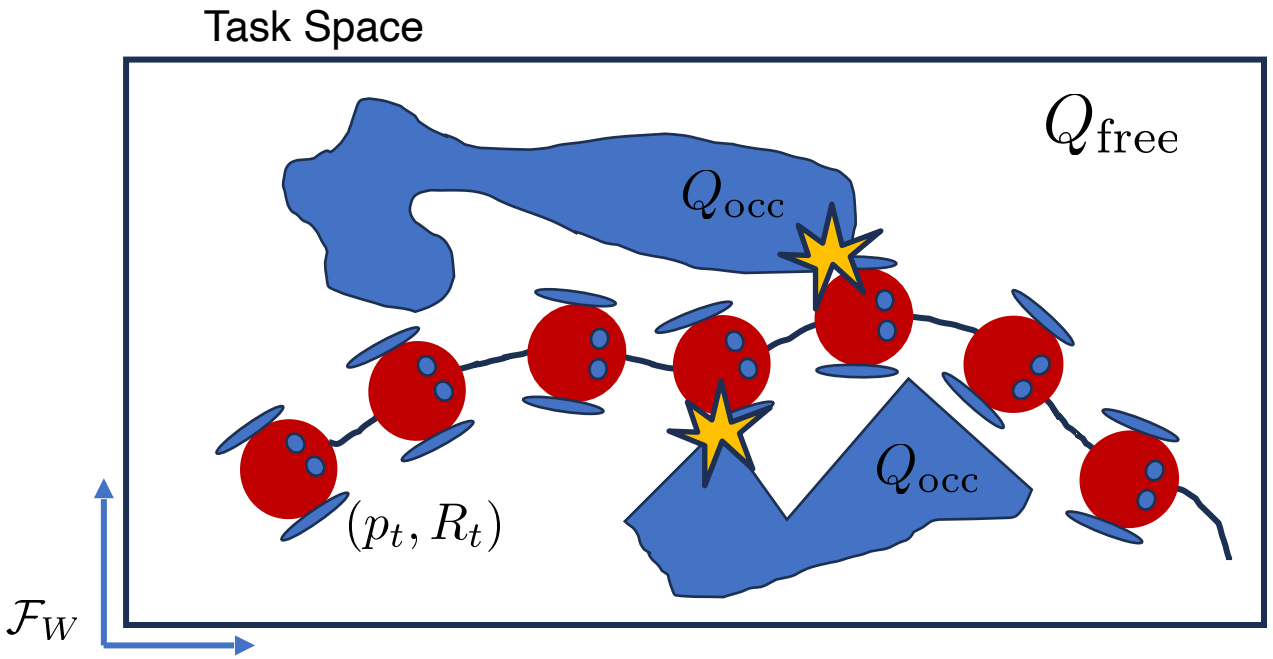
Robot body in world frame:

$$B(p_t, R_t) = \{R_t b + p_t \mid b \in B\}$$

Collision check: $B(p_t, R_t) \cap Q_{occ} = \emptyset$

Can be computationally challenging!

Collisions and C-Space



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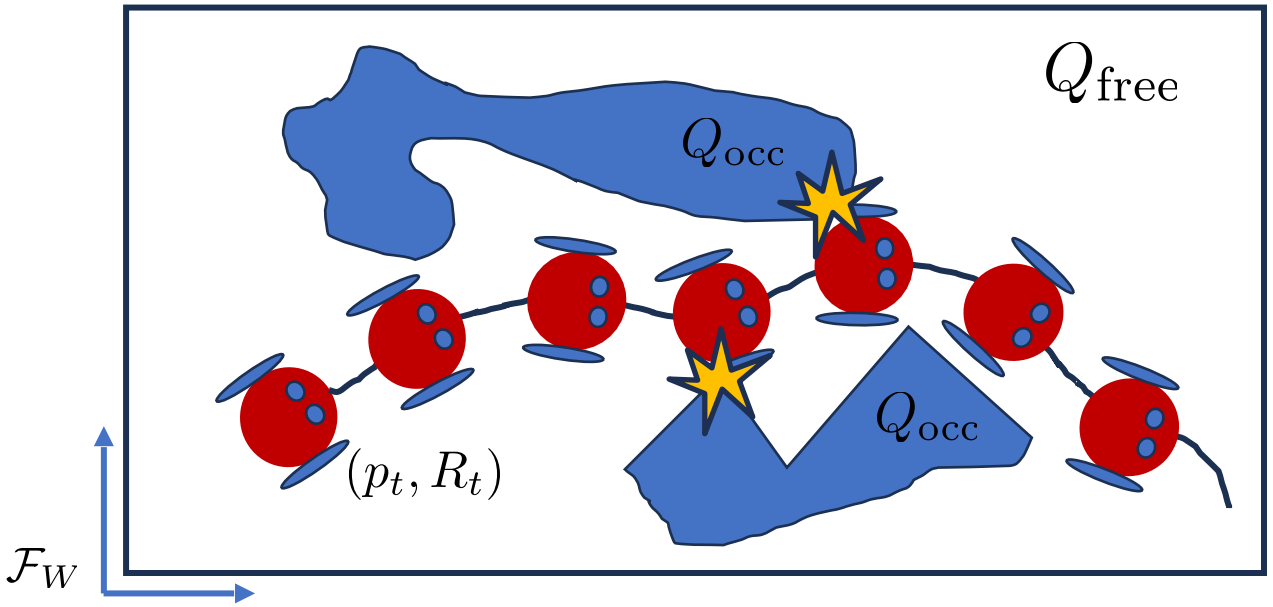
Collisions and C-Space

Task Space



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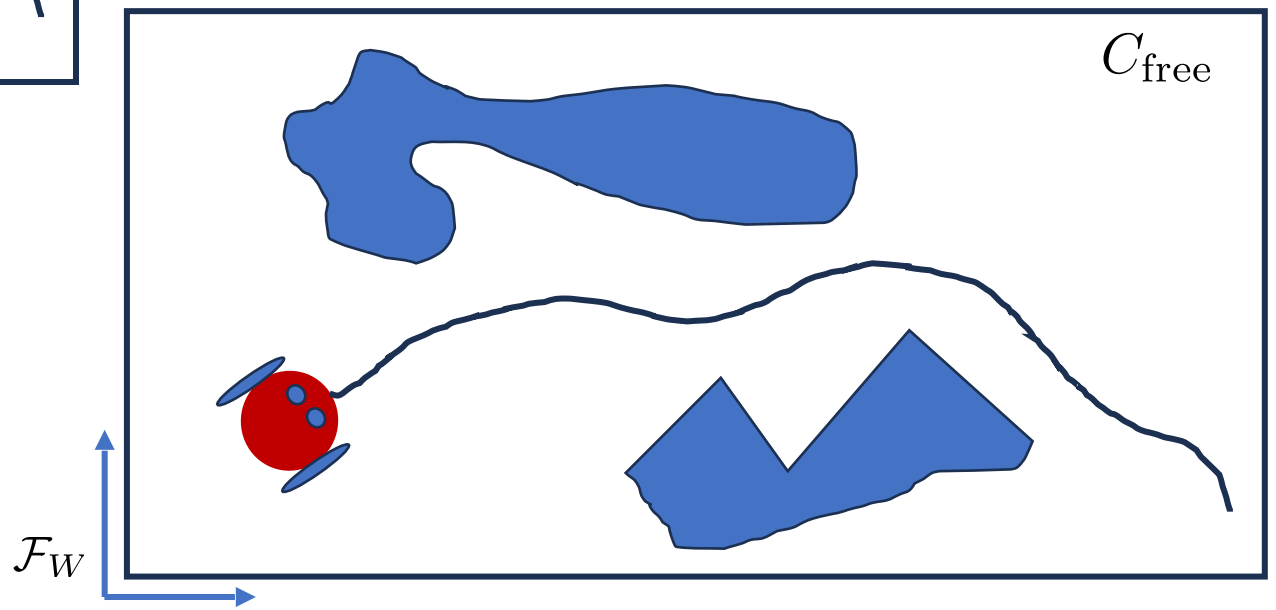
Configuration Space (C-Space)



Collisions and C-Space

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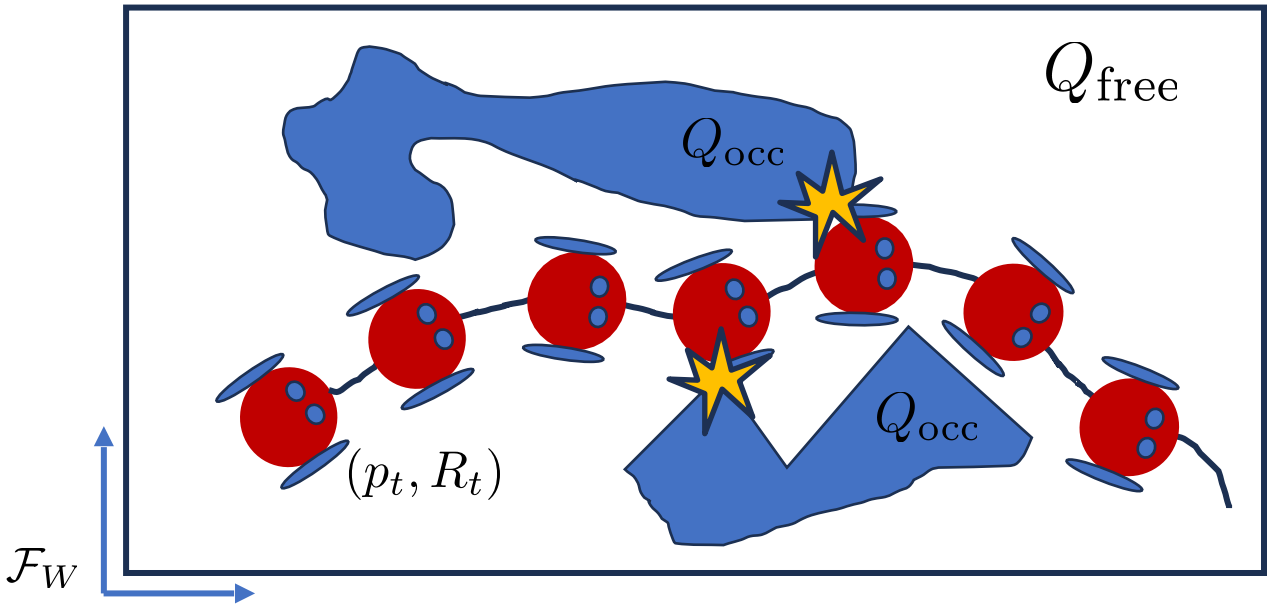

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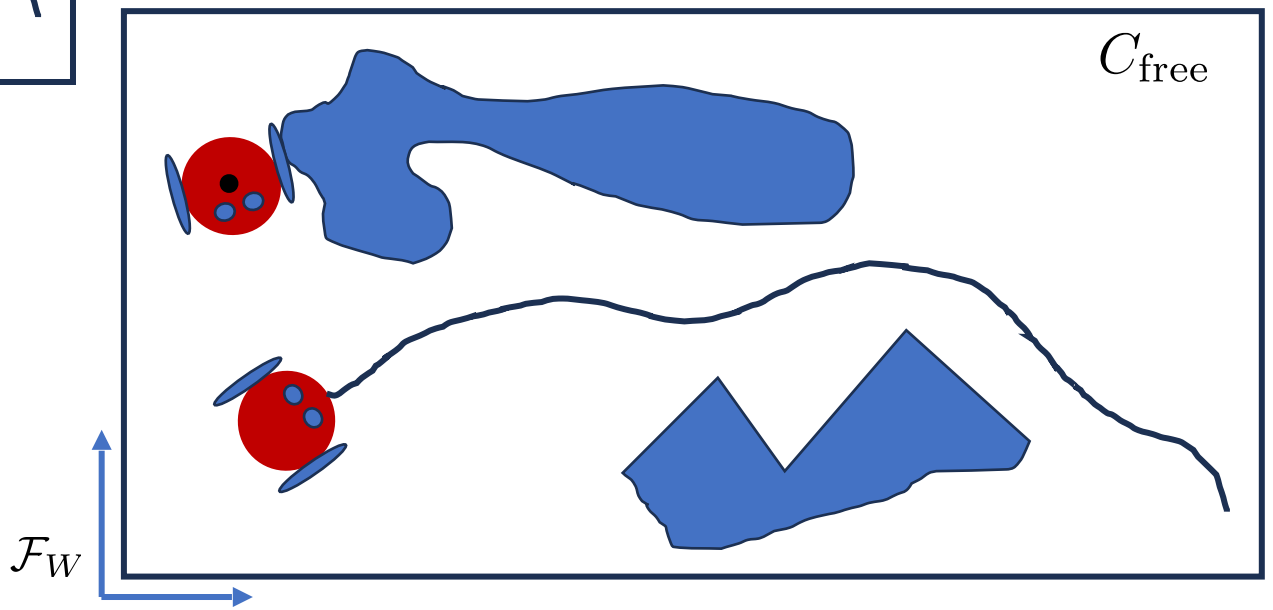

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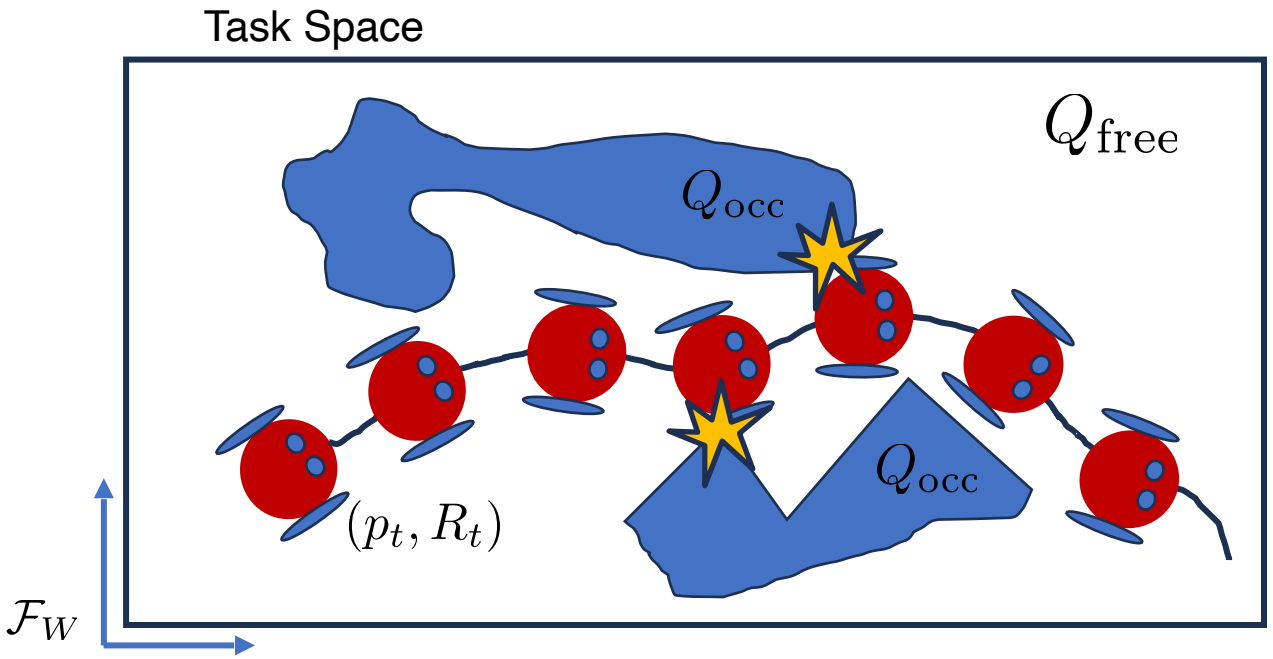

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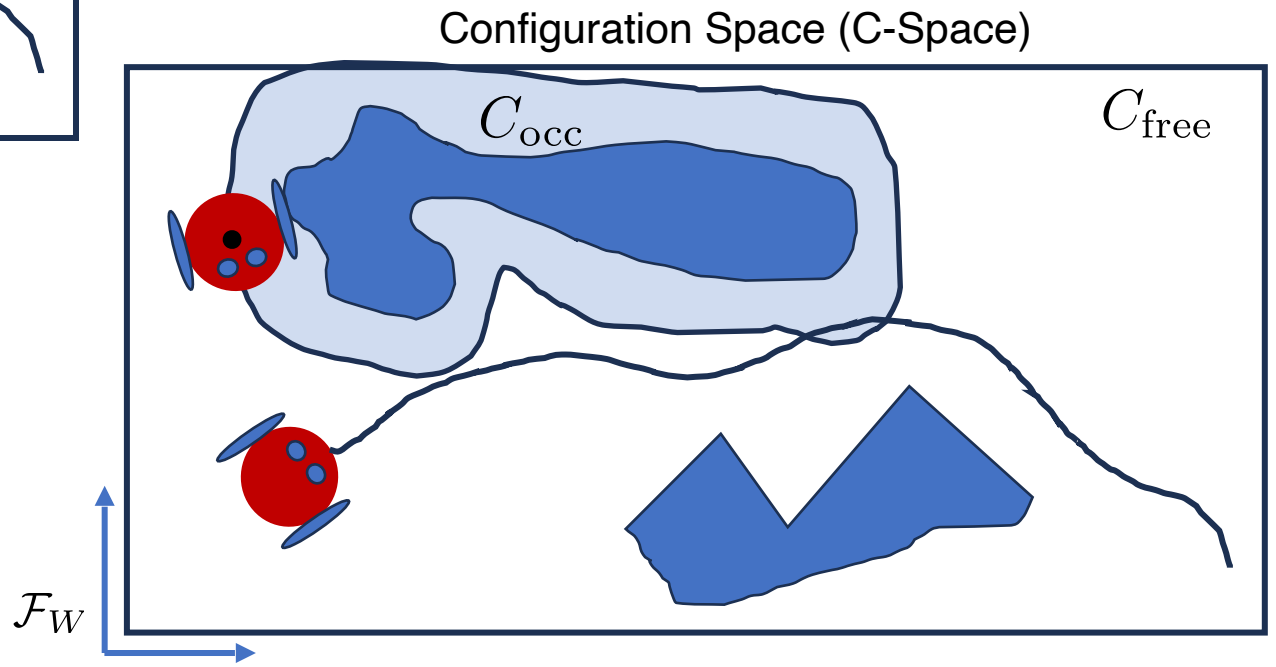
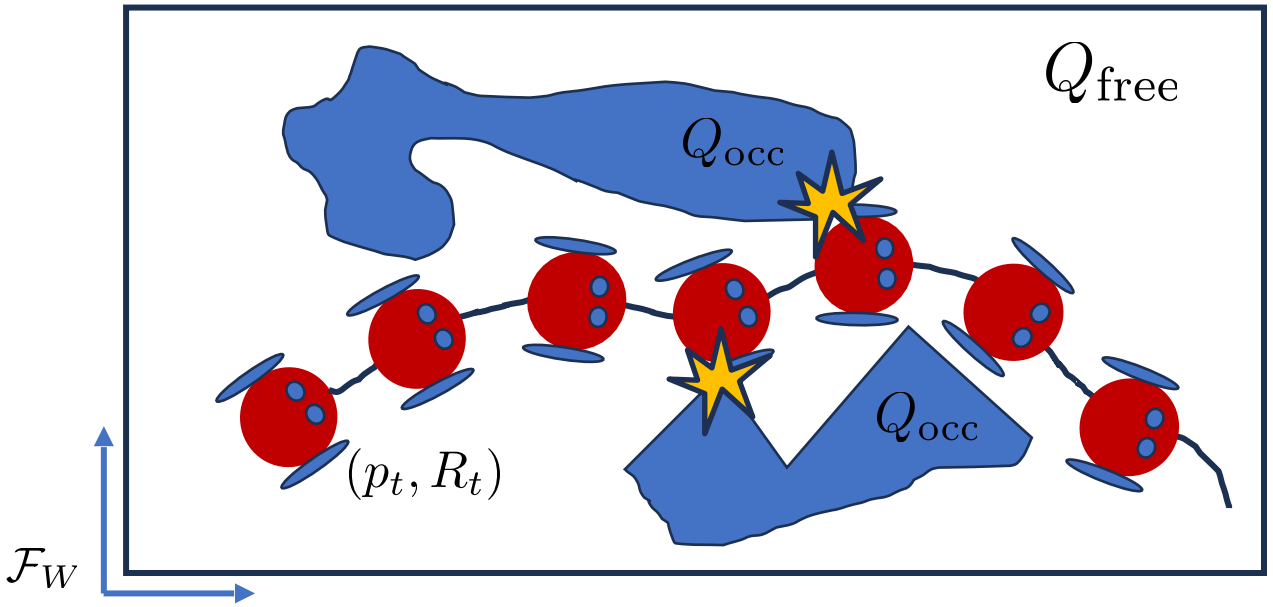


Collisions and C-Space

Task Space



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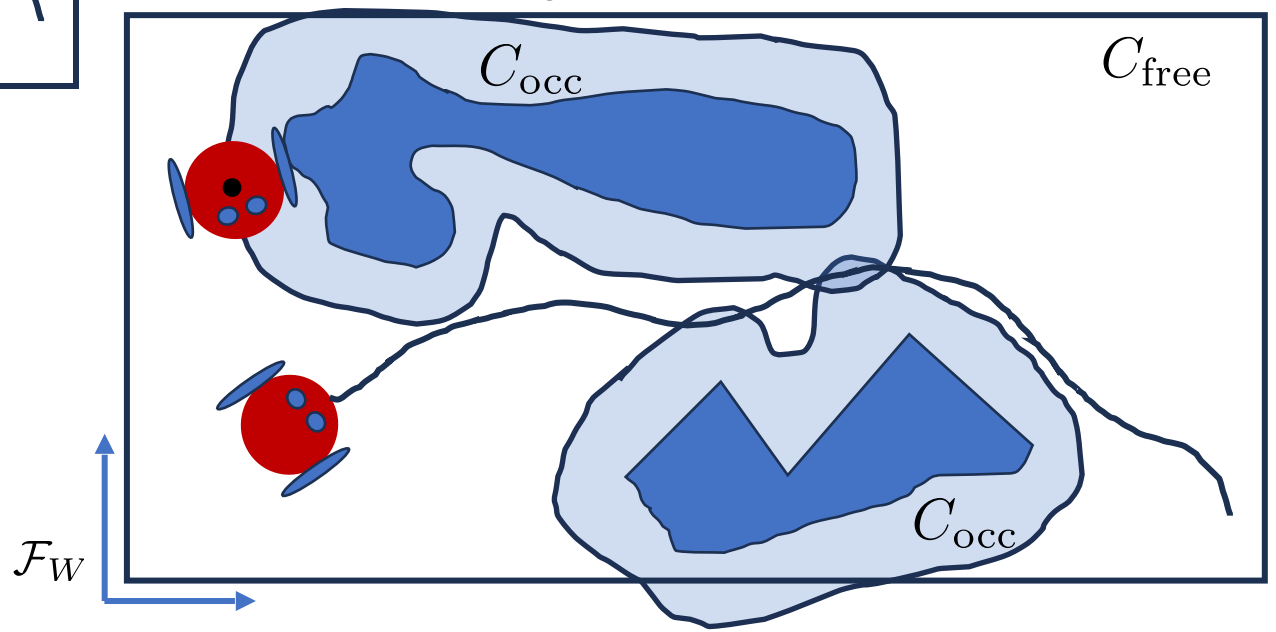
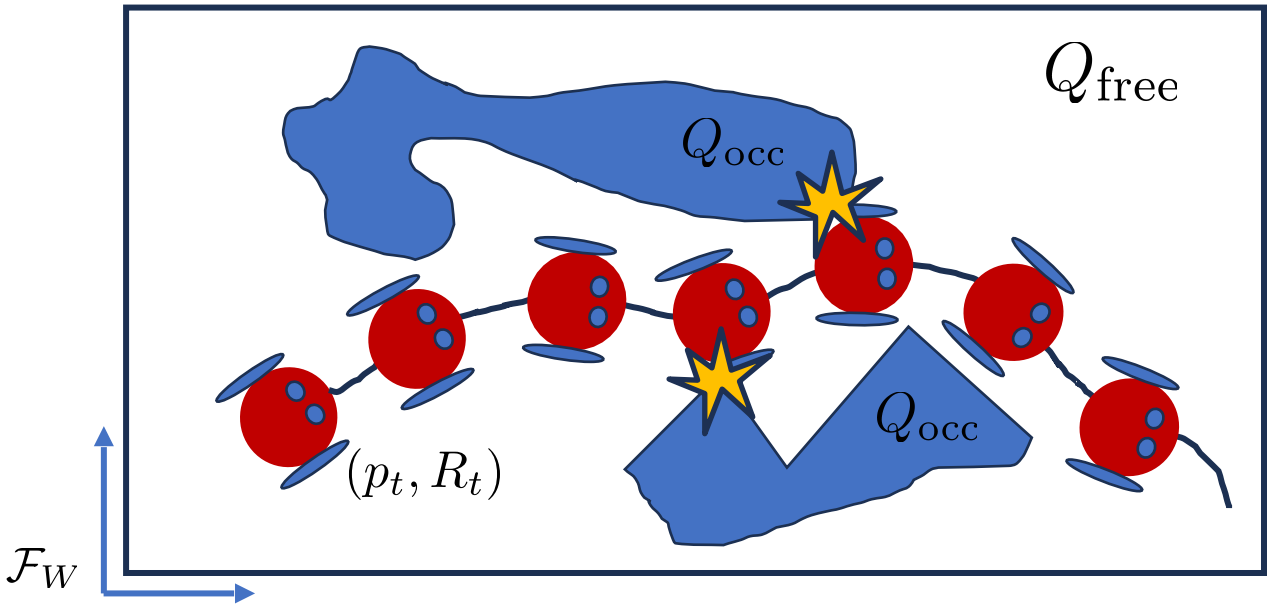


Collisions and C-Space

Task Space



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More rigorously, set of robot poses:

$$C_{\text{free}} = \{(p, \theta) \mid B(p, \theta) \cap Q_{\text{occ}} = \emptyset\}$$

$$C_{\text{occ}} = \{(p, \theta) \mid \exists b \in B(p, \theta) \text{ s.t. } b \in Q_{\text{occ}}\}$$

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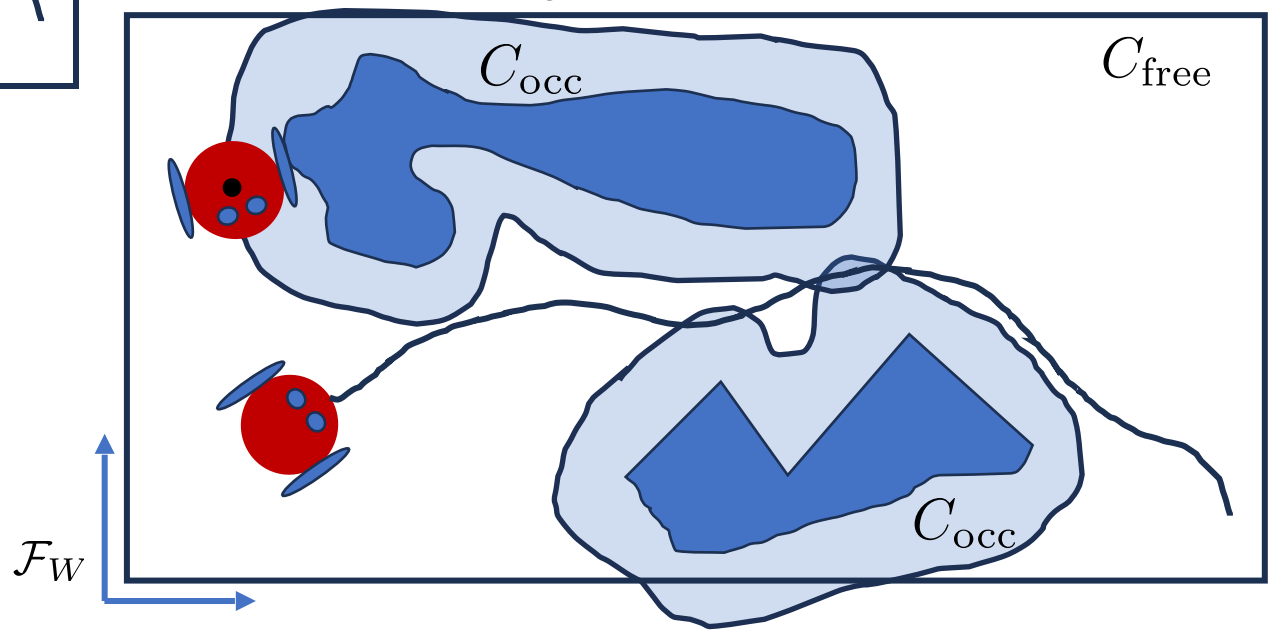
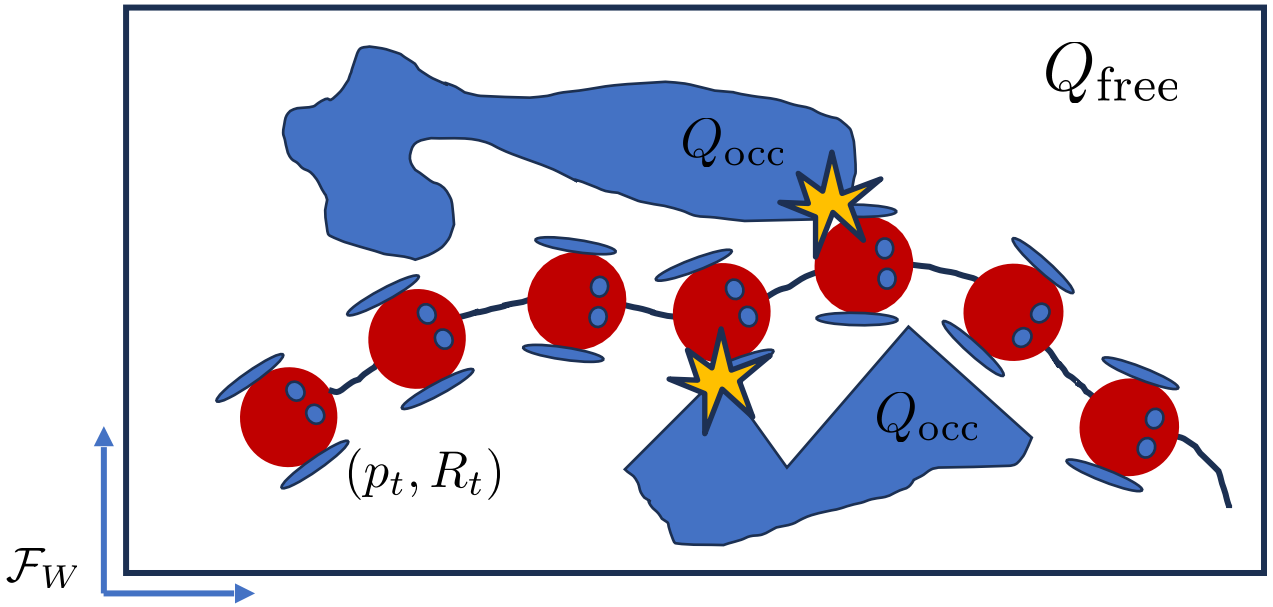


Collisions and C-Space

Task Space



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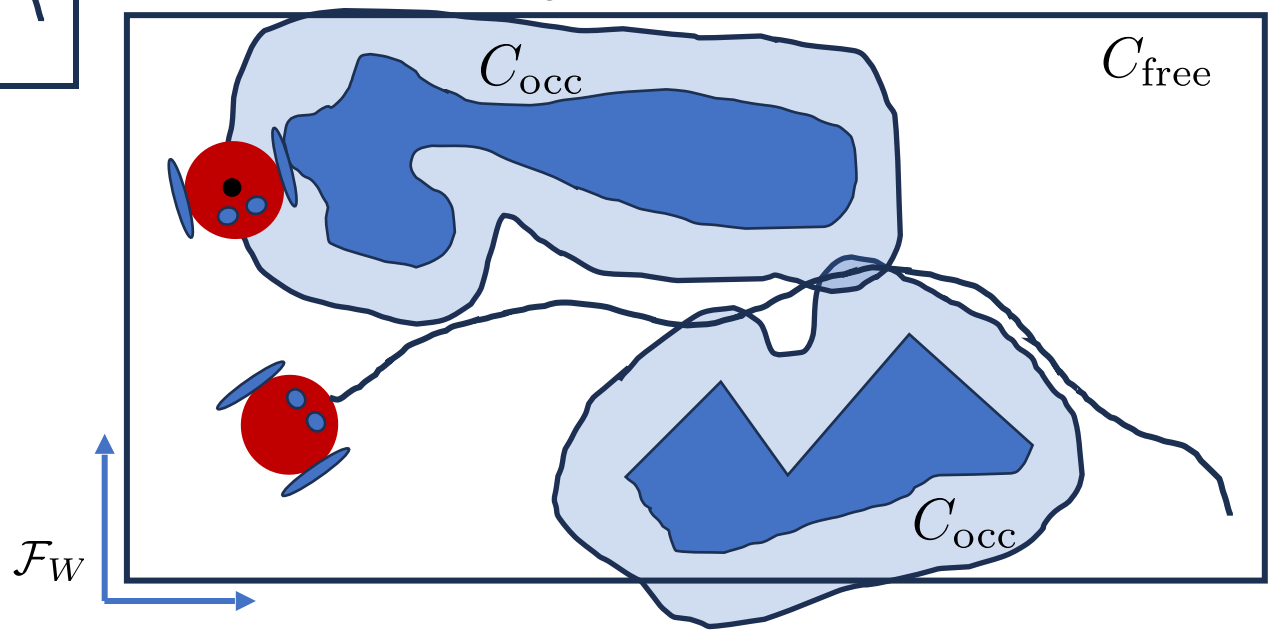
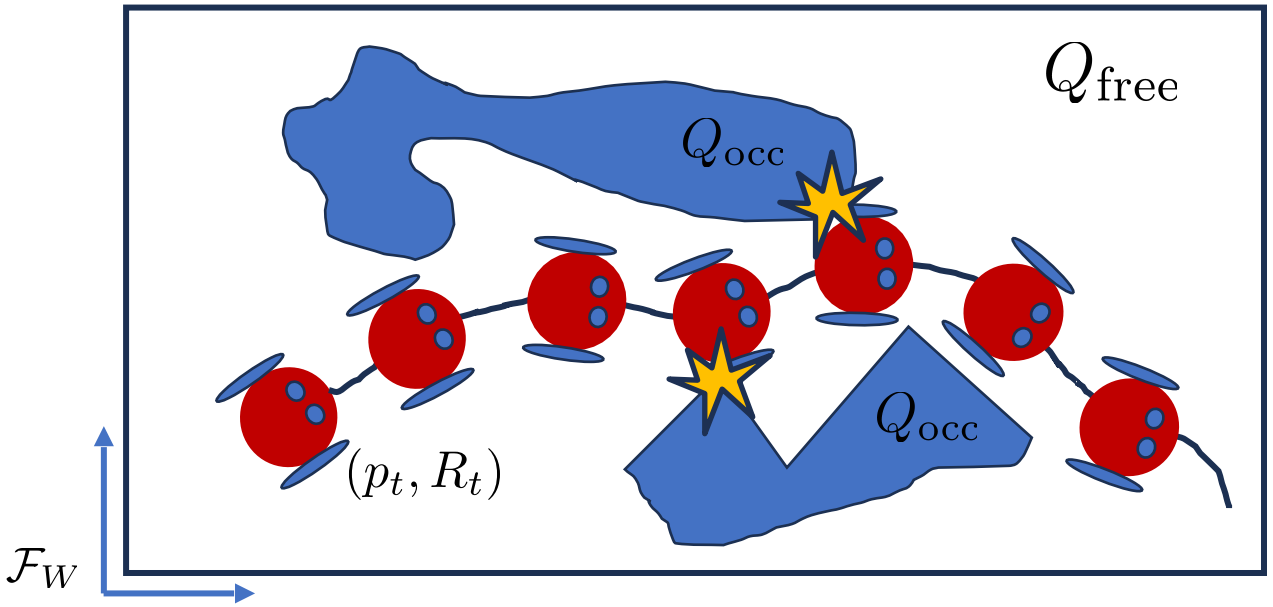
Can simplify to: $C_{\text{occ}} = \{(p, \theta) \mid p \in Q_{\text{occ}} \oplus -B(0, \theta)\}$

Configuration Space (C-Space)



Collisions and C-Space

Task Space



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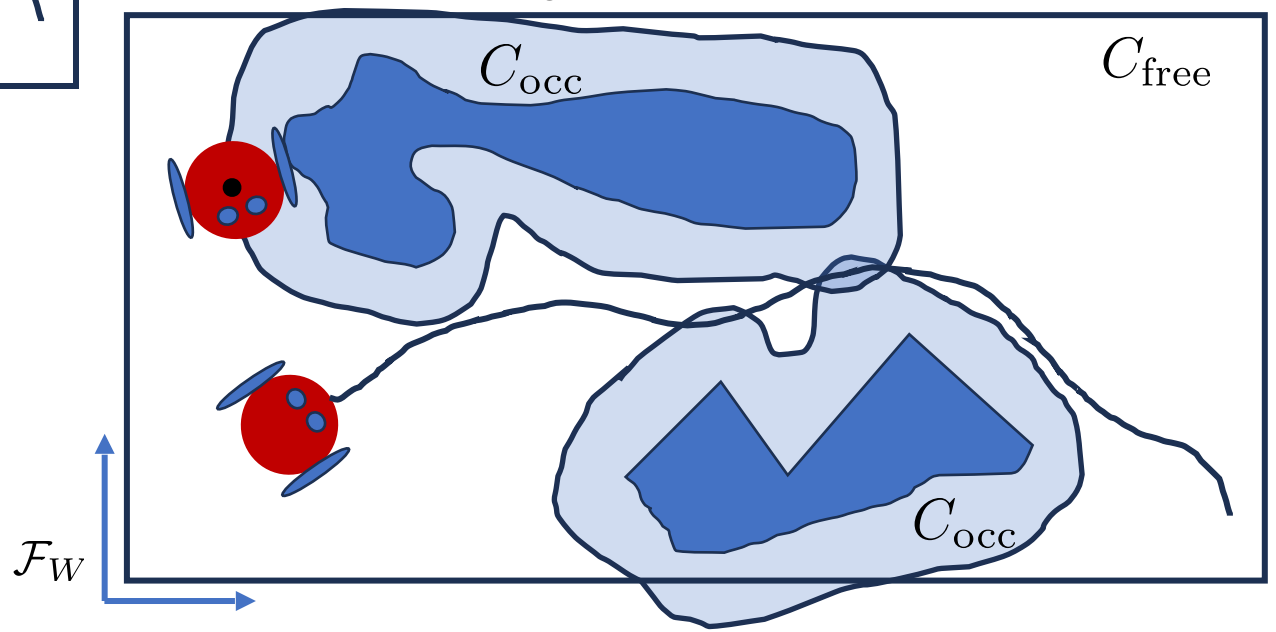
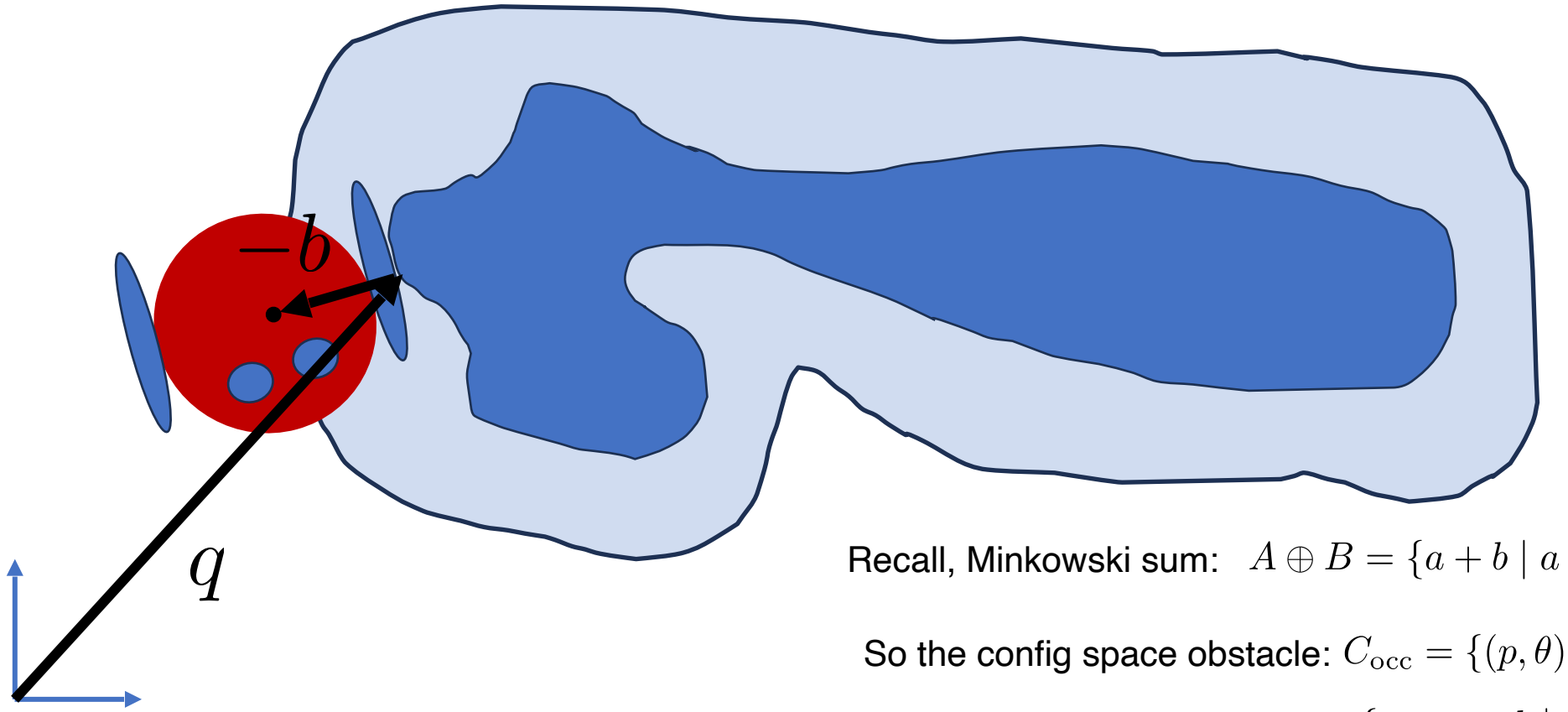
Collision check: $\boxed{(p, \theta) \in C_{\text{occ}}}$

Configuration Space (C-Space)



Challenges and Practicalities

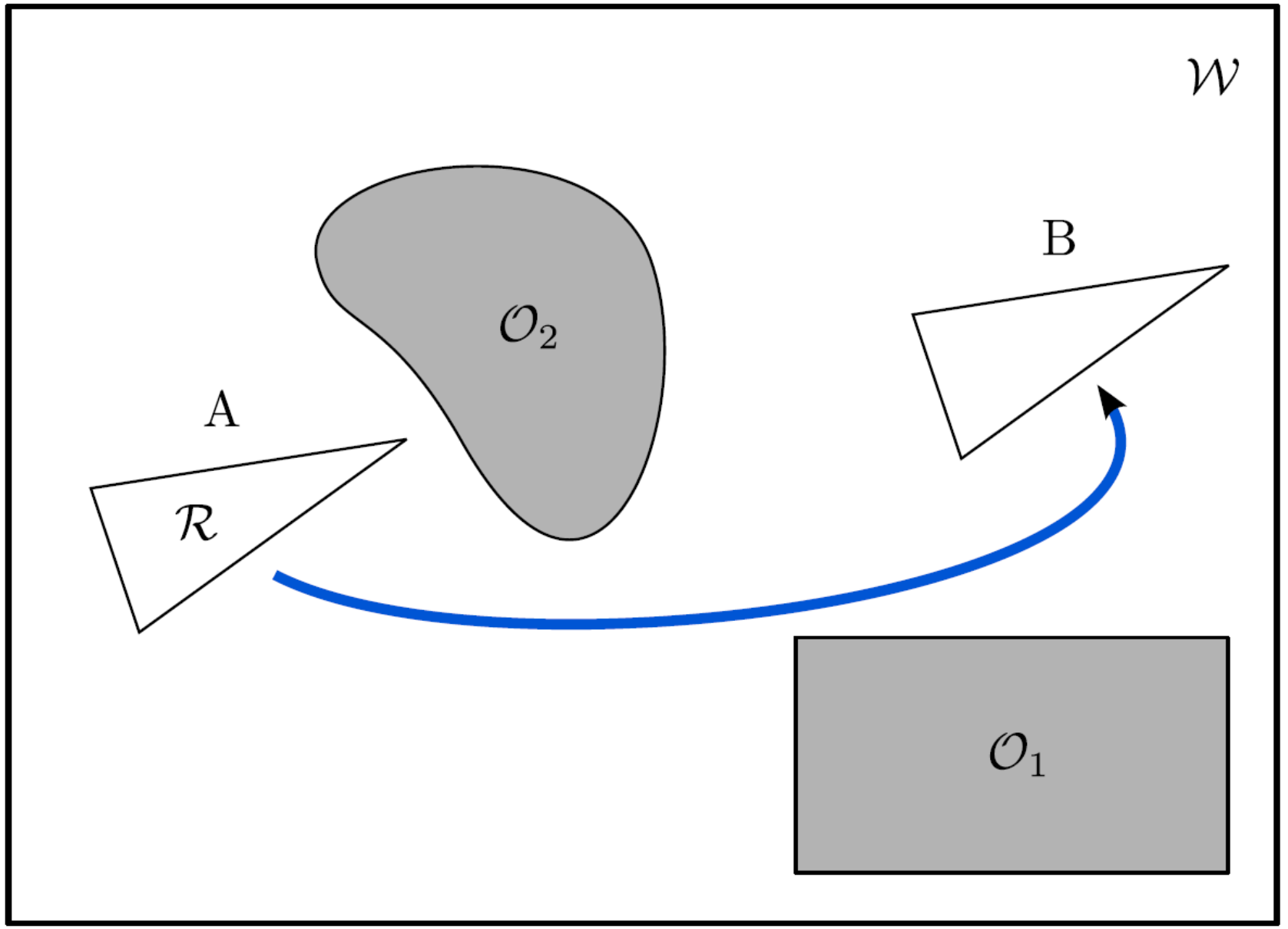


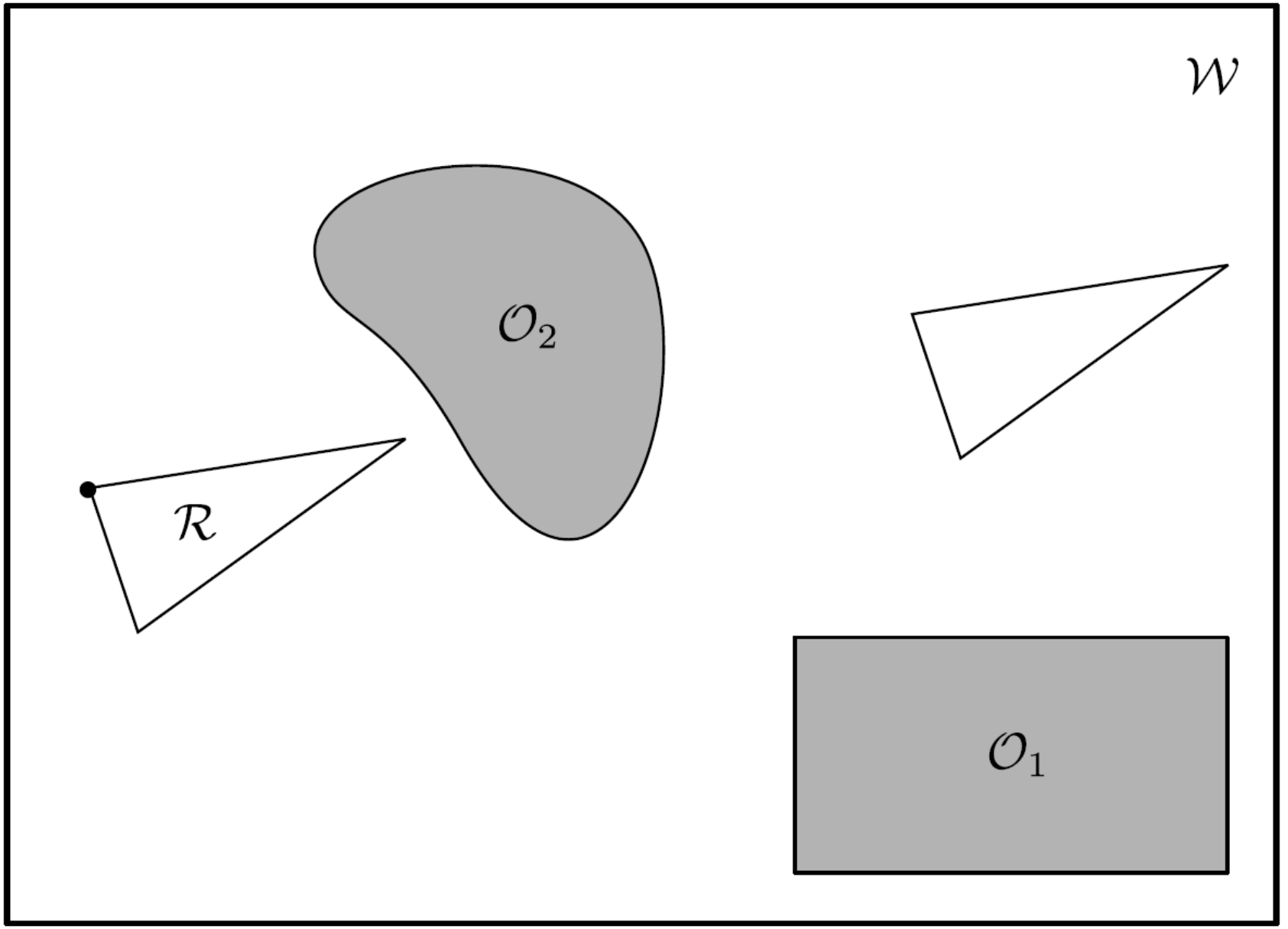
Recall, Minkowski sum: $A \oplus B = \{a + b \mid a \in A, b \in B\}$

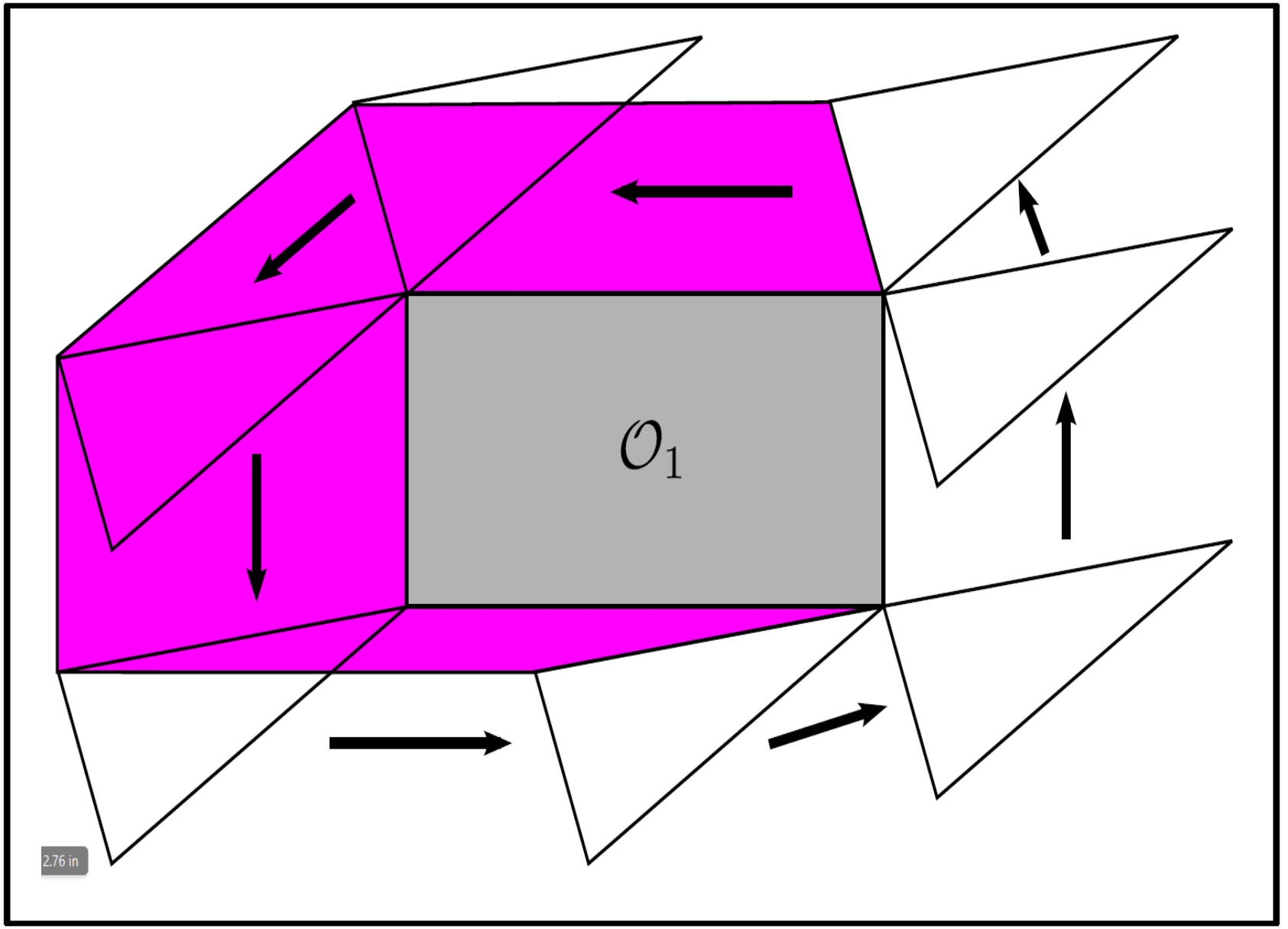
So the config space obstacle: $C_{\text{occ}} = \{(p, \theta) \mid p \in Q_{\text{occ}} \oplus -B(0, \theta)\}$

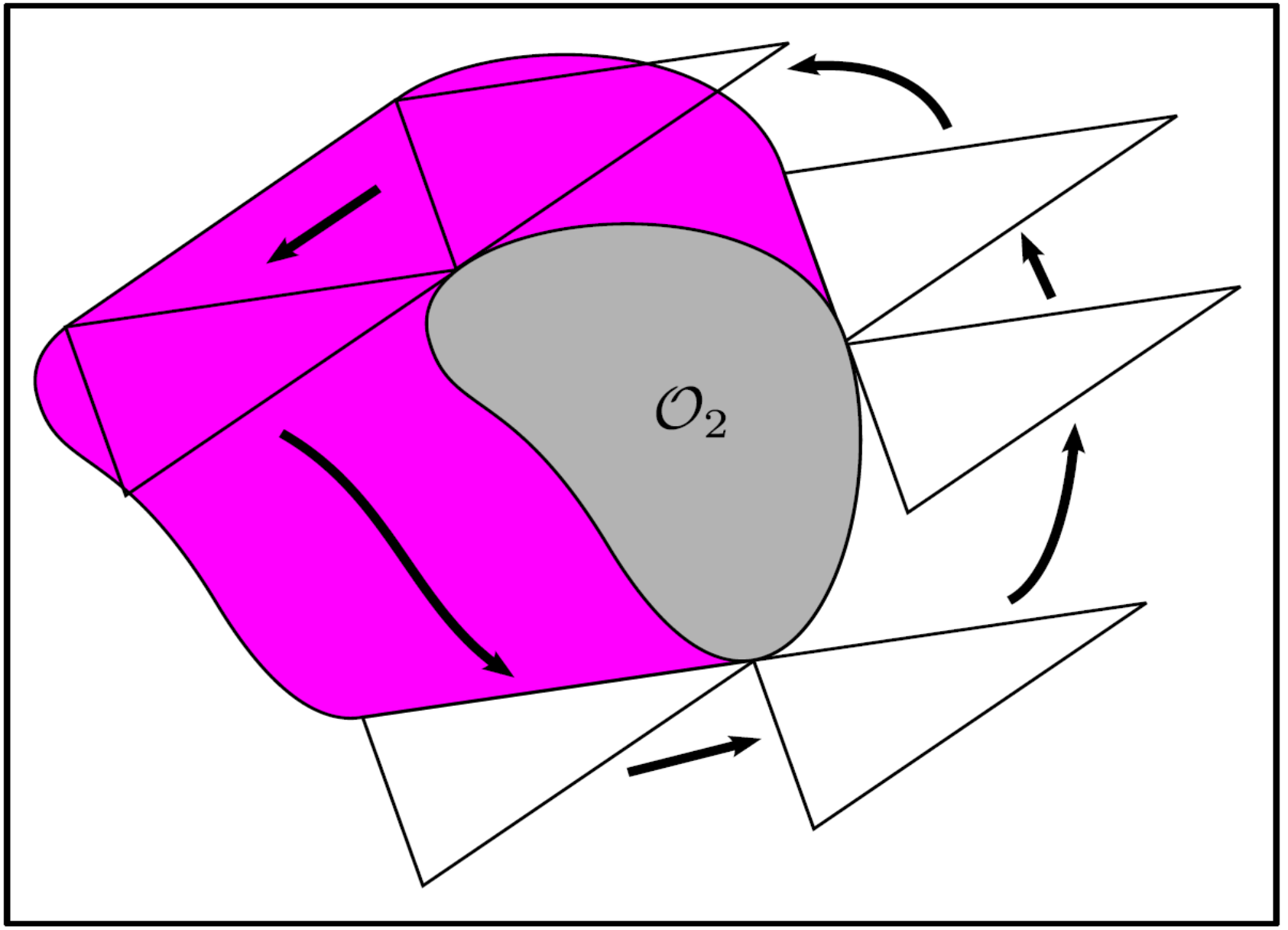
Just means: $\{p = q - b \mid q \in Q_{\text{occ}}, b \in B(0, \theta)\}$

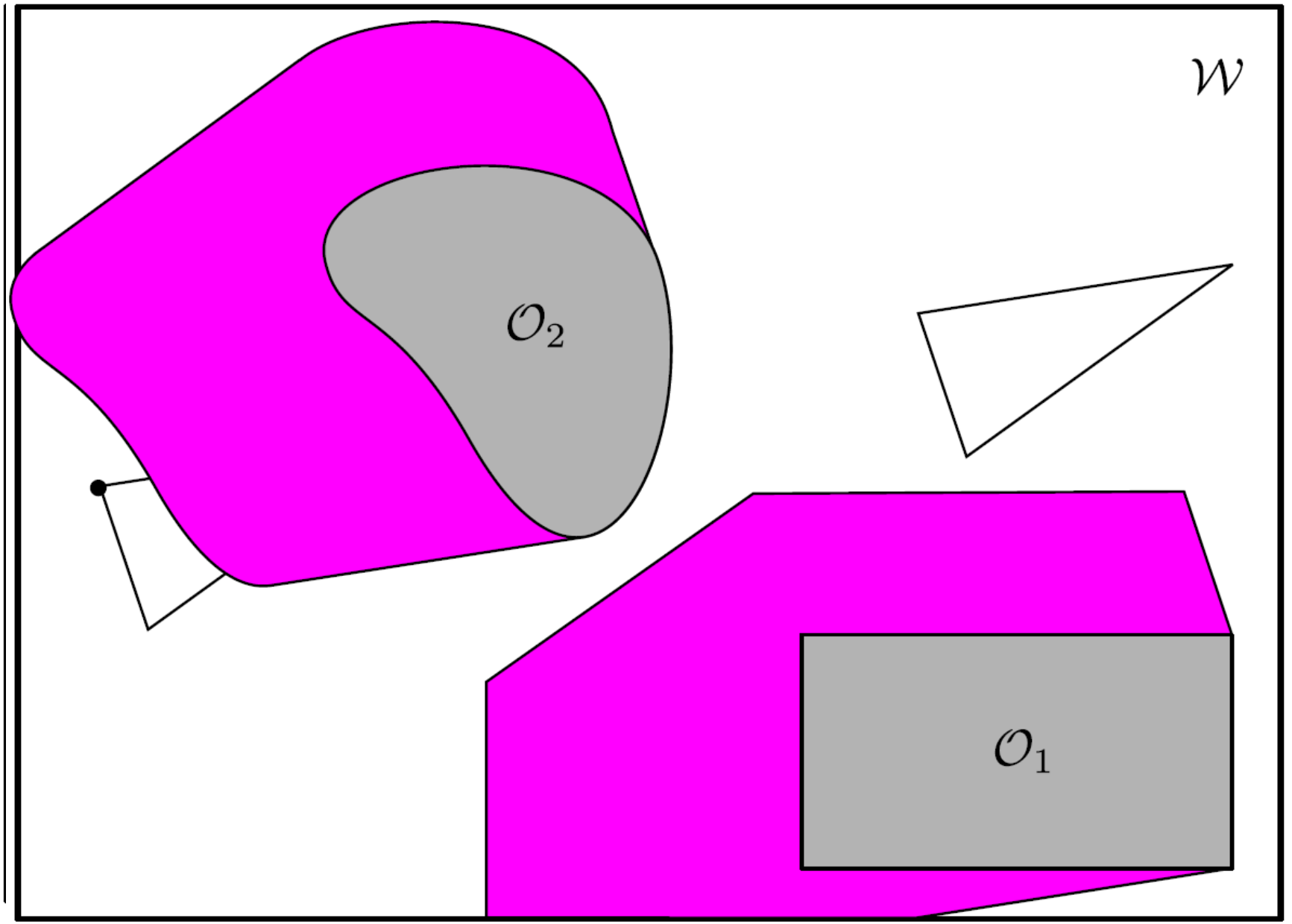
Useful conceptual tool, but In practice this is all very difficult to compute! We usually approx. B as circle or sphere, or use specialized collisions checking algorithms.


$\mathcal{W}$
$\mathcal{O}_2$
A
B
$\mathcal{R}$
$\mathcal{O}_1$


$\mathcal{W}$
$\mathcal{O}_2$
$\mathcal{R}$
$\mathcal{O}_1$


$\mathcal{O}_1$
2.76 in


$\mathcal{O}_2$


$\mathcal{W}$
$\mathcal{O}_2$
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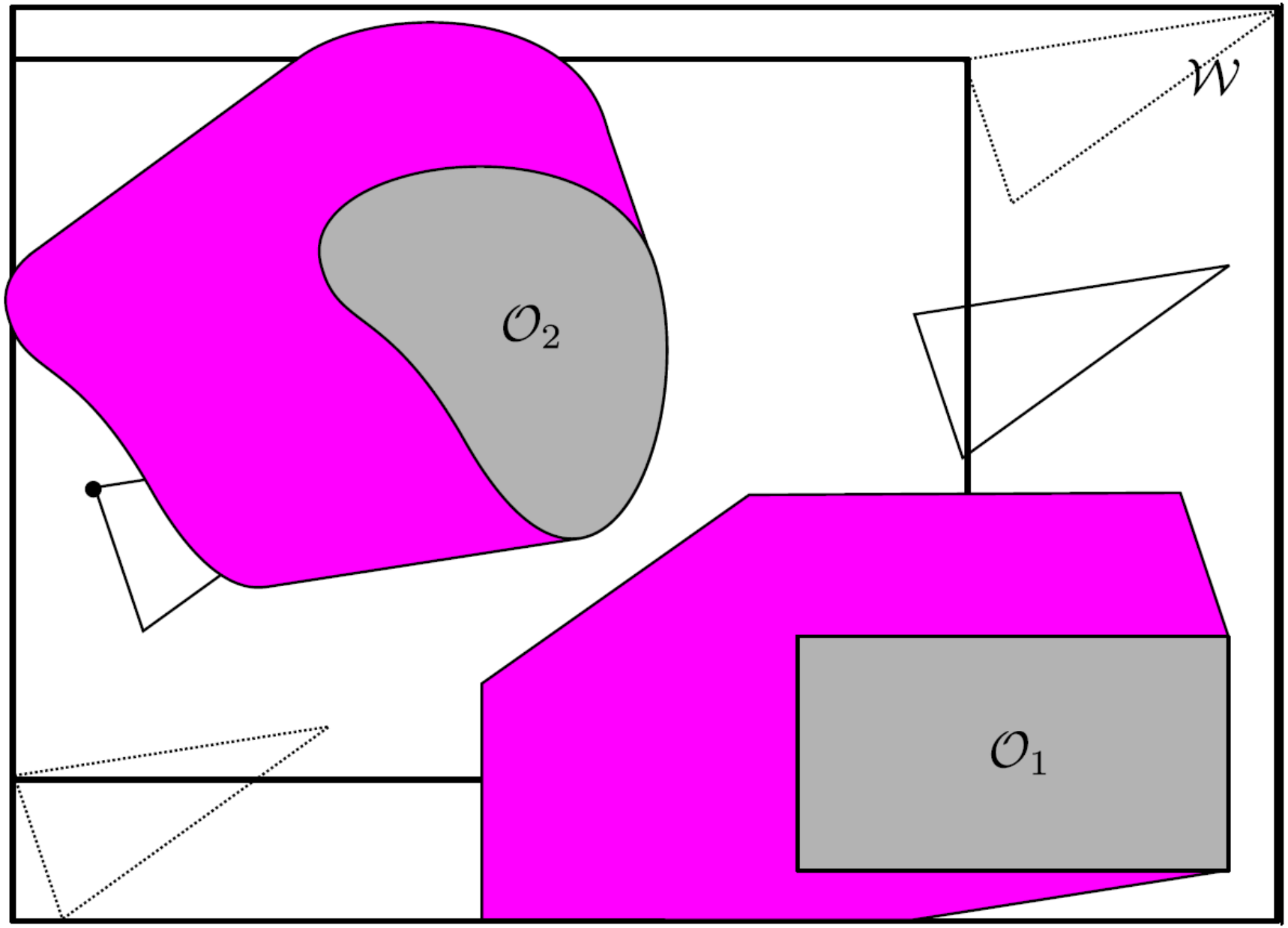
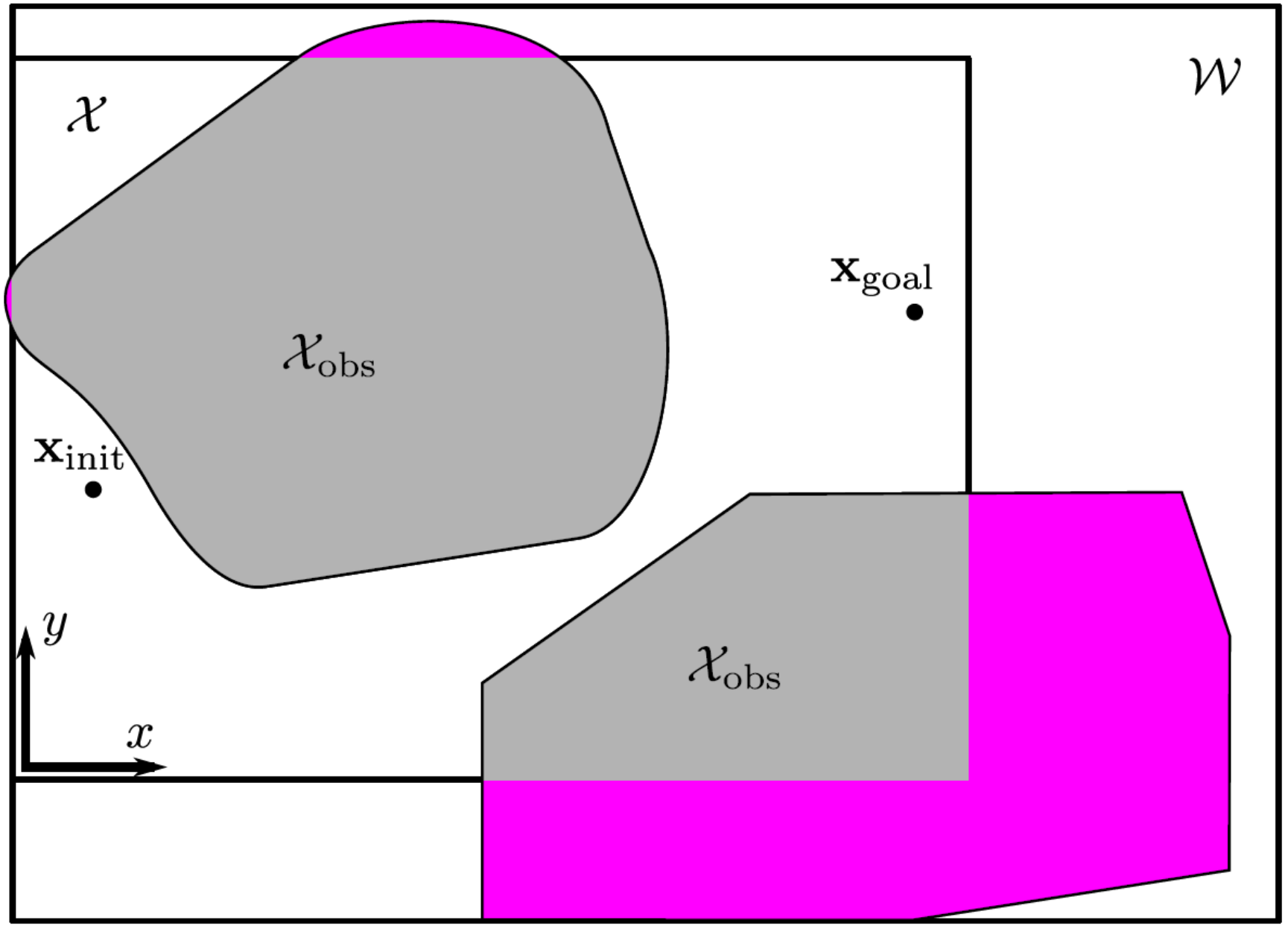
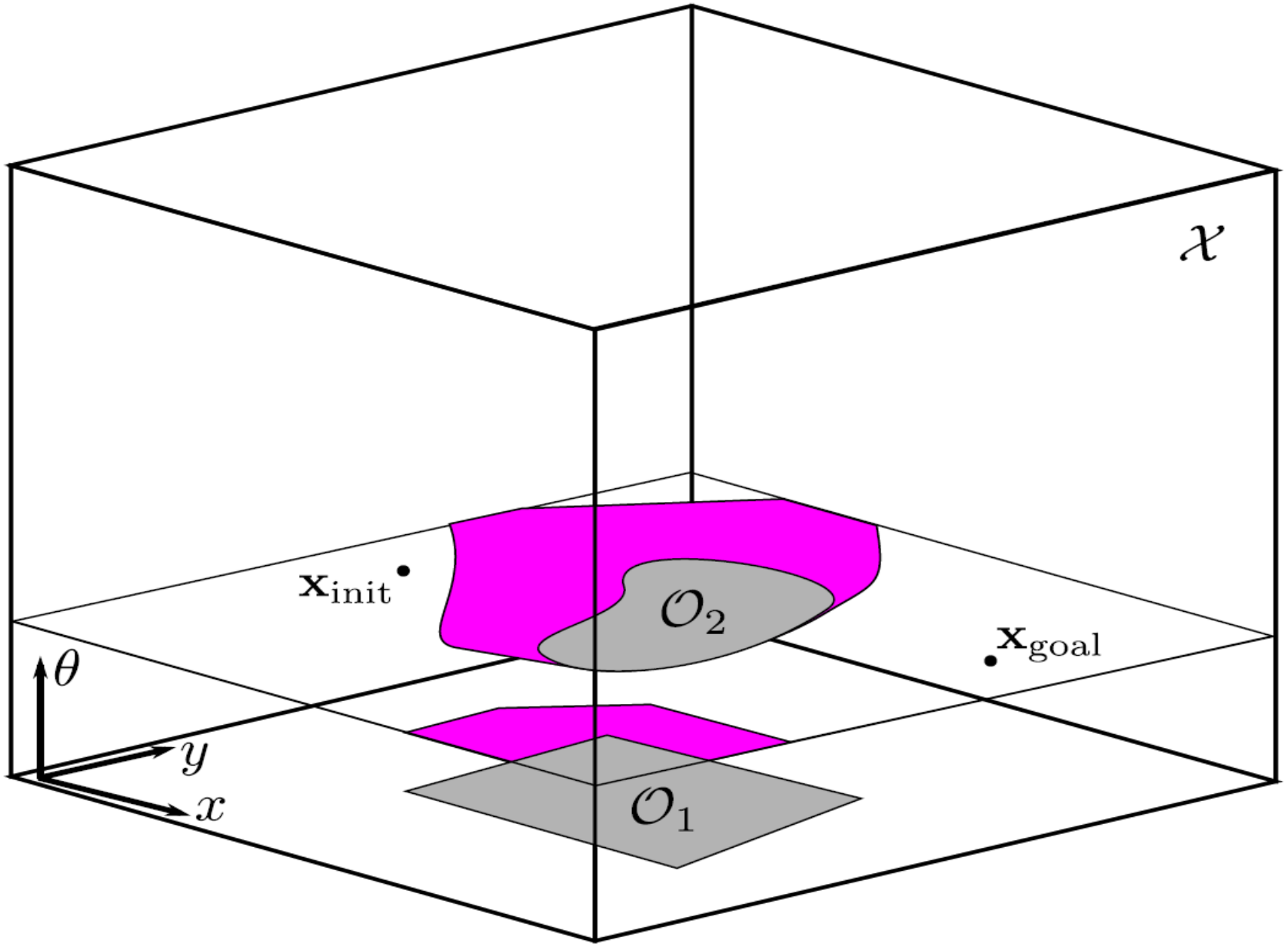
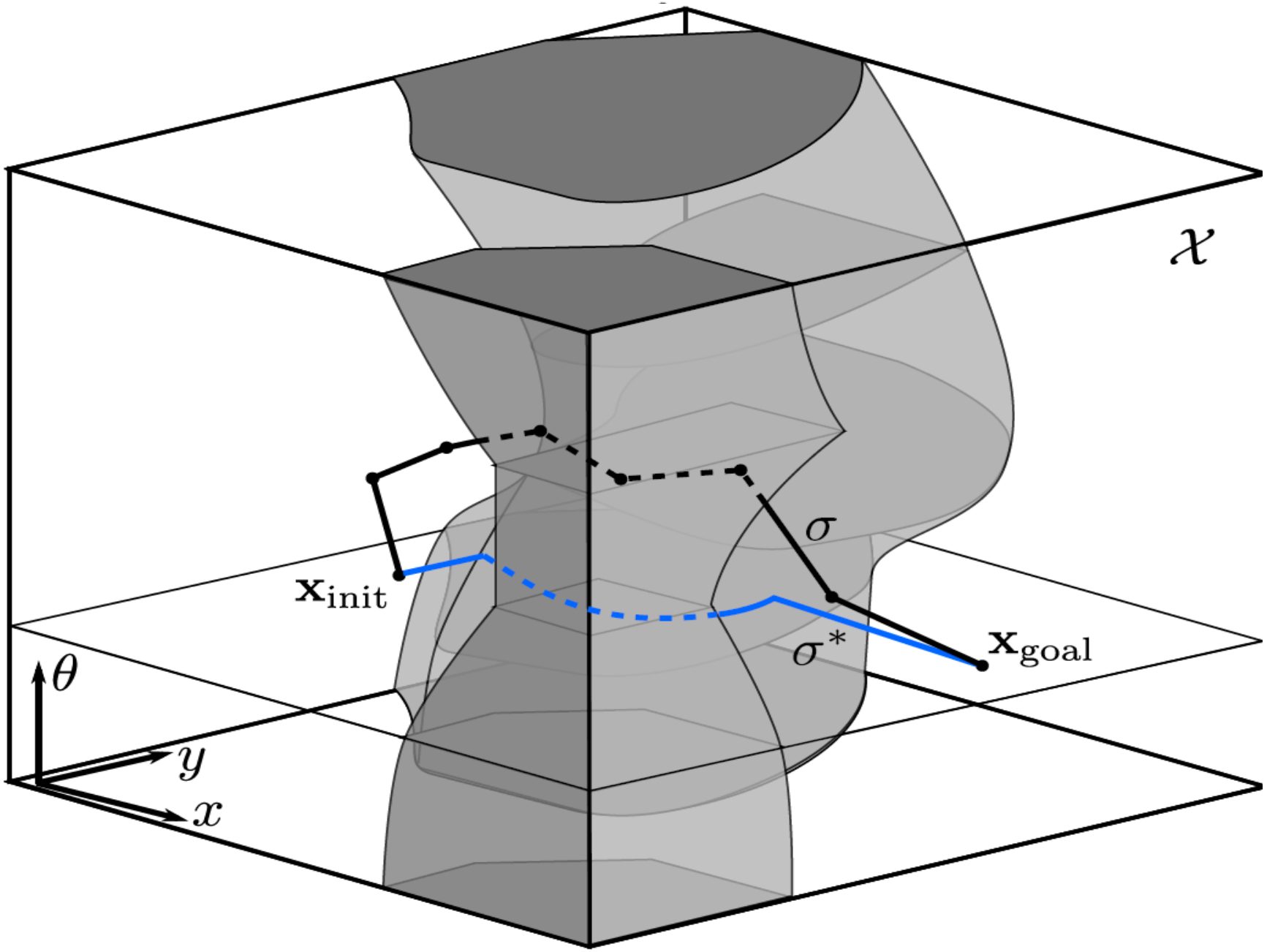
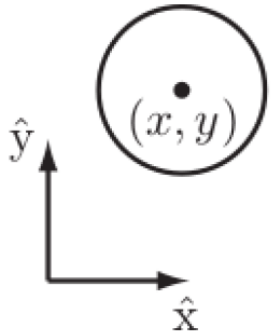

$\mathcal{W}$
$\mathcal{O}_2$
$\mathcal{O}_1$


$\mathcal{X}$
$\mathcal{W}$
$\mathcal{X}_{\text{obs}}$
$\mathbf{x}_{\text{goal}}$
$\mathbf{x}_{\text{init}}$
$\mathcal{X}_{\text{obs}}$
y
x

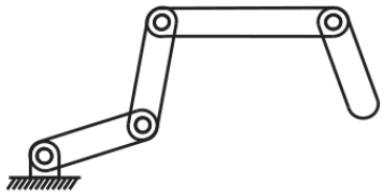

$\mathcal{X}$
$\mathbf{x}_{\text{init}}$
$\mathcal{O}_2$
$\mathbf{x}_{\text{goal}}$
θ
y
x
$\mathcal{O}_1$


$\mathcal{X}$
σ
$\mathbf{x}_{\text{init}}$
σ^*
$\mathbf{x}_{\text{goal}}$
θ
y
x

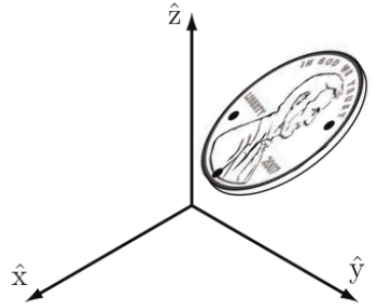
Configuration space examples



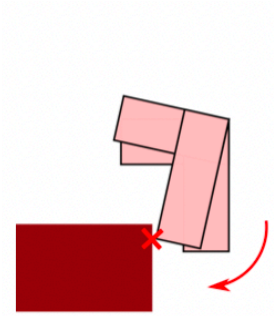
$d = 2$

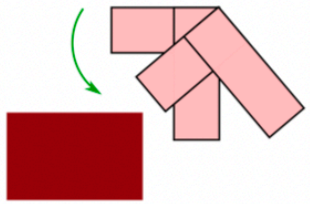


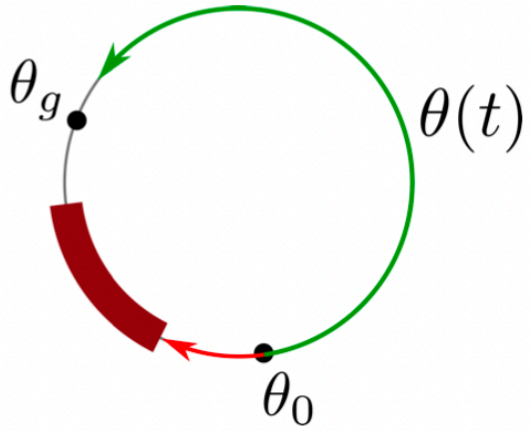
$d = 4$



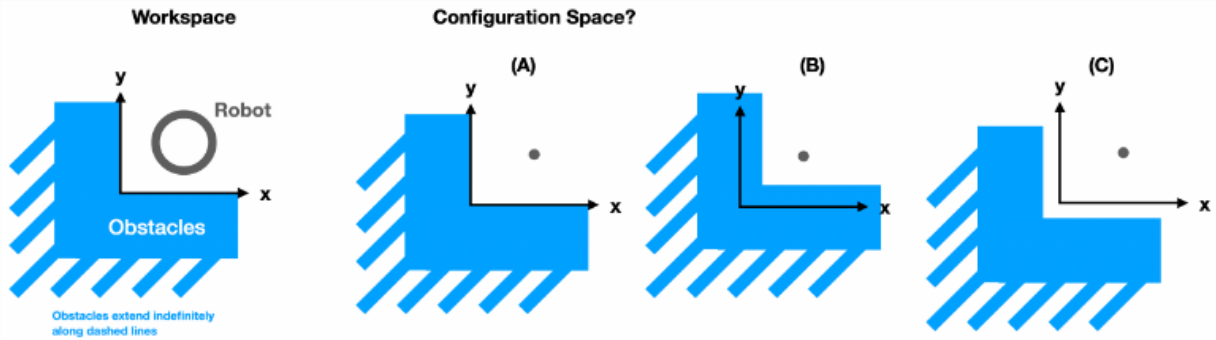
$d = 6$







For the robot workspace on the left with blue obstacles, what is the correct configuration space?



(A)

(B)

(C)

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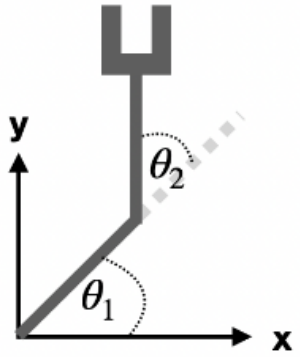
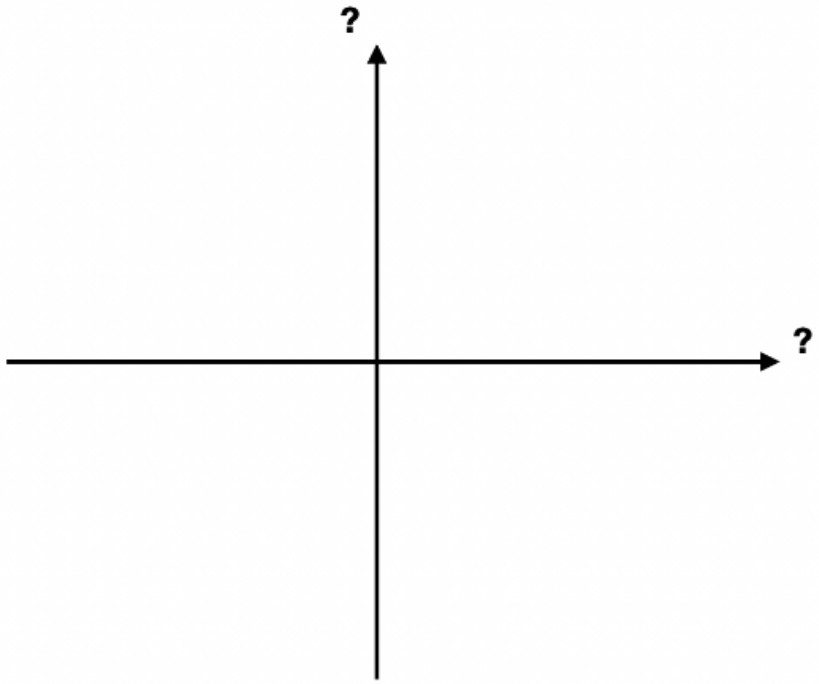
For the current state of the two-link robot robot arm on the left, click on the corresponding point in the configuration space on the right





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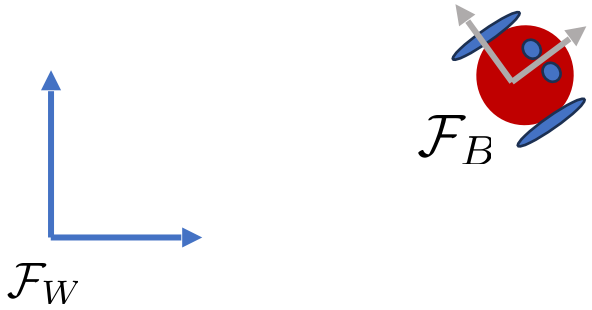
Robot Kinematics and Dynamics



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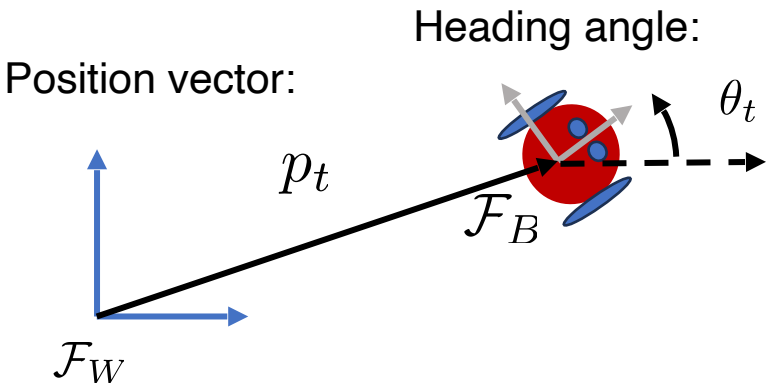

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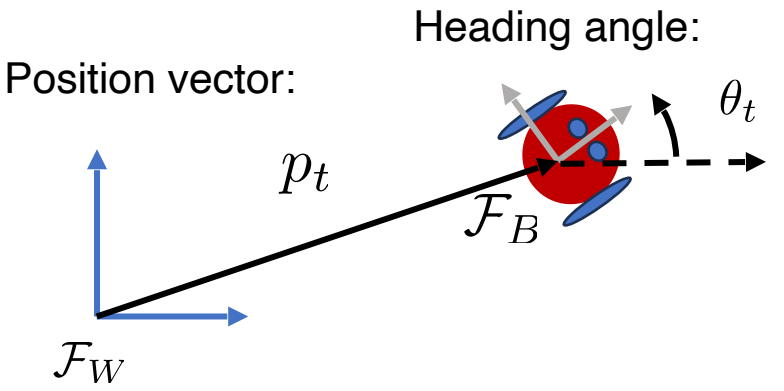
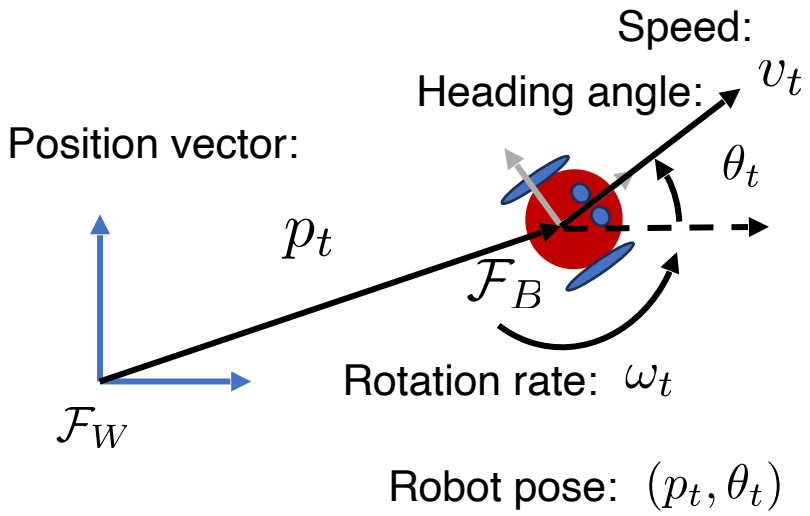
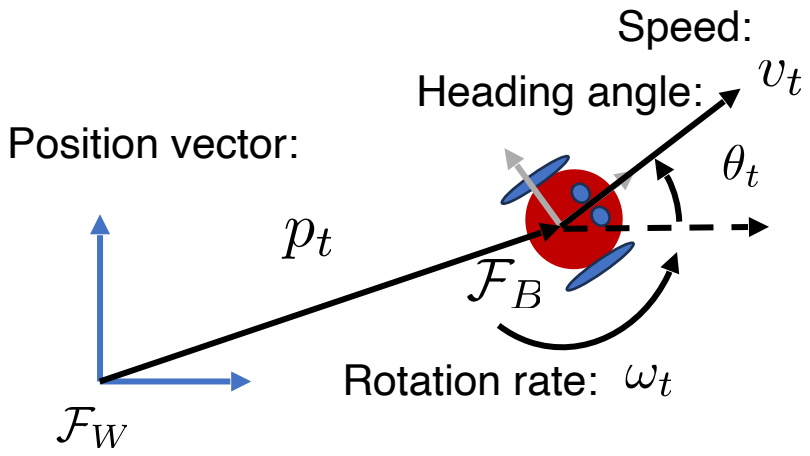


Robot pose: (p_t, θ_t)

Robot Kinematics and Dynamics



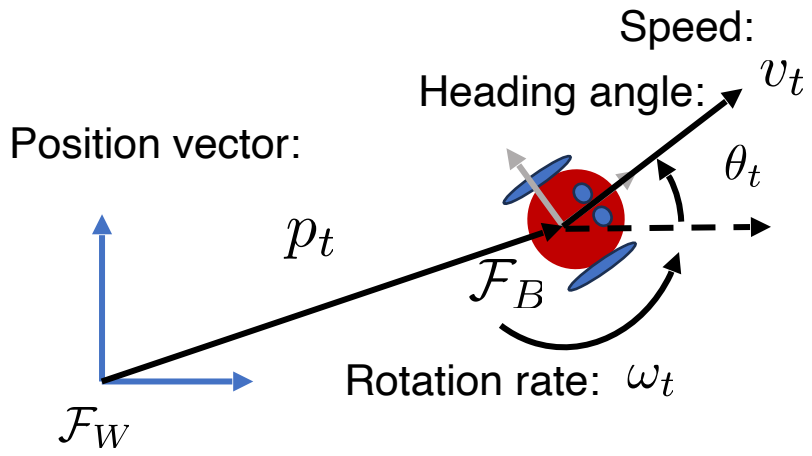
Robot Kinematics and Dynamics



Robot pose: (p_t, θ_t)

Control inputs: (v_t, ω_t)

Robot Kinematics and Dynamics

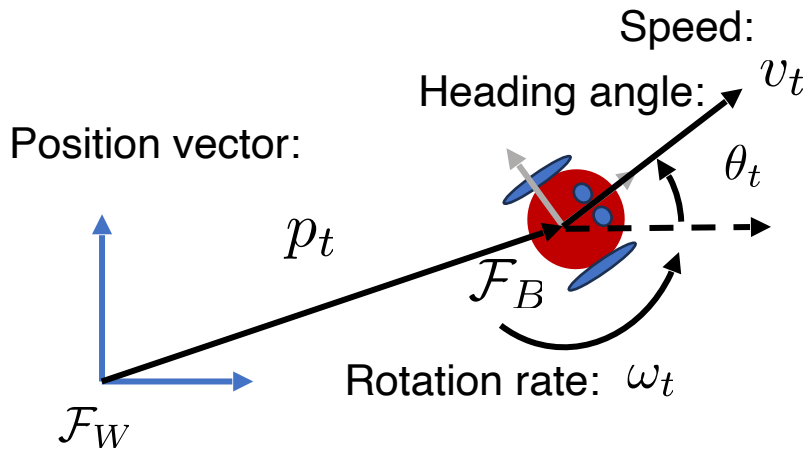


Robot pose: (p_t, θ_t)

Control inputs: (v_t, ω_t)

Differential Drive Robot (continuous time):

Robot Kinematics and Dynamics



Robot pose: (p_t, θ_t)

Control inputs: (v_t, ω_t)

Differential Drive Robot (continuous time):

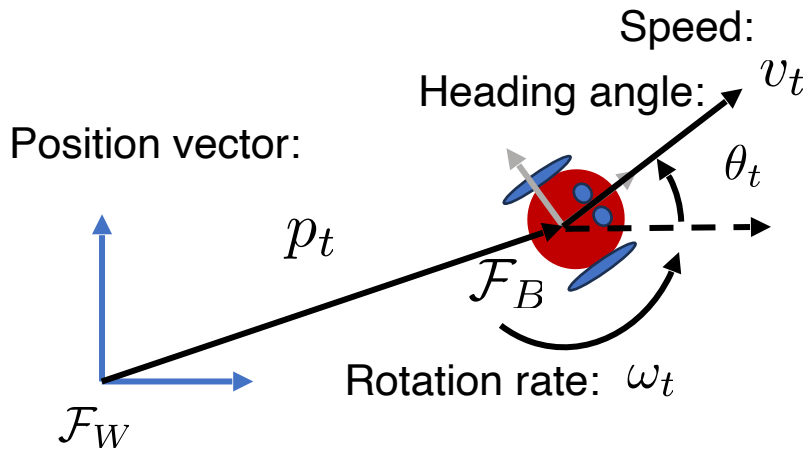
Recall, first order Euler time discretization:

$$\dot{x} = f(x, u)$$

$$\frac{x_{t+1} - x_t}{\delta t} \approx f(x_t, u_t)$$

$$x_{t+1} = x_t + f(x_t, u_t)\delta t$$

Robot Kinematics and Dynamics



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Differential Drive Robot (kinematic/dynamic):

$$x^p_{t+1} = x^p_t + v_t \cos(\theta_t)\delta t$$
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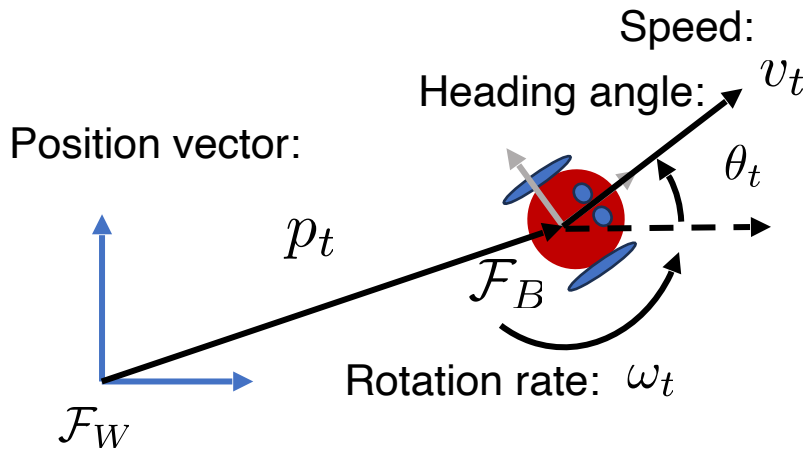
Robot Kinematics and Dynamics



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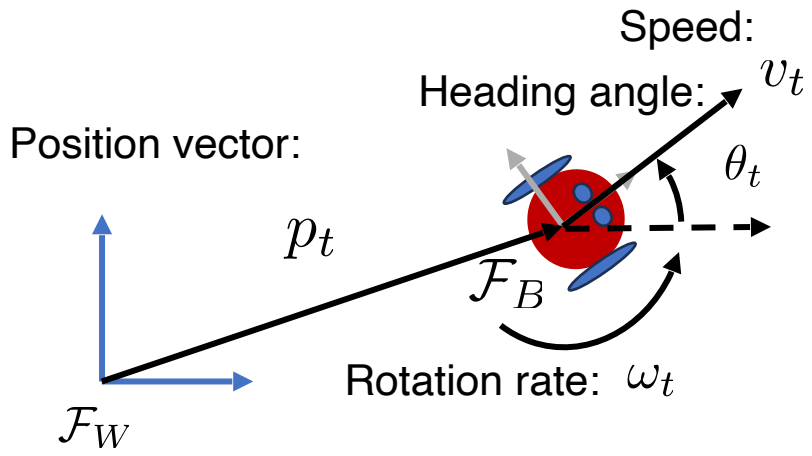
Velocities

Robot Kinematics and Dynamics



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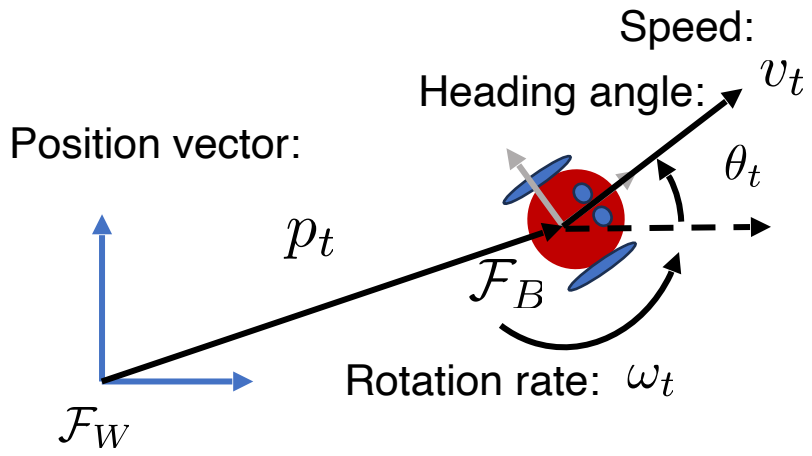
Velocities

Accelerations (or forces/torques)

Robot Kinematics and Dynamics



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Velocities

Accelerations (or forces/torques)

Nonholonomic: constraint on velocity vector
Nonlinear

Models: Robot Kinematics and Dynamics

Concept of state:

Kinematic model:

$$
\begin{aligned}
x^p_{t+1} &= x^p_t + v_t \cos(\theta_t)\delta t \\
y^p_{t+1} &= y^p_t + v_t \sin(\theta_t)\delta t \\
\theta_{t+1} &= \theta_t + \omega_t \delta t
\end{aligned}
$$

Kinematic state (a.k.a pose):

$$
x_t = \begin{bmatrix} x^P_t \\ y^p_t \\ \theta_t \end{bmatrix}
$$

Dynamic model:

$$
\begin{aligned}
x^p_{t+1} &= x^p_t + v_t \cos(\theta_t)\delta t \\
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\end{aligned}
$$

Dynamic state (pose+velocities):

$$
x_t = \begin{bmatrix} x^P_t \\ y^p_t \\ \theta_t \\ v_t \\ \omega_t \end{bmatrix}
$$

Models: Robot Kinematics and Dynamics

Concept of state:

Kinematic model:

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State: All information required to predict trajectory forward given an input signal

Models: Robot Kinematics and Dynamics

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Dynamic state (pose+velocities):

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x_t = \begin{bmatrix} x^P_t \\ y^p_t \\ \theta_t \\ v_t \\ \omega_t \end{bmatrix}
$$

State: All information required to predict trajectory forward given an input signal

General nonlinear ss model:

State space models:

$$x_{t+1} = f(x_t, u_t)$$

state (n x 1)

dynamics

Input (m x 1)

Models: Robot Kinematics and Dynamics

Concept of state:

Kinematic model:

$$x_{t+1}^p = x_t^p + v_t \cos(\theta_t)\delta t$$
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Dynamic model:

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Dynamic state (pose+velocities):

$$x_t = \begin{bmatrix} x_t^P \\ y_t^p \\ \theta_t \\ v_t \\ \omega_t \end{bmatrix}$$

State: All information required to predict trajectory forward given an input signal

State space models:

General nonlinear ss model:

$$x_{t+1} = f(x_t, u_t)$$

state (n x 1) — dynamics — Input (m x 1)

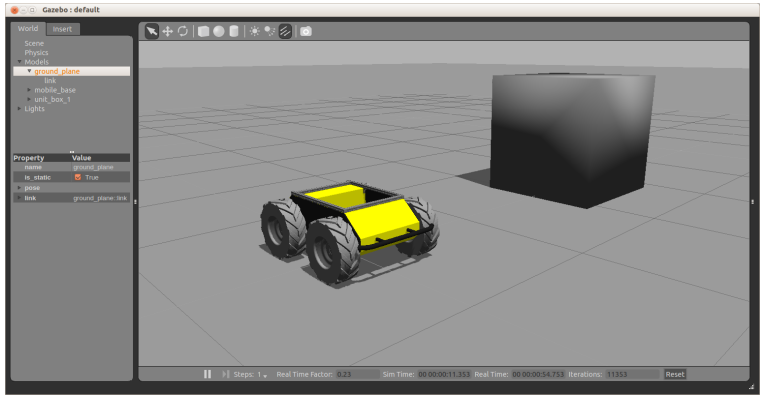
Linear ss model:

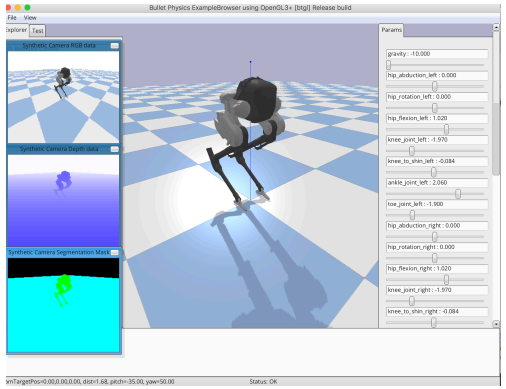
$$x_{t+1} = Ax_t + Bu_t$$

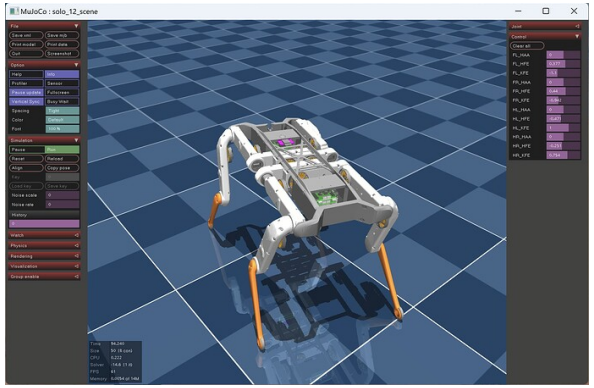
dynamics matrix (n x n) — control matrix (n x m)

Robot Simulators

Purpose: To automate all of this (geometry, collision checking, coordinate transforms, kinematics/dynamics). To scale up to systems too complex to model by hand.







GAZEBO

BULLET PHYSICS LIBRARY

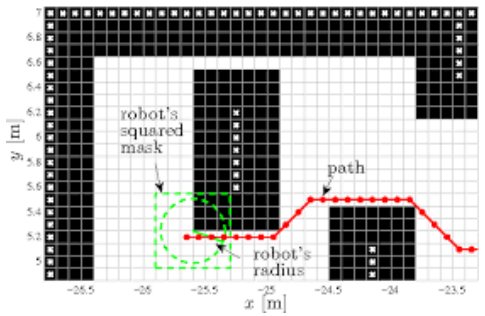
MuJoCo

Advanced physics simulation

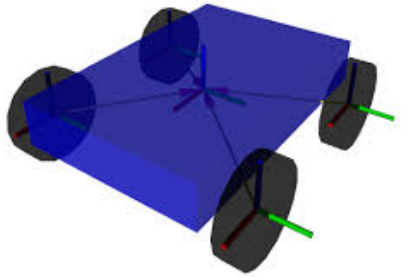
Many, many more: IsaacLab, Drake, Webots, DoJo, TiaChi …

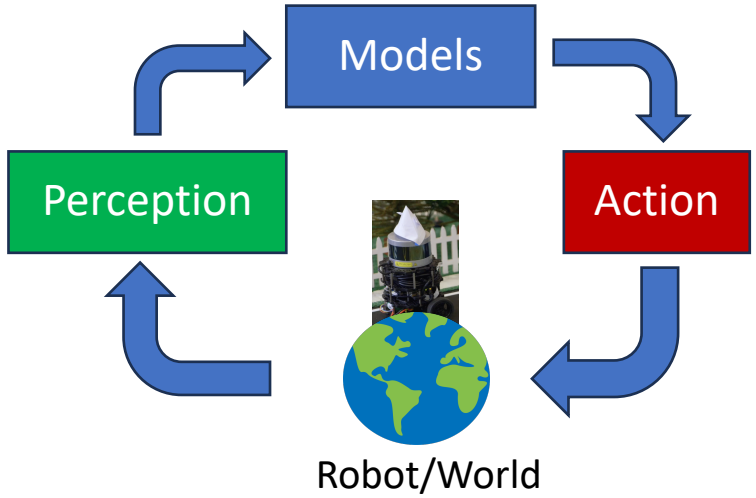
Models: Environment and Robot Representations

Maps:



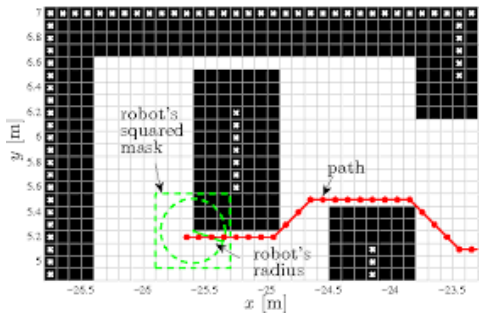
Robot models
or simulators:





Models: Environment and Robot Representations

Maps:



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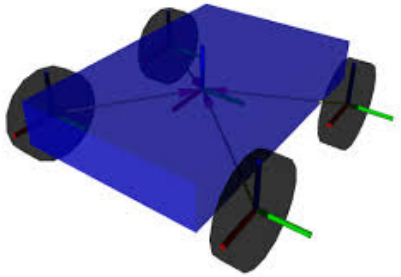
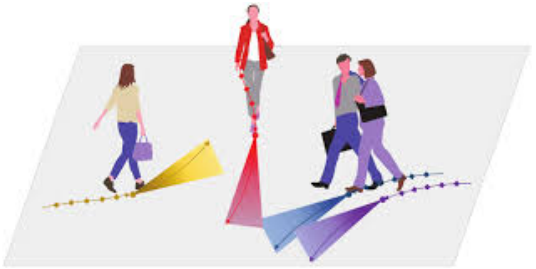
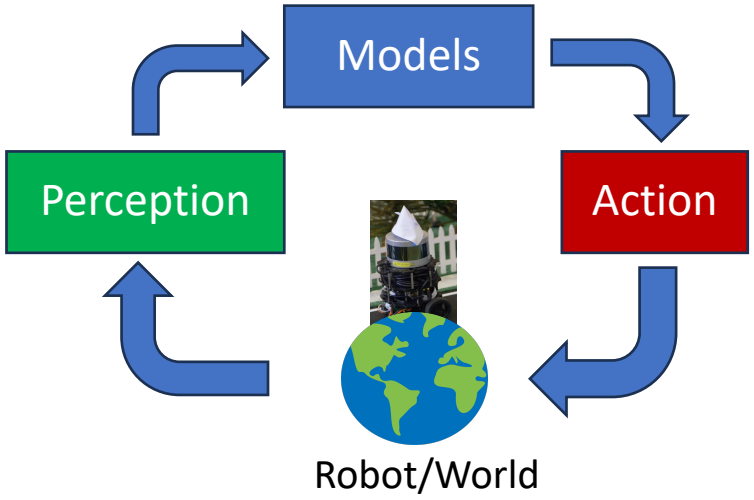


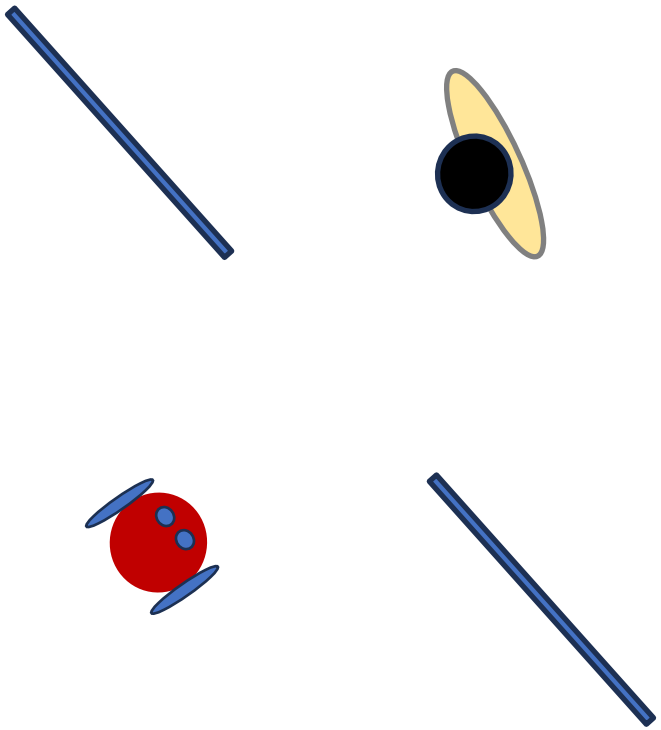
Agent models or
simulators:





Agent Models

I. Predict then plan



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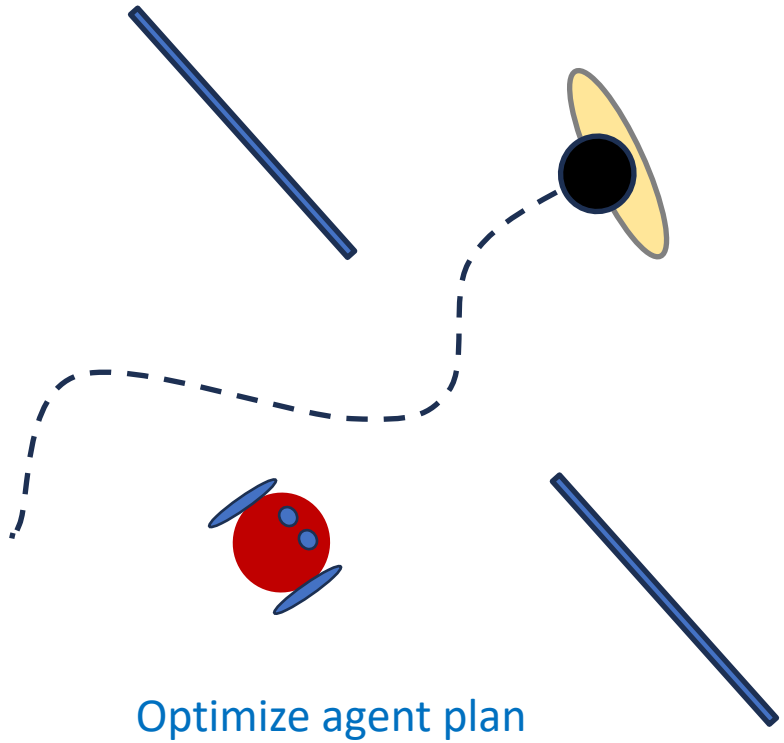

Optimize agent plan
first, then robot

Agent Models

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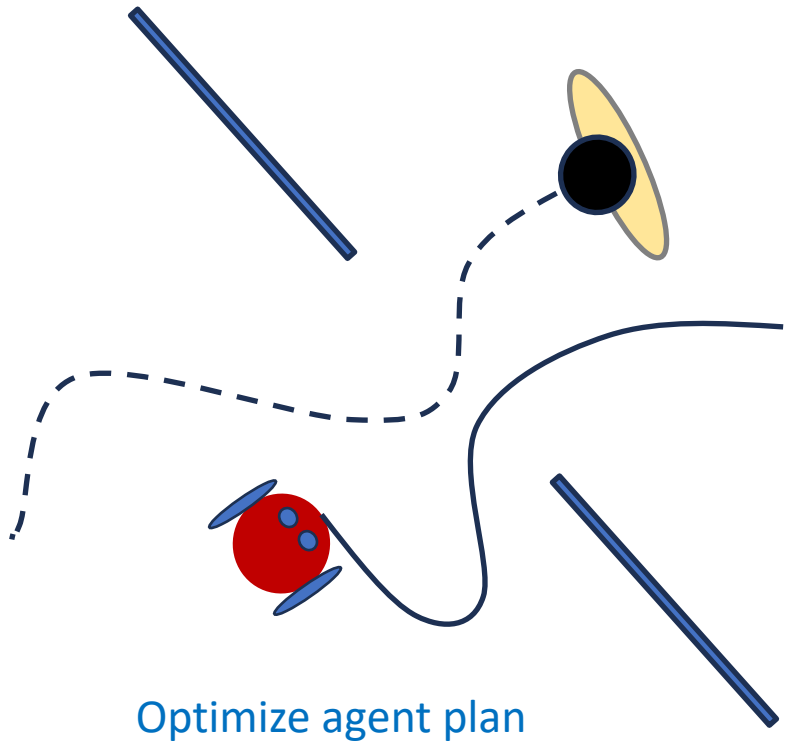

Optimize agent plan
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Agent Models

Model the agent → infer intent → predict a trajectory → plan the robot

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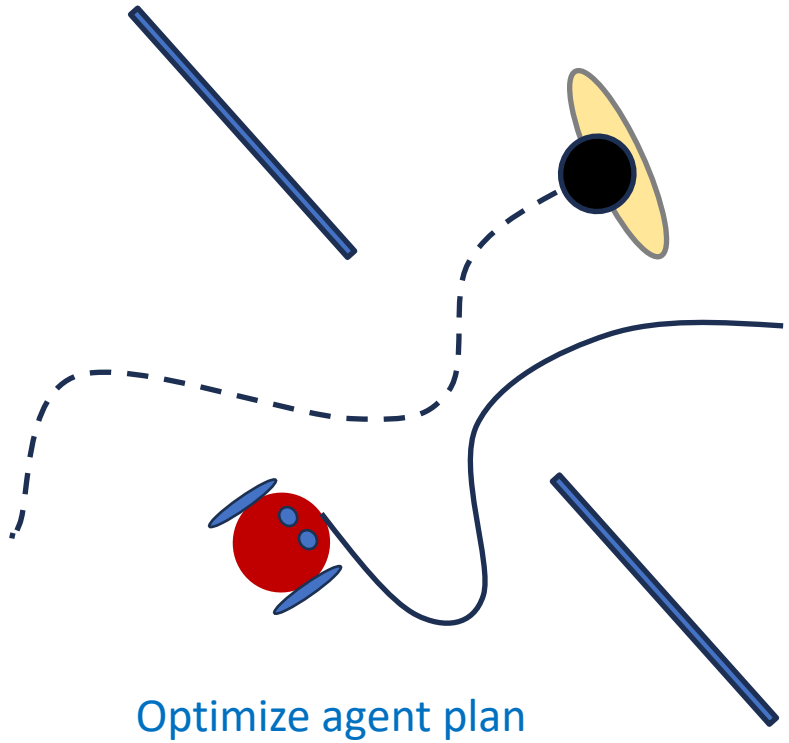

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Agent Models

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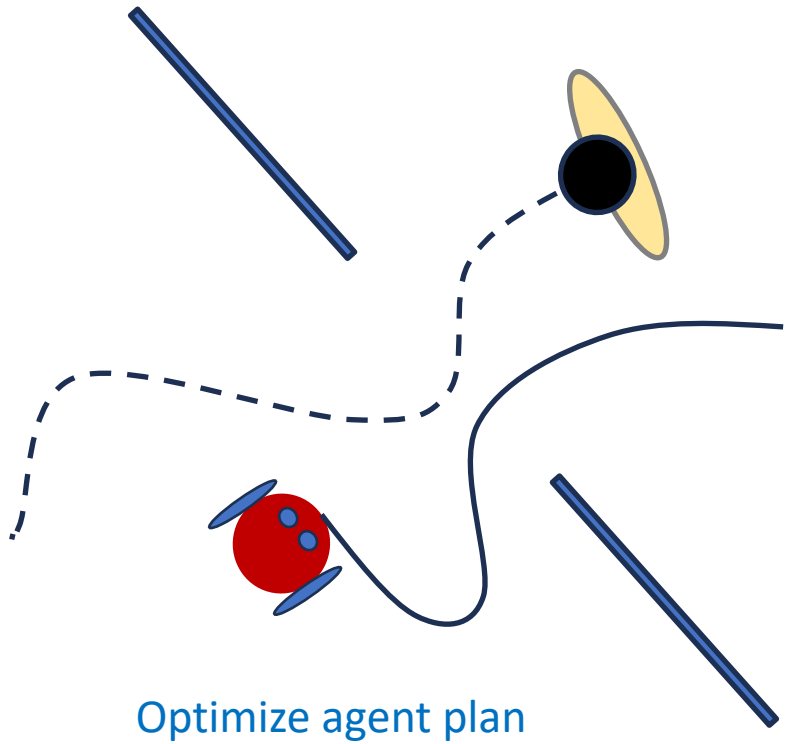


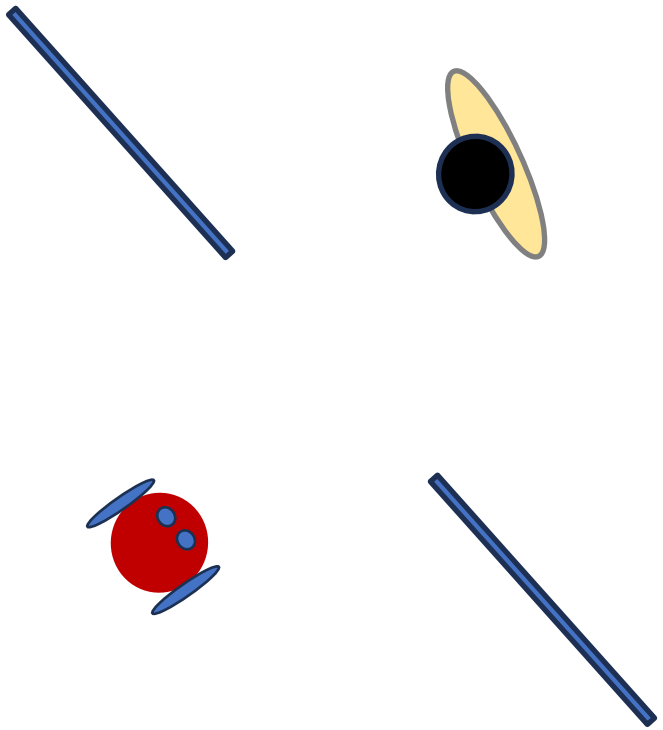
Optimize agent plan first, then robot

Model the agent → infer intent → predict a trajectory → plan the robot

- Run a planning-and-control stack for the agent, often at lower fidelity than the robot's.
- Specify behavior through dynamics, intent, and a decision model. Examples:
  - car-following with IDM: maps speed, headway, relative speed to a longitudinal acceleration
  - a rational planning agent: agent plans an optimal trajectory to its goal
  - Boltzmann-rational action – makes higher value actions more probable with a rationality parameter that controls variability
- Treat the resulting trajectories as moving obstacles in the robot planner.

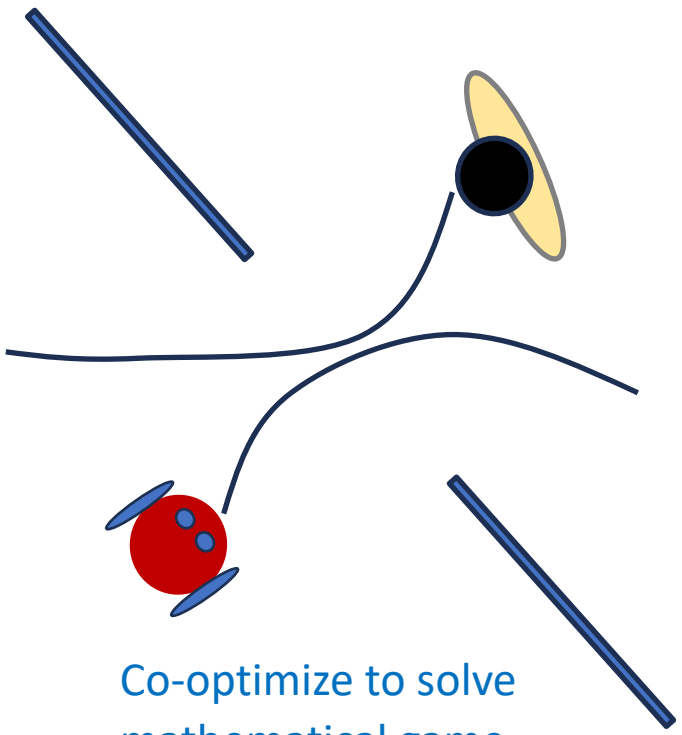
Agent Models

II. Game theoretic



Agent Models

II. Game theoretic

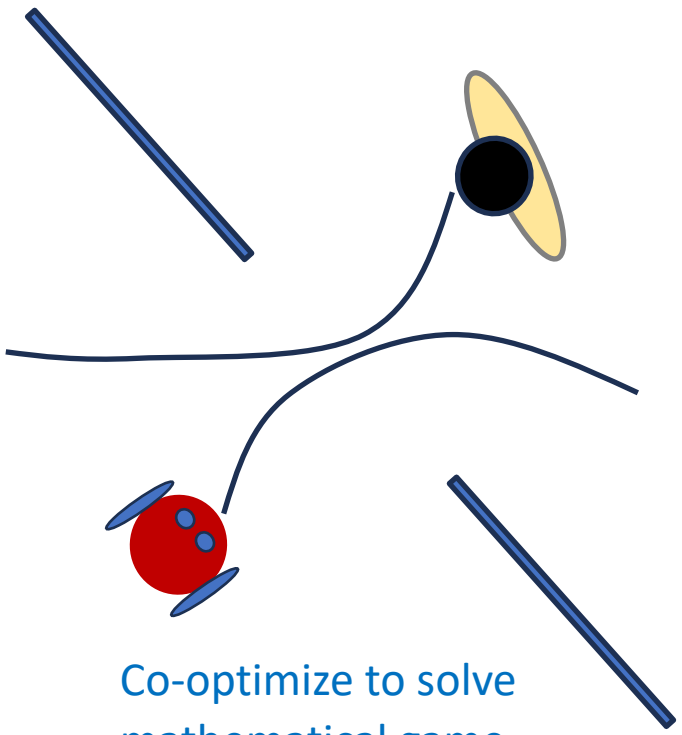


Co-optimize to solve mathematical game

Agent Models

II. Game theoretic

Choose robot and agent actions jointly



Co-optimize to solve mathematical game

Agent Models

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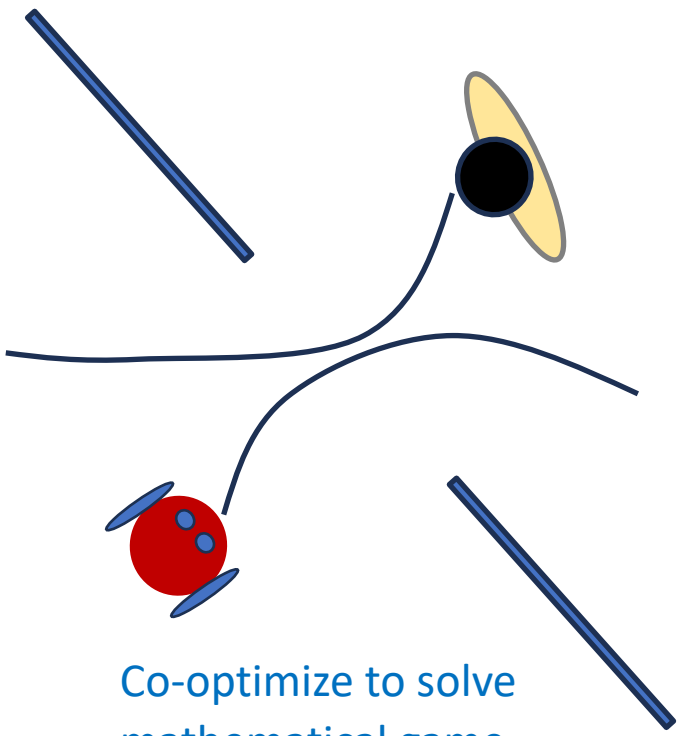


Co-optimize to solve mathematical game

Choose robot and agent actions jointly

- Each agent optimizes an objective while anticipating how the other will respond. Goal is to find equilibrium plans.

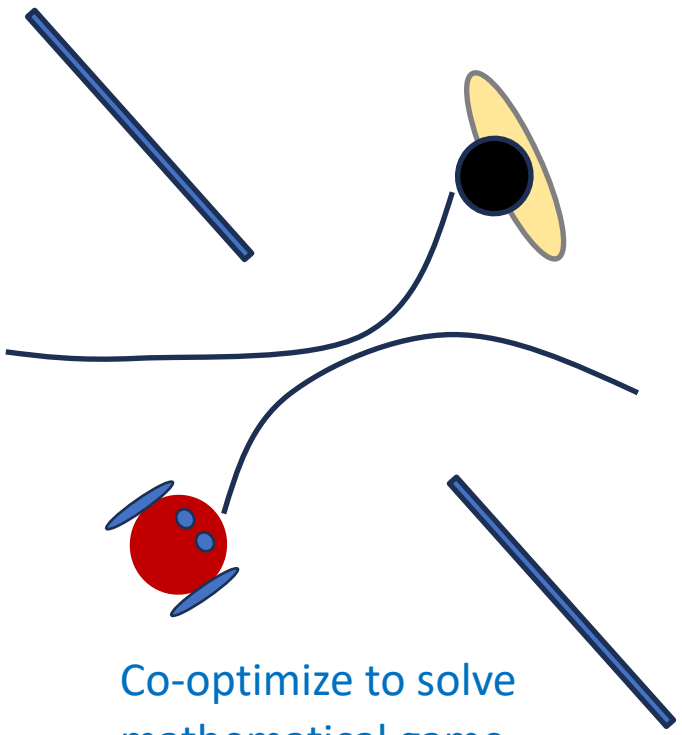
- Captures negotiation in merging, yielding, overtaking, and shared-space motion.

- Can produce more realistic behavior than treating predictions as fixed.

- Equilibria couple the robot's plan to the other agent's best response.

Agent Models

II. Game theoretic



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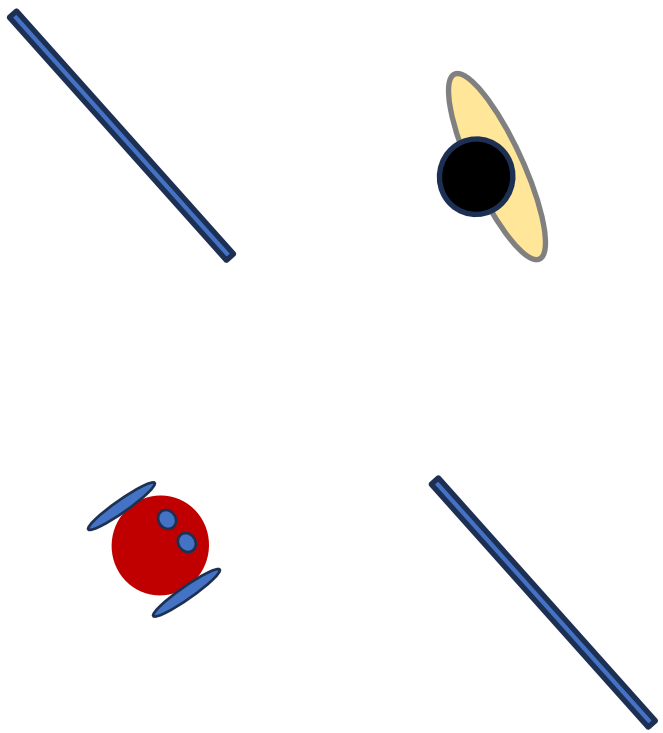
- Can produce more realistic behavior than treating predictions as fixed.
- Equilibria couple the robot's plan to the other agent's best response.

Often rely on explicit costs, dynamics, information structure, and rationality assumptions.

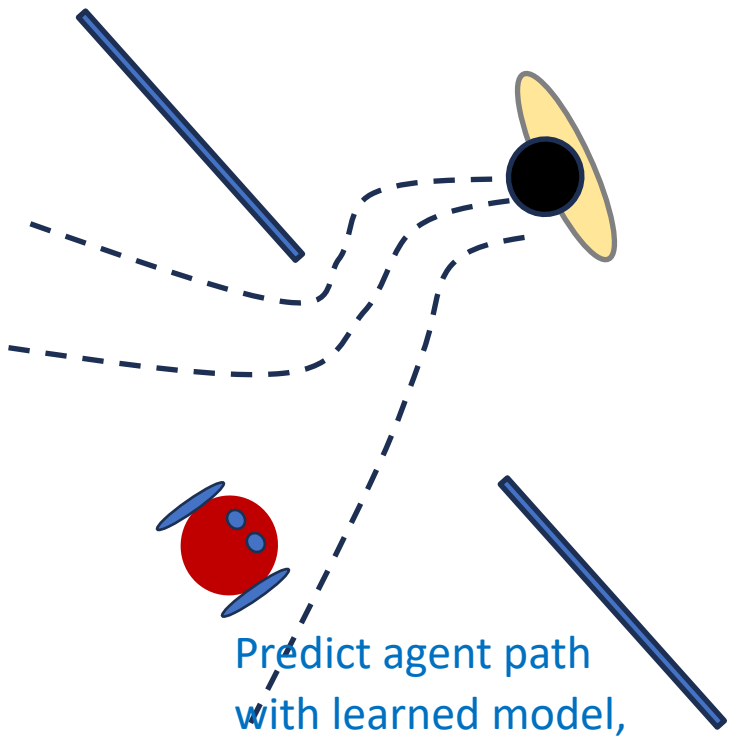
Agent Models

III. Learning based



Agent Models

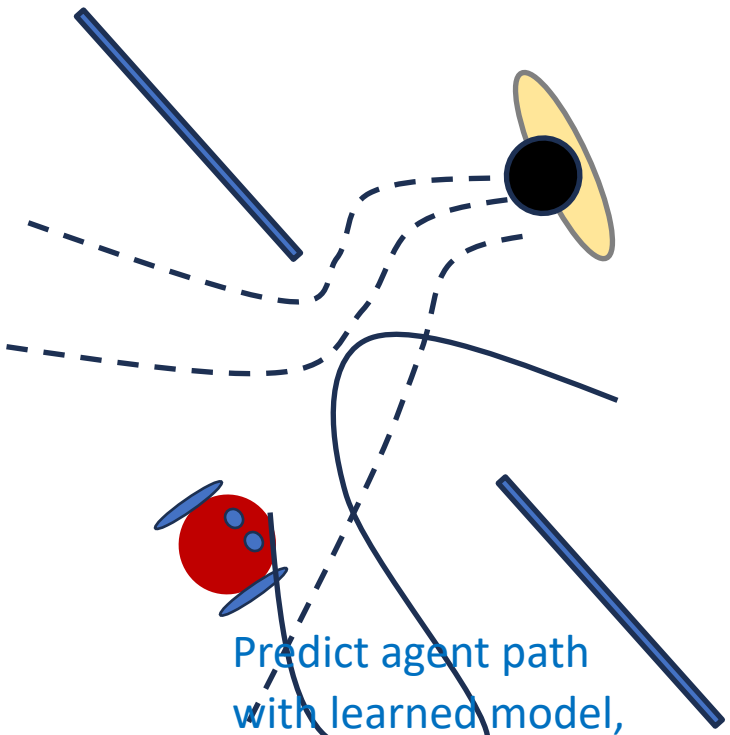
III. Learning based



Predict agent path with learned model, that plan robot

Agent Models

III. Learning based

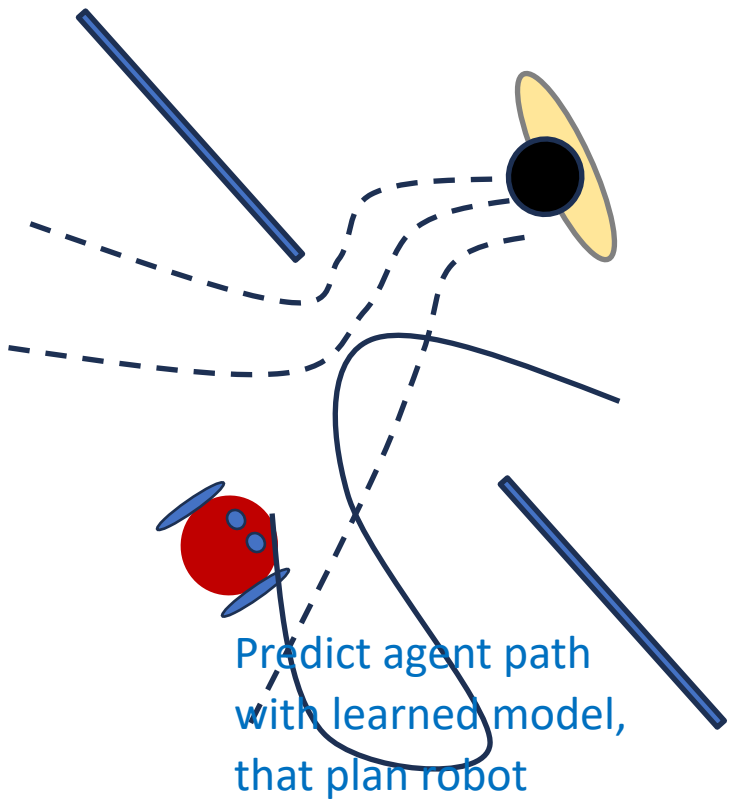


Predict agent path with learned model, that plan robot

Agent Models

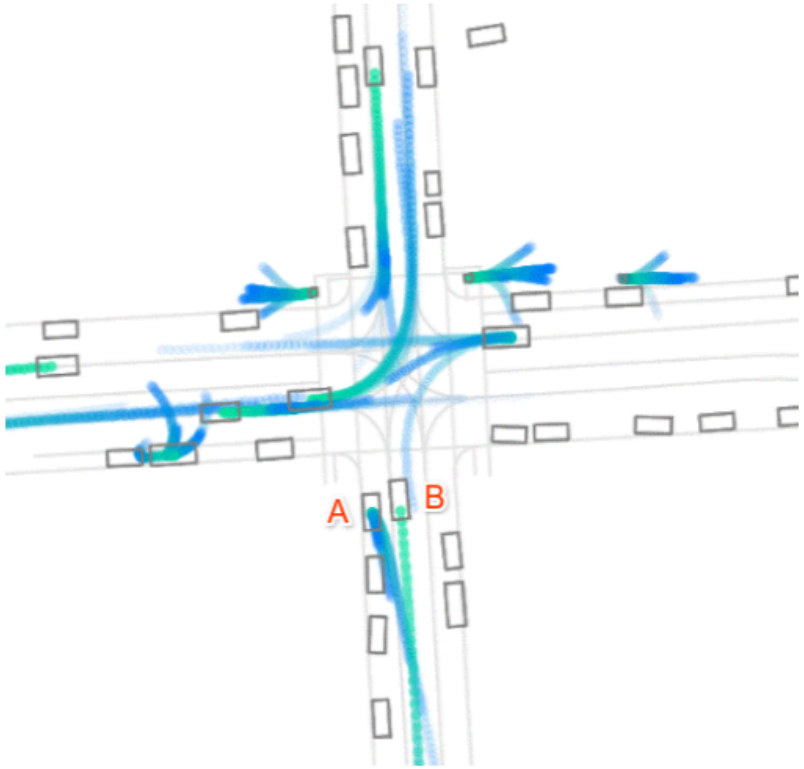
III. Learning based

Learn a distribution over plausible futures from data



Predict agent path with learned model, that plan robot

Agent Models



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Learn a distribution over plausible futures from data

- Dominant paradigm for motion forecasting in complex environments.
- Learns scene context and behaviors such as late merges or red-light violations.
- Represents inherent uncertainty through multiple modes, probabilities, or samples.
- Can predict interacting agents jointly rather than one at a time.

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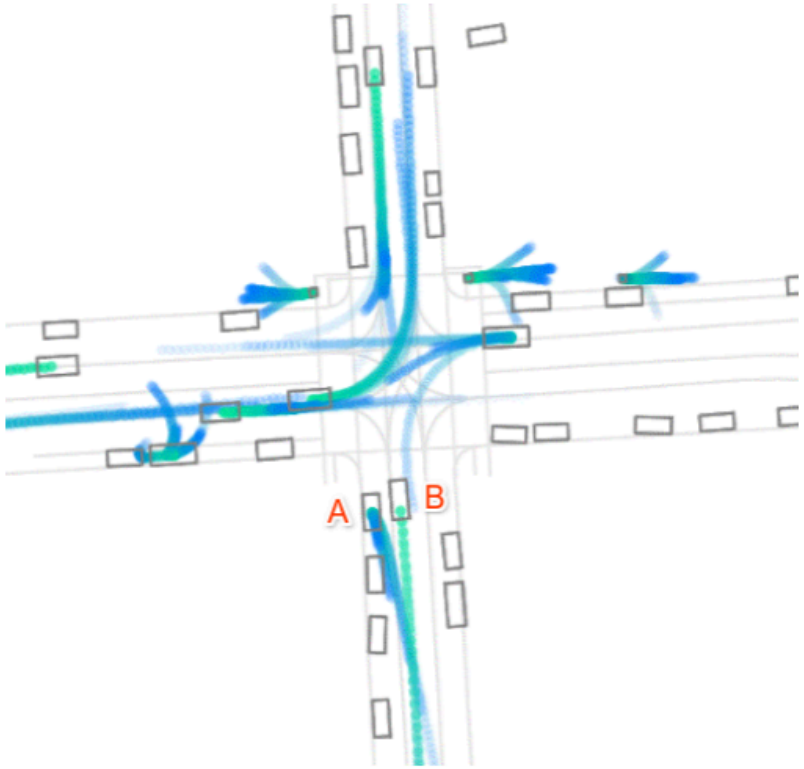

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- Dominant paradigm for motion forecasting in complex environments.
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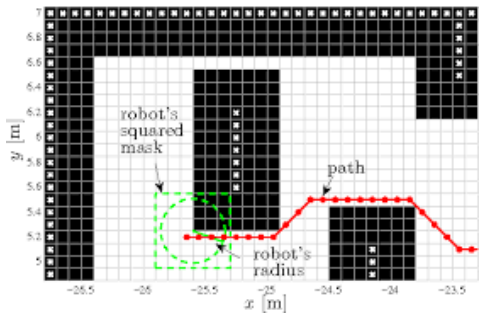
- Represents inherent uncertainty through multiple modes, probabilities, or samples.
- Can predict interacting agents jointly rather than one at a time.

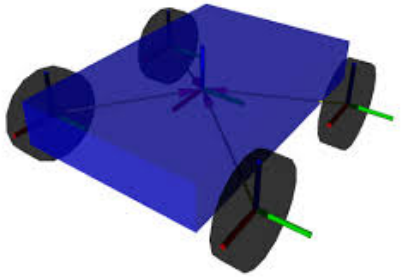
Powerful but opaque: distribution shift and miscalibration can lead to silent failure.

Models: Environment and Robot Representations

Maps:



Robot models
or simulators:



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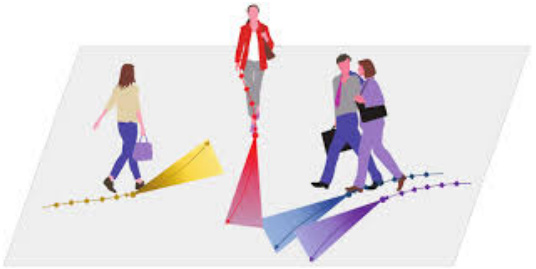
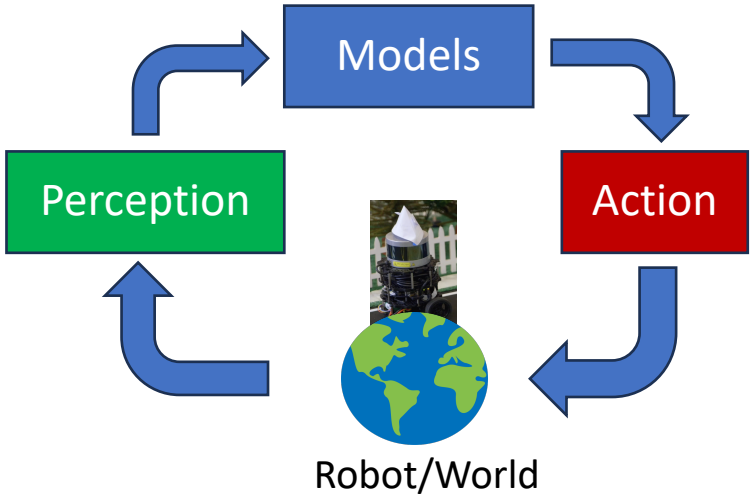
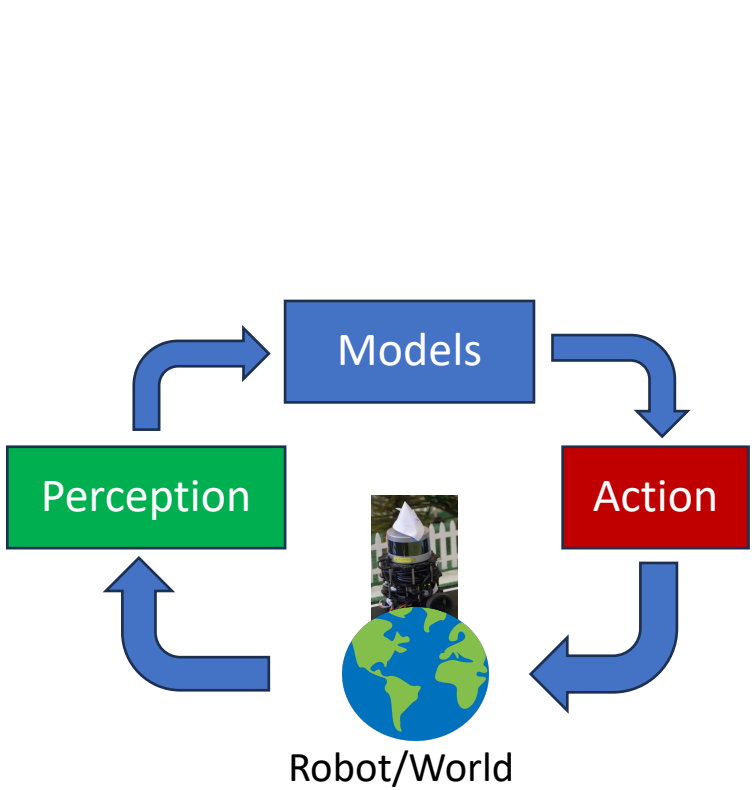




Action: Planning and Control Stack

Use the models to move the robot



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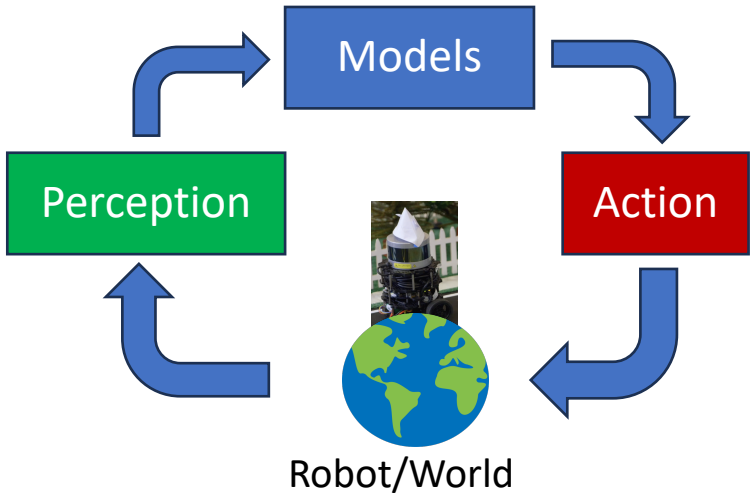
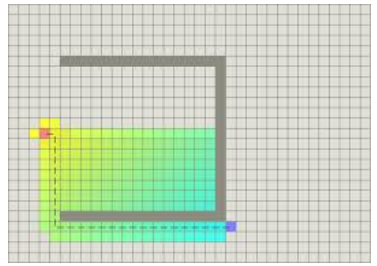
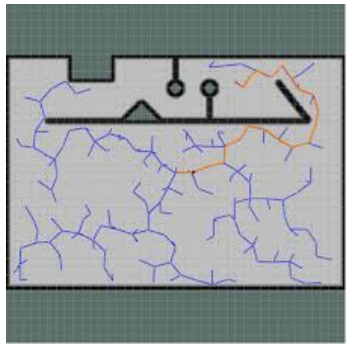


Path planning:

- A* planning on a graph
- RRT, RRT*





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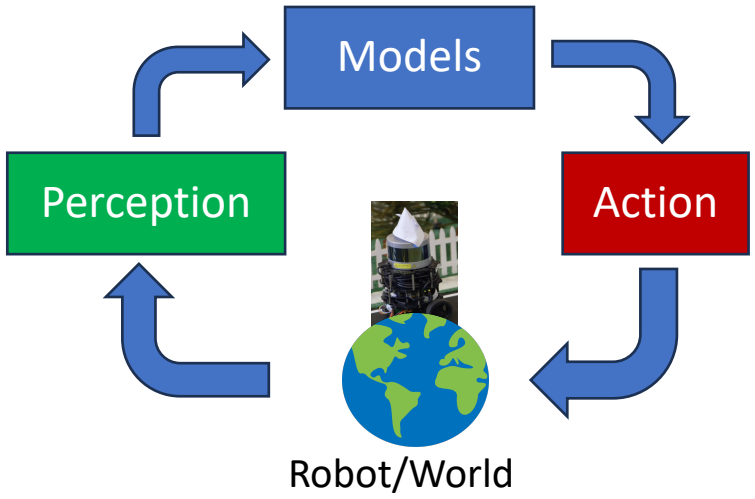

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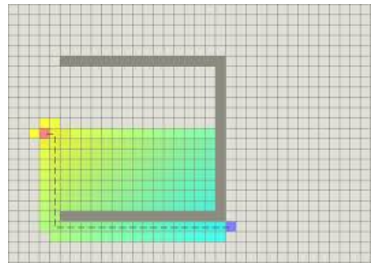
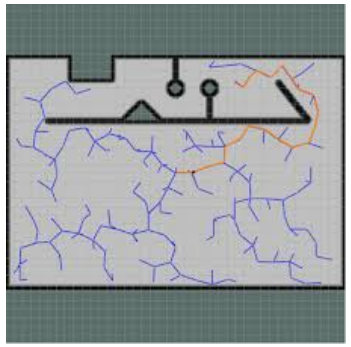
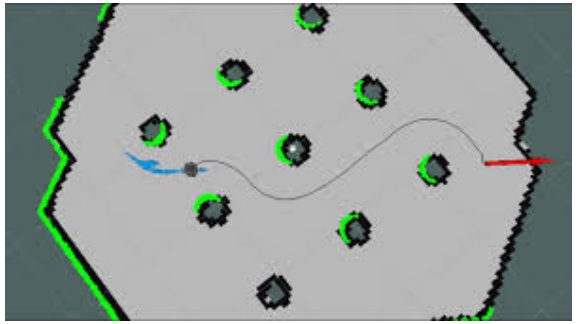




Trajectory optimization:

- Optimal control
- Numerical solvers



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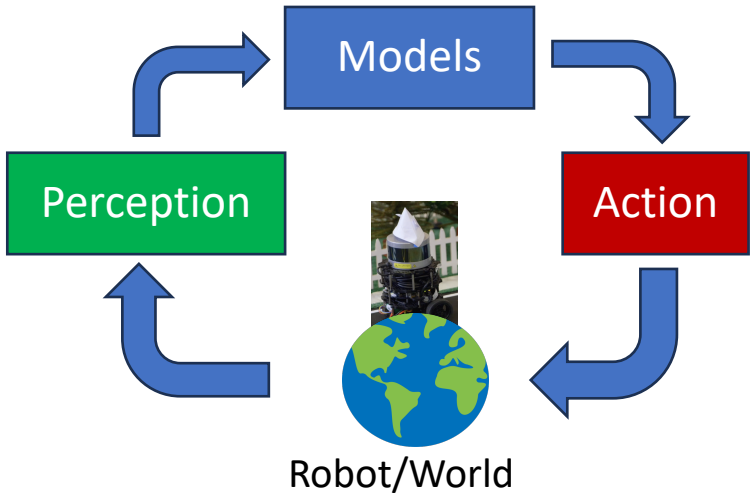

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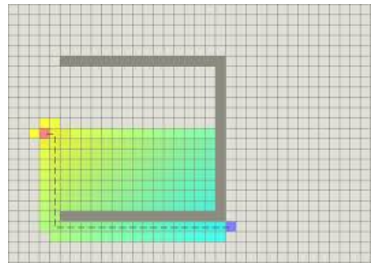

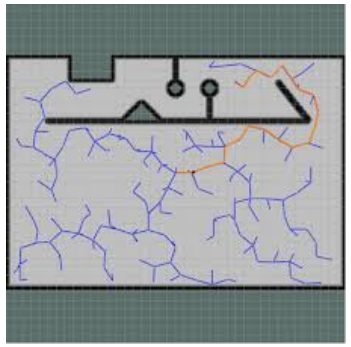

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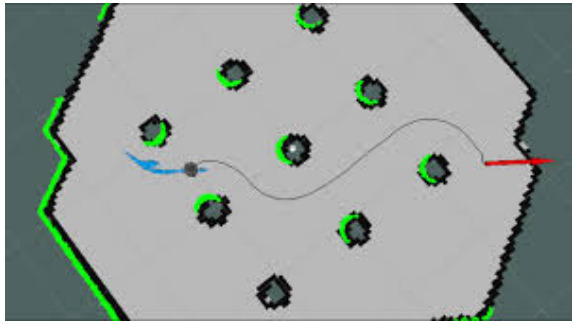
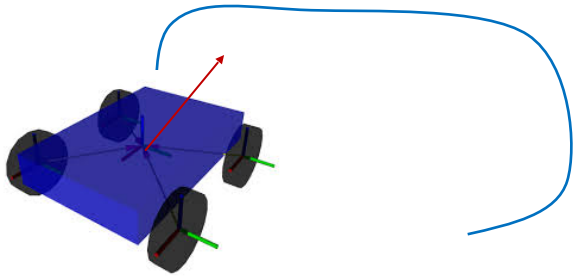


Trajectory following:

- Kinematic/dynamic models
- PID control
- LQR, gain scheduled LQR



Path Planning

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Map representation

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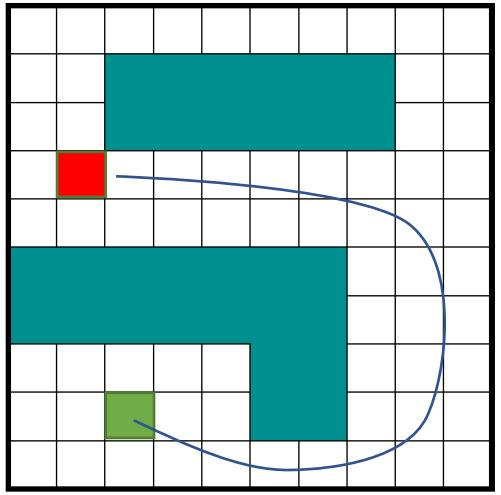
Graph search planning, A*



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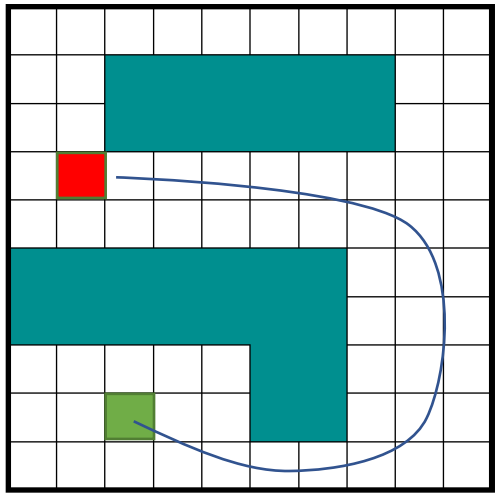
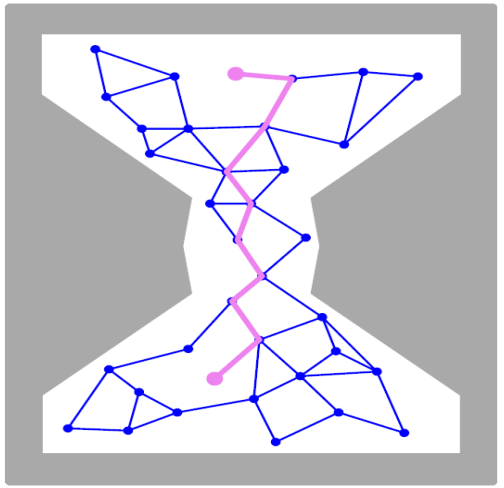


Planning on random graph, RRT*, PRM



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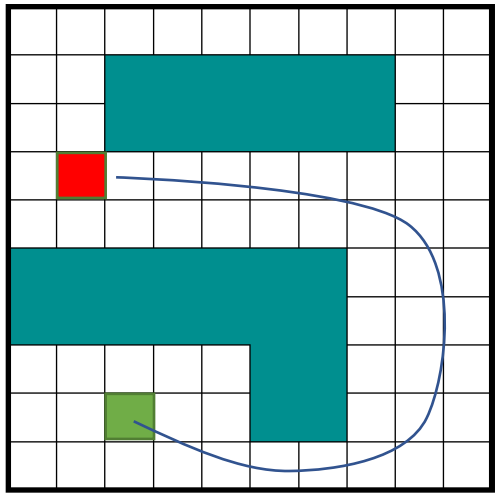

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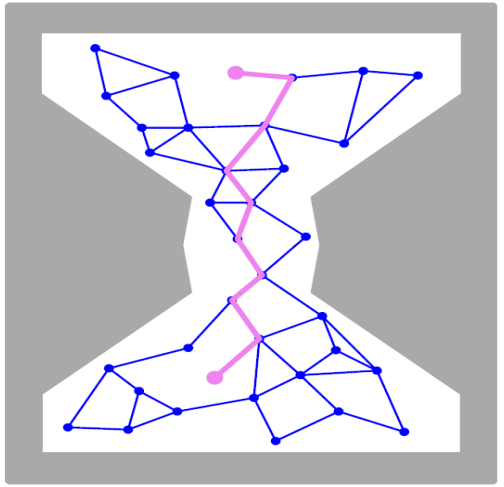
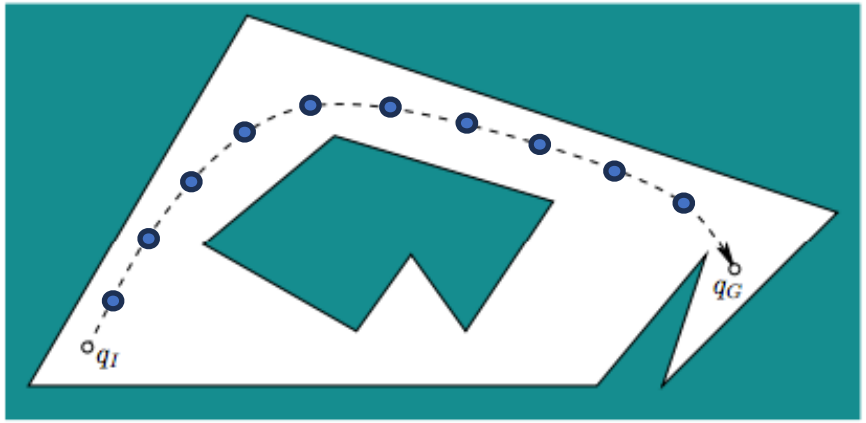


Goal: Find a path so the robot doesn't hit things!

Trajectory Optimization



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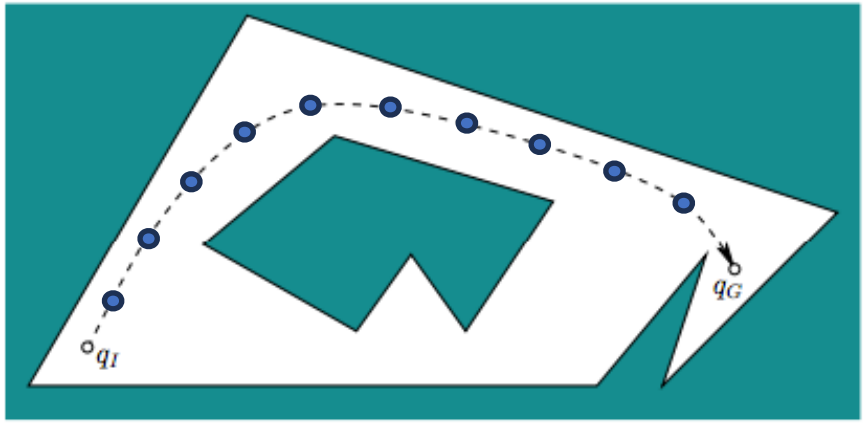

Trajectory optimization problem:

$$\min_{u_{0:T}} \sum_{t=0}^{T} c_t(x_t, u_t)$$

$$\text{s.t.} \quad x_{t+1} = f(x_t, u_t)$$

$$x_t \in \mathbb{X}_{\text{Free}}$$

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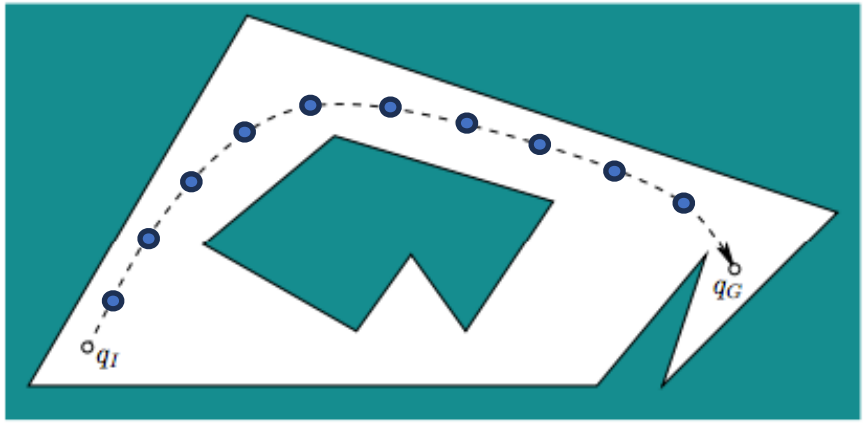

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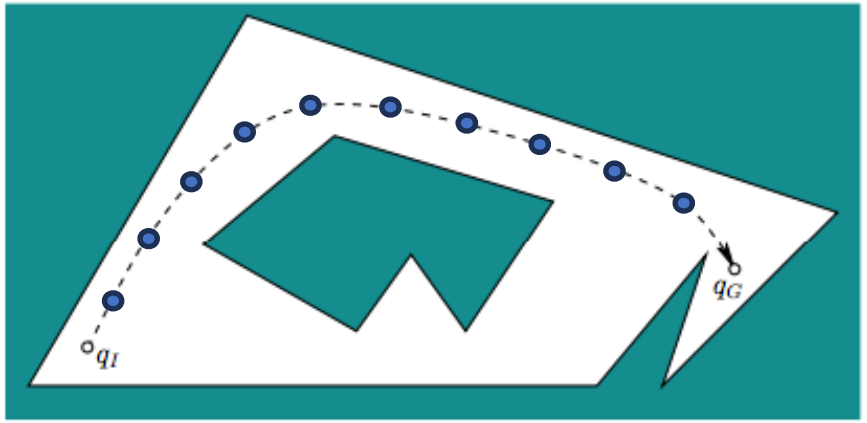

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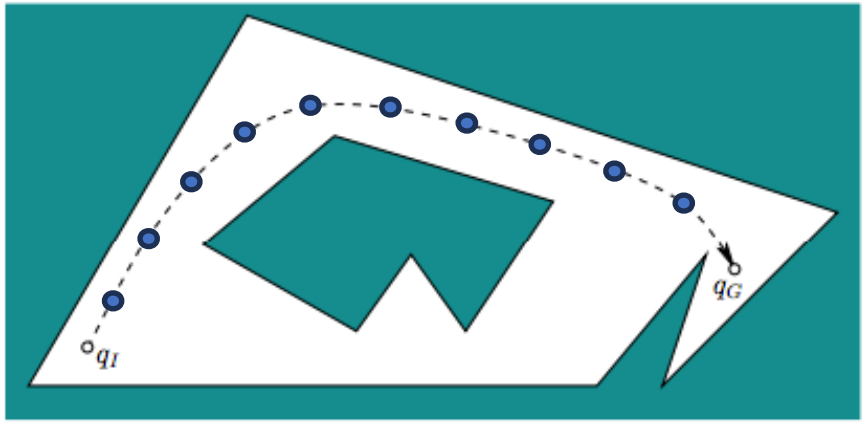

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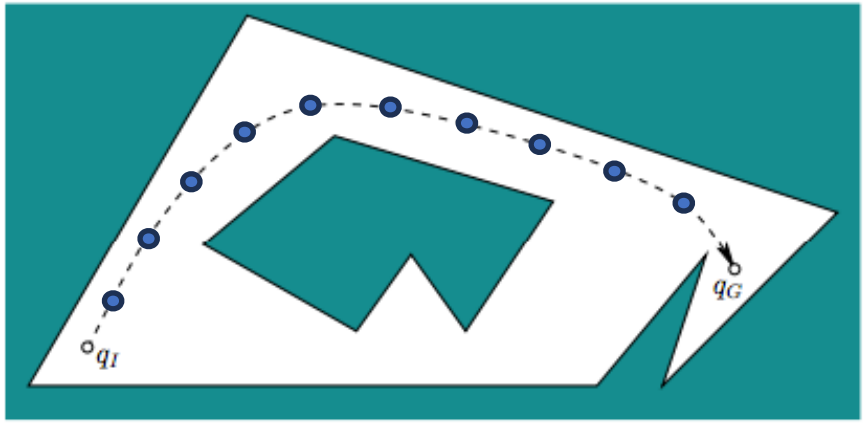

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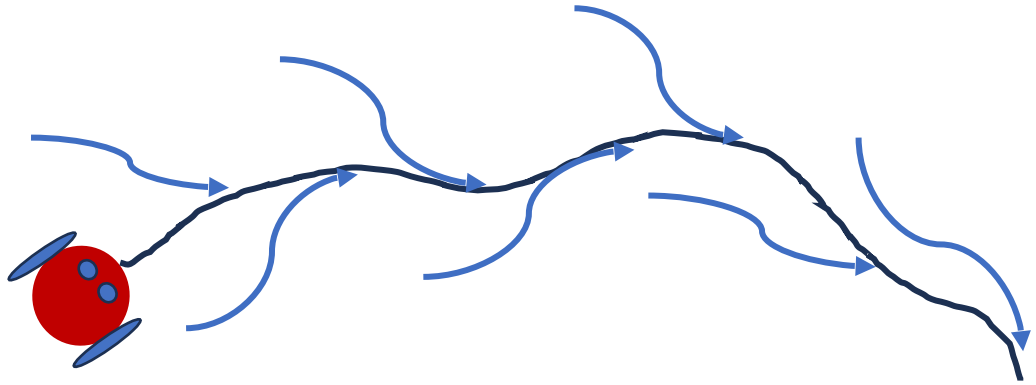
$$\text{s.t.} \quad x_{t+1} = f(x_t, u_t)$$ Dynamic constraints

$$x_t \in \mathbb{X}_{\text{Free}}$$ Collision constraints

Goal: Find a dynamically feasible trajectory (close to the path) so the robot doesn't hit things!

Trajectory Following



Motion model:

$$x_{t+1} = f(x_t, u_t)$$

Feedback control law:

$$u_t = h(x_t, x_t^*)$$

planned traj waypoints

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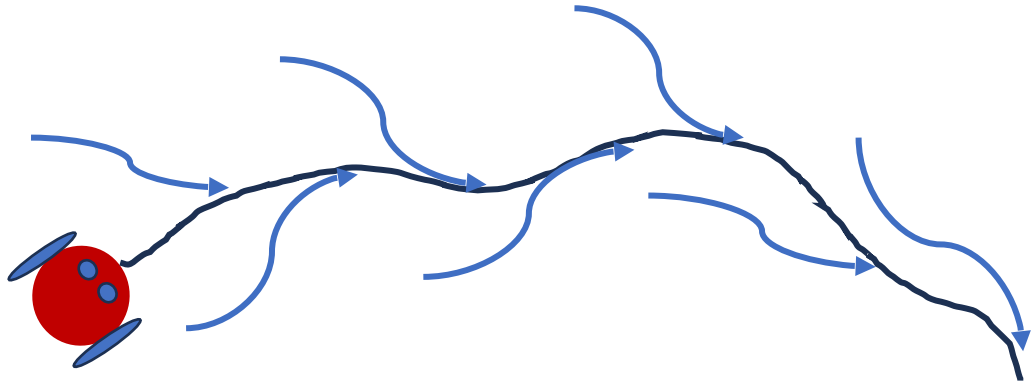

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Feedback control law:

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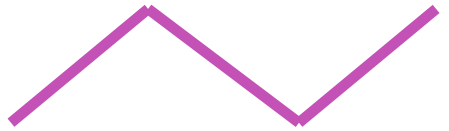
planned traj waypoints

Goal: Use closed loop feedback control to track desired trajectory, rejecting disturbances and being robust to model errors.

Path Planning vs Trajectory Planning

A* path plan:

(i) Global search
(ii) Collision free
(iii) Not dynamically feasible



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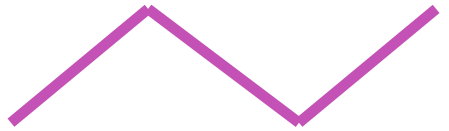

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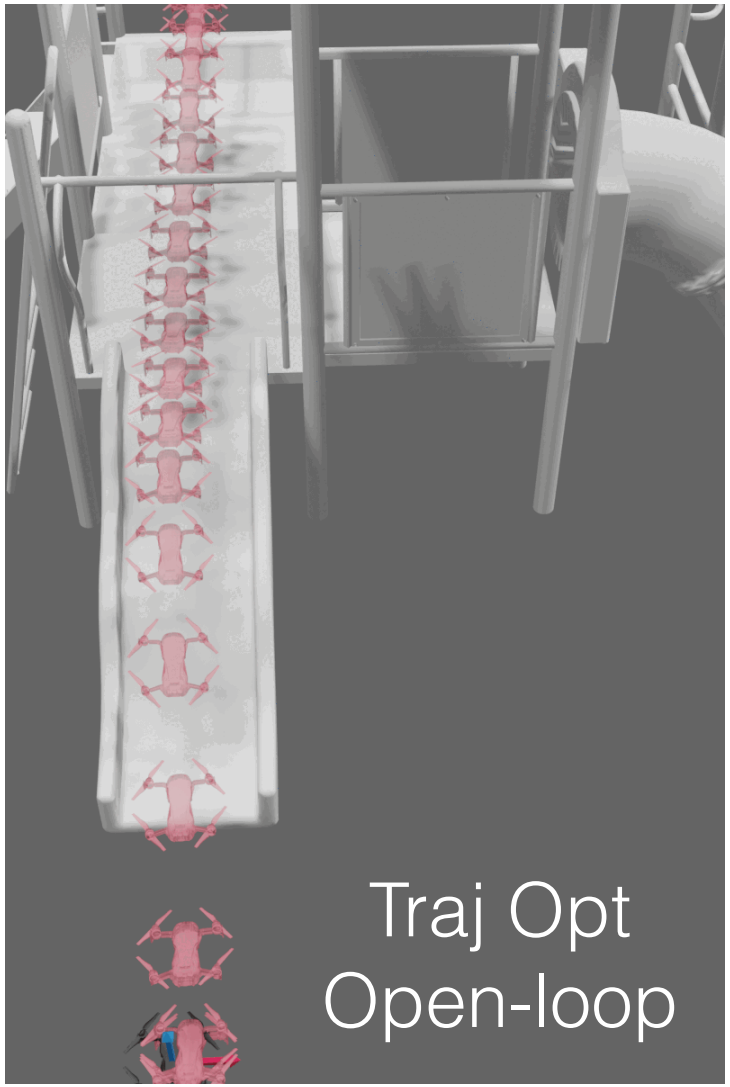

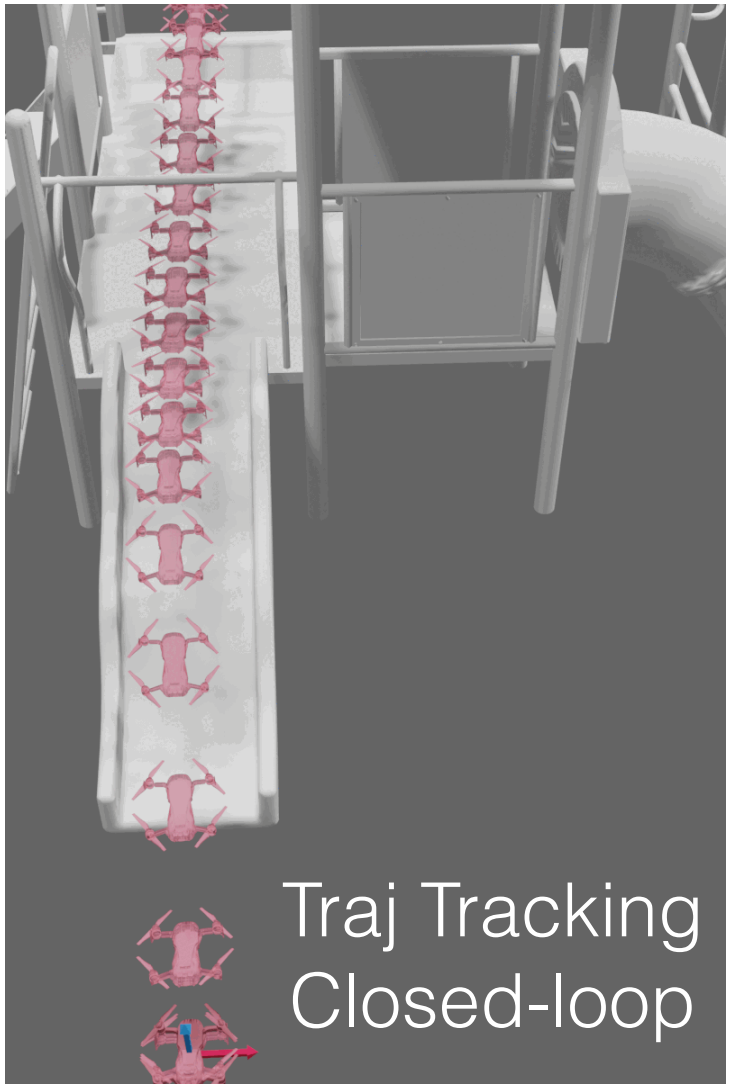

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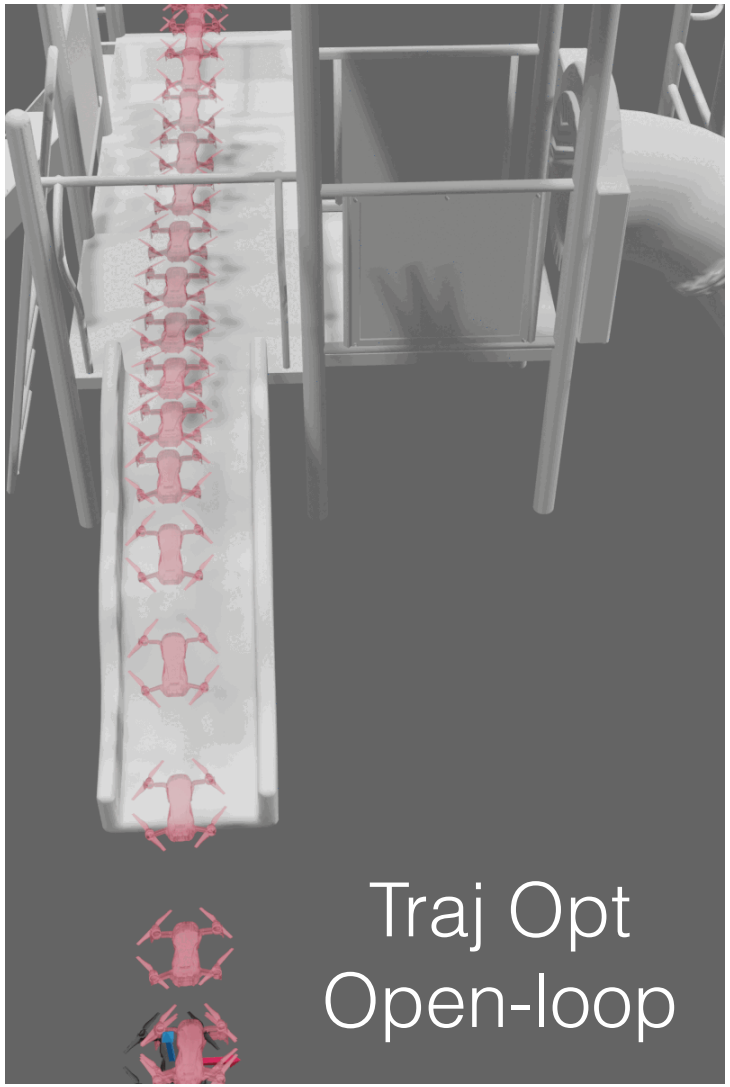
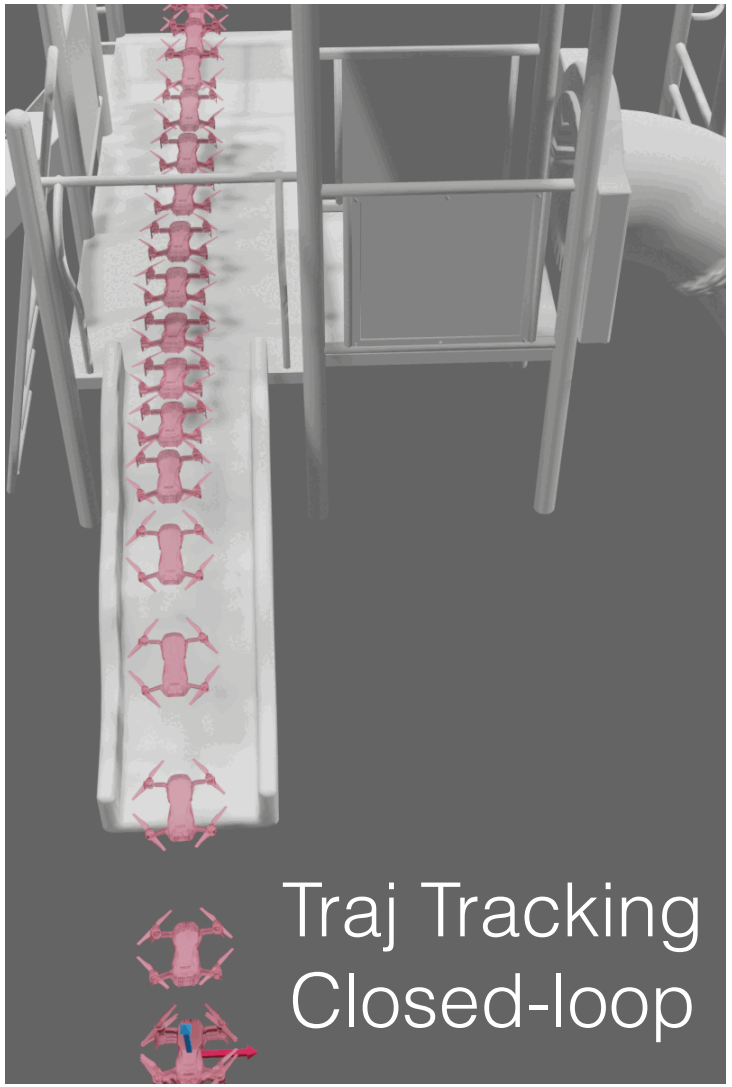
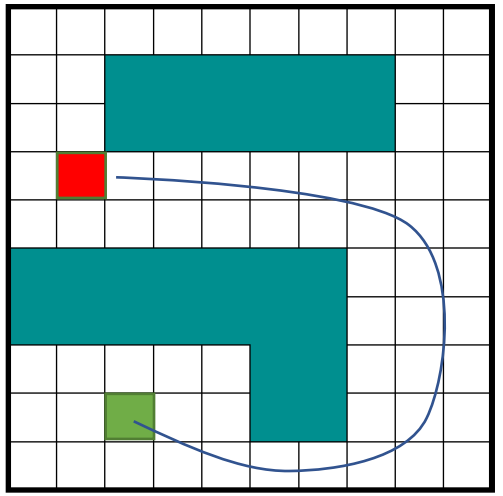




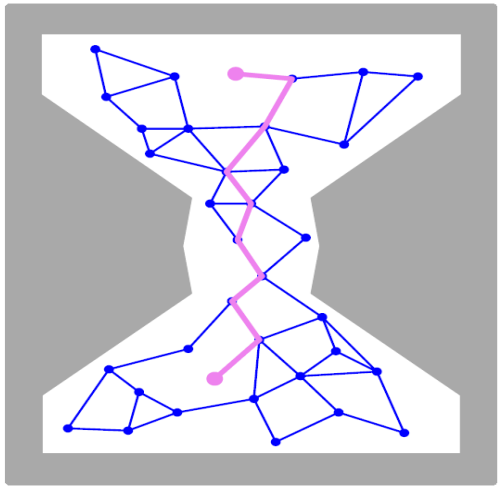
Path Planning

Map representation

Graph search planning



Planning on random graph



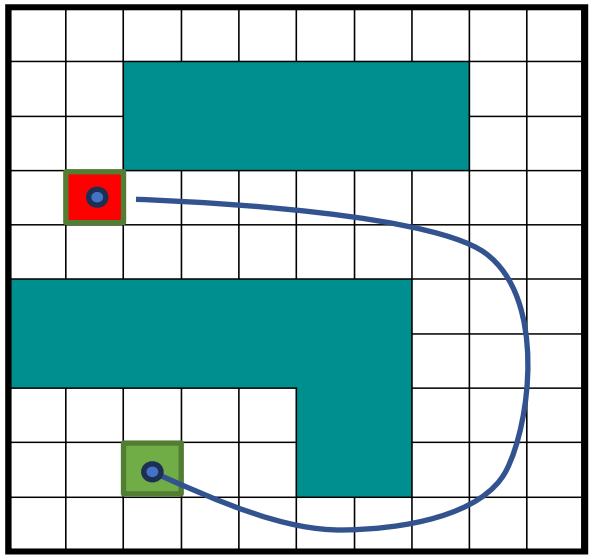
Goal: Find a path so the robot doesn't hit things!

History

- *Potential fields* [Khatib ‘85, Rimon & Koditschek ‘92]: create forces on the robot that pull it toward the goal and push it away from obstacles
- *Grid-based planning* [Stentz ‘94]: discretizes problem into grid and runs a graph-search algorithm (Dijkstra, A*, …)
- *Combinatorial planning* [LaValle ‘06]: constructs structures in the configuration (C-) space that completely capture all information needed for planning
- *Sampling-based planning* [Kavraki et al ‘96, LaValle & Kuffner ‘06, Karaman and Frazzoli, ‘11]: uses collision detection algorithms to probe and incrementally search the C-space for a solution, rather than completely characterizing all of the C_{free} structure. Asymptotic optimality with RRT*/PRM*.
- Current research: inclusion of differential and logical constraints, planning under uncertainty, parallel implementation, deep learning warm starting

Graph search planning

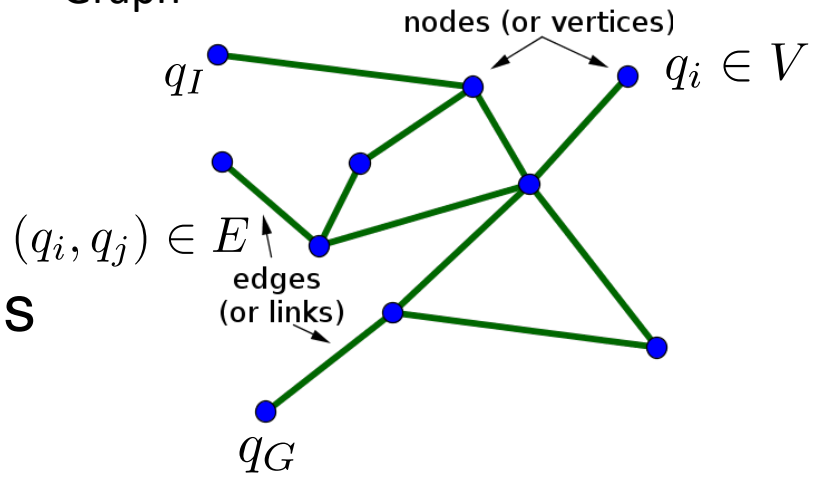
Map



- Discretize robot's **configuration space** into a grid
  - Each grid cell is either free or occupied
  - Robot moves between adjacent free cells
  - **Goal:** find shortest sequence of free cells from start to goal

- Mathematically, this corresponds to pathfinding in a discrete graph $G = (V, E)$
  - Each vertex $q_i \in V$ represents a free cell
  - Edges $(q_i, q_j) \in E$ connect adjacent grid cells

Graph



Graph search algorithms

- Having determined decomposition, how to find “best” path?
- **Label-Correcting Algorithms**: $C(q)$: *cost-to-come* from q_I to q
- Set to ∞ for all nodes, except start node to 0, frontier queue has only start node

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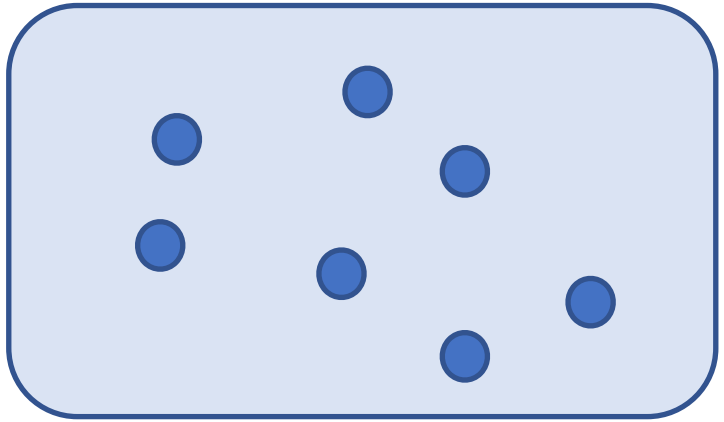

FRONTIER/ALIVE/PRIORITY QUEUE

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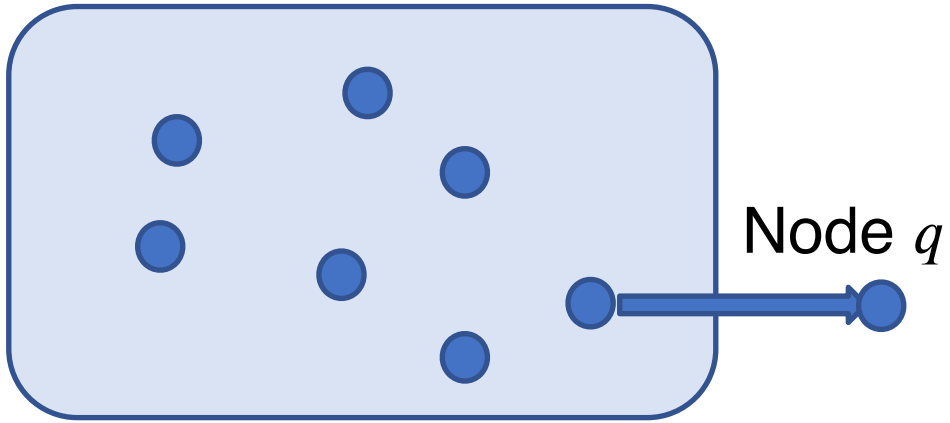

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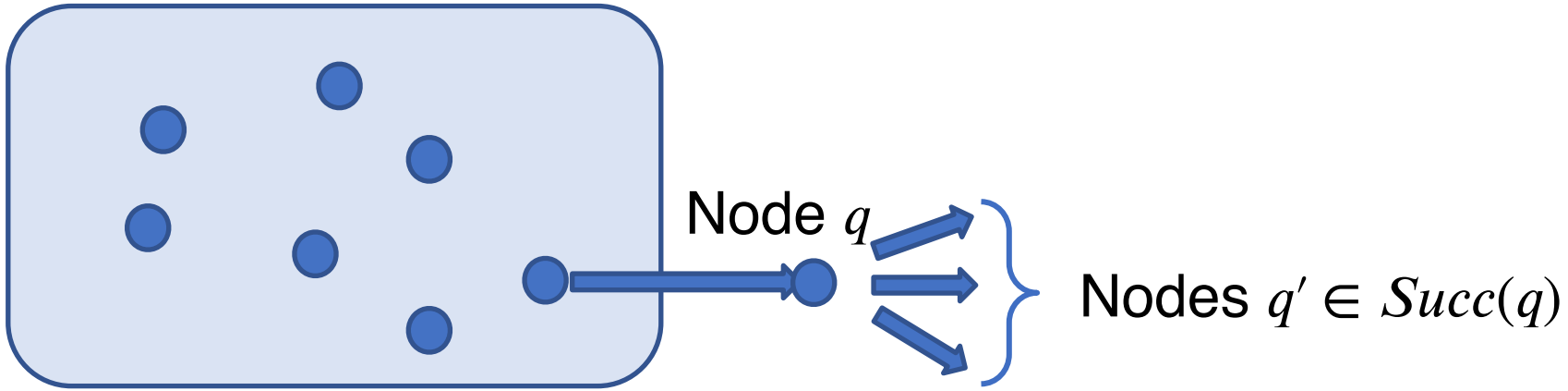

FRONTIER/ALIVE/PRIORITY QUEUE

Graph search algorithms

- Having determined decomposition, how to find “best” path?
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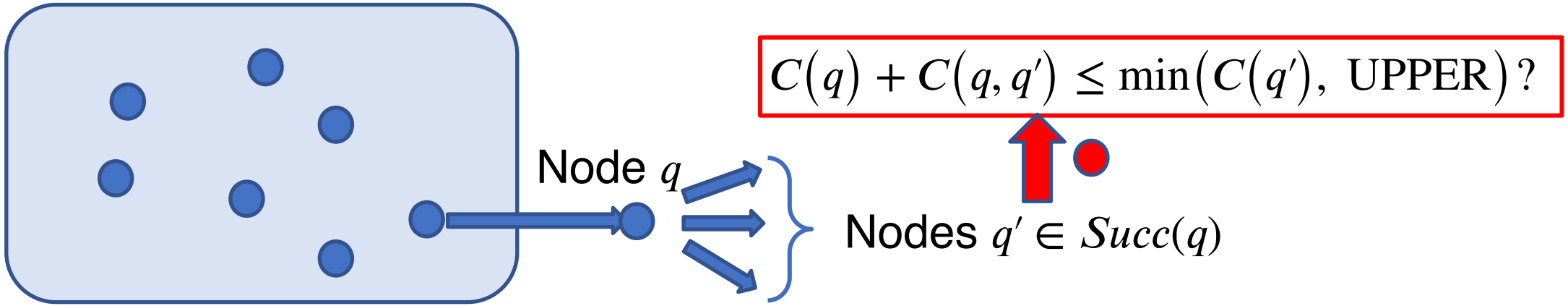

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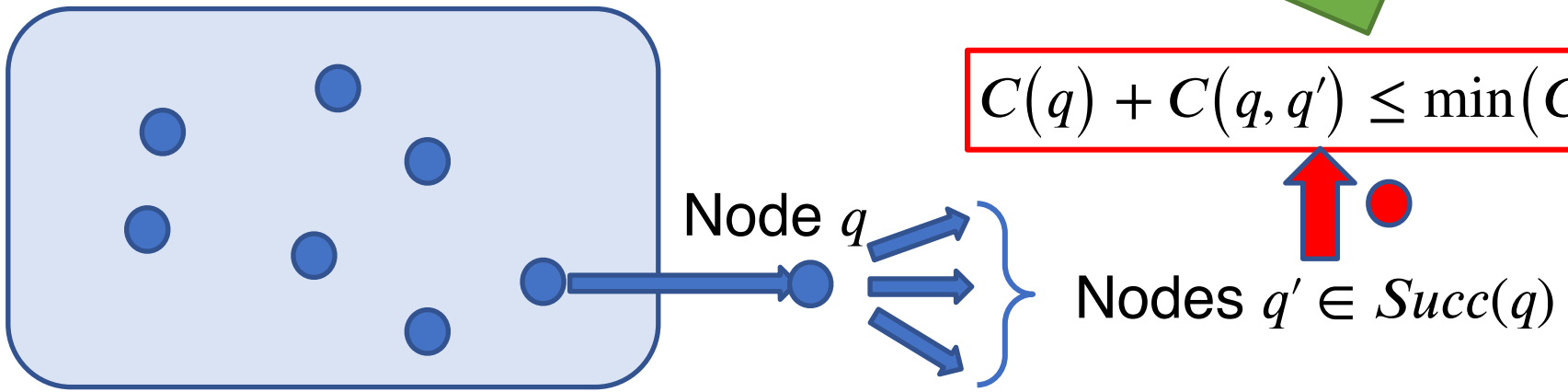
Yes $\Rightarrow C(q') := C(q) + C(q, q')$
Parent $(q') := q$

$C(q) + C(q, q') \leq \min(C(q'),\ \text{UPPER})$?



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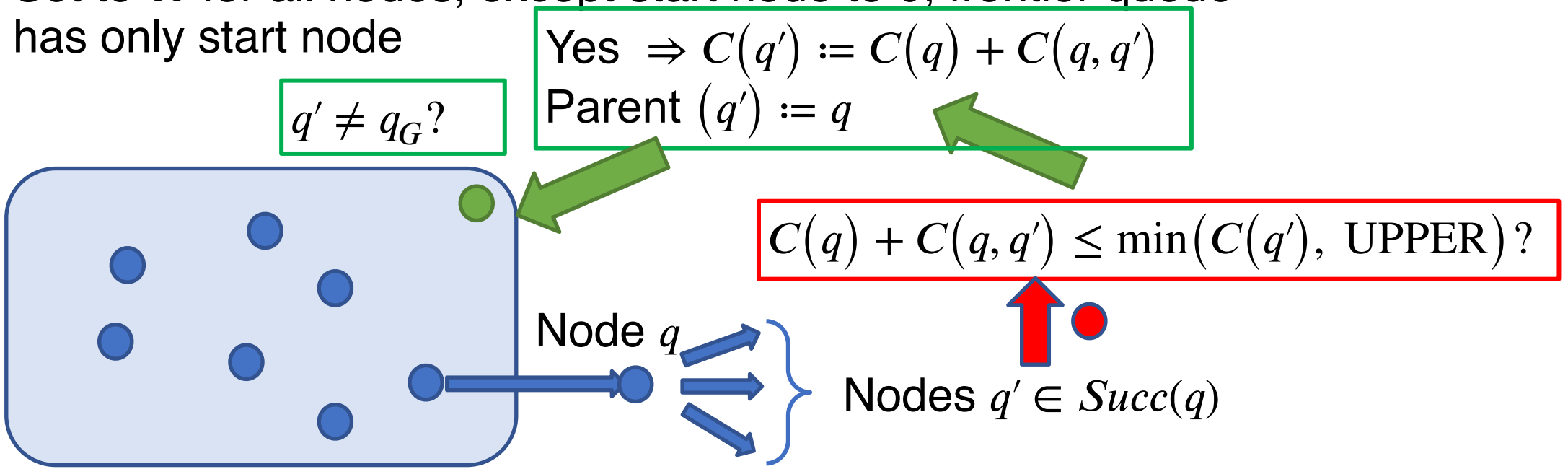

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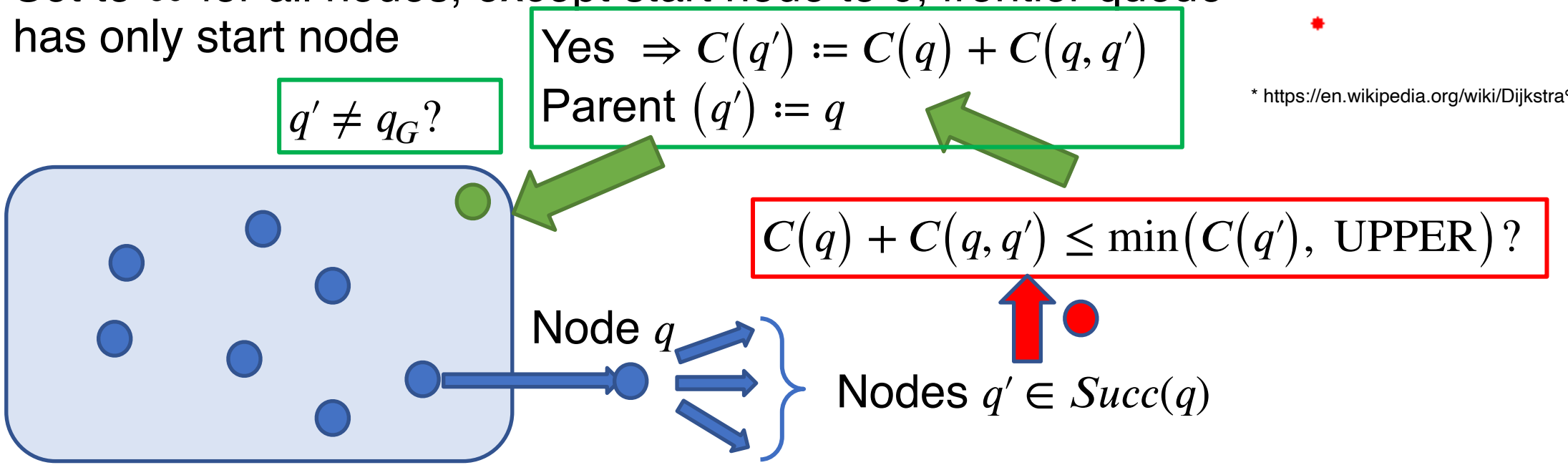


* https://en.wikipedia.org/wiki/Dijkstra%27s_algorithm

FRONTIER/ALIVE/PRIORITY QUEUE

Label correcting algorithm

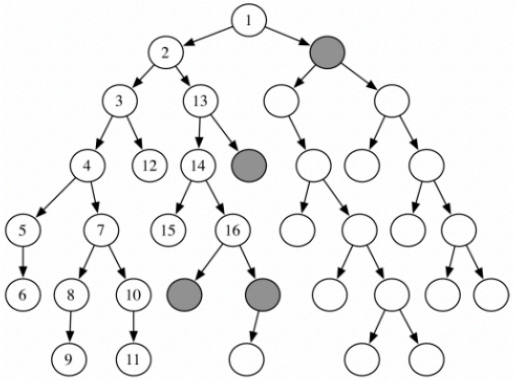
Initialization: set the labels of all nodes to ∞, except for the label of the origin node, which is set to 0, frontier queue has only start node.

Step 1. Remove a node q from frontier queue Q and for each child q' of q, execute step 2

Step 2. If $C(q) + C(q, q') \leq \min(C(q'), \text{ UPPER})$, set $C(q') := C(q) + C(q, q')$ and set q to be the parent of q'. In addition, if $q' \neq q_G$, place q' in the frontier queue if it is not already there, while if $q' = q_G$, set UPPER to the new value $C(q) + C(q, q_G)$

Step 3. If the frontier queue is empty, terminate, else go to step 1

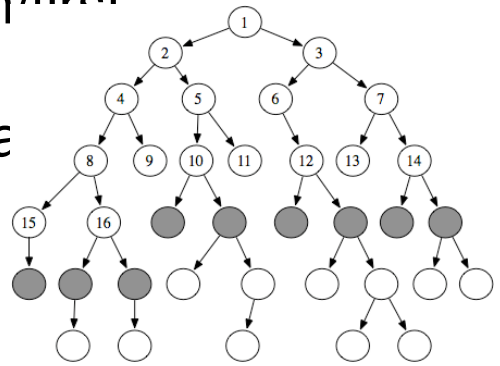
How to prioritize the queue?



Depth-First-Search (DFS): Maintain Q as a **stack** – Last in/first out

- Lower memory requirement (only need to store part of graph)

Breadth-First-Search (BFS, Bellman-Ford): Maintain Q as a **list** – First in/first out



- Update cost for all edges up to current depth before proceeding to greater depth
- Can deal with negative edge (transition) costs

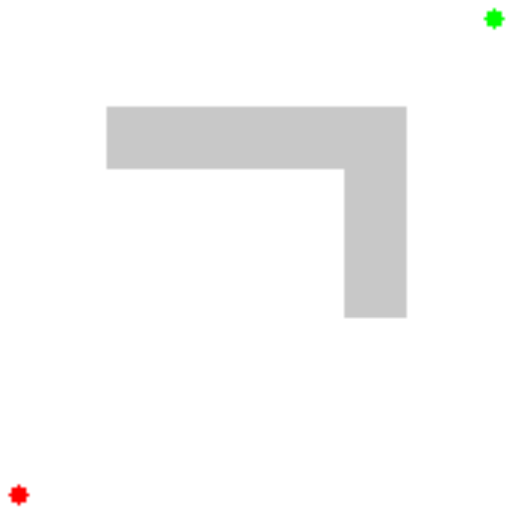
Best-First (BF, Dijkstra): Greedily select next q: $q = \operatorname{argmin}_{q \in Q} C(q)$

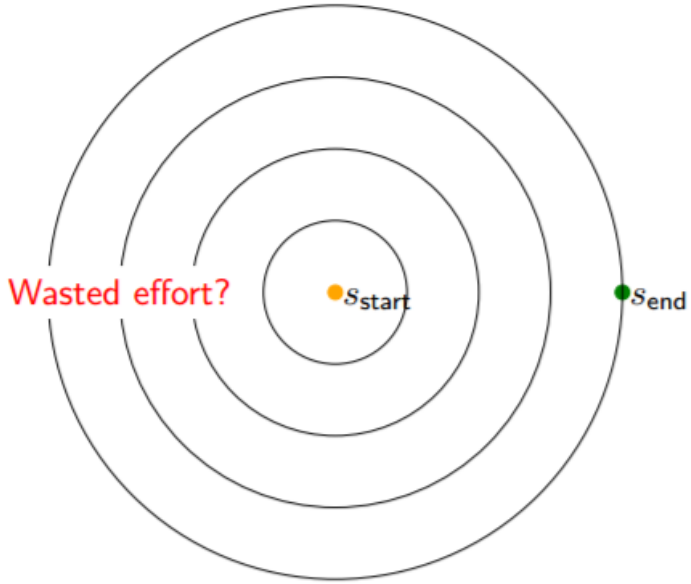
- Node will enter the frontier queue at most *once*
- Requires costs to be non-negative

Correctness and improvements

Theorem

If a feasible path exists from q_I to q_G, then algorithm terminates in finite time with $C(q_G)$ equal to the optimal cost of traversal, $C^*(q_G)$.





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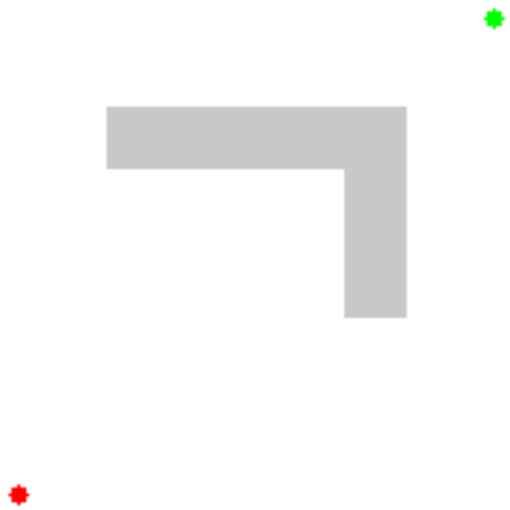

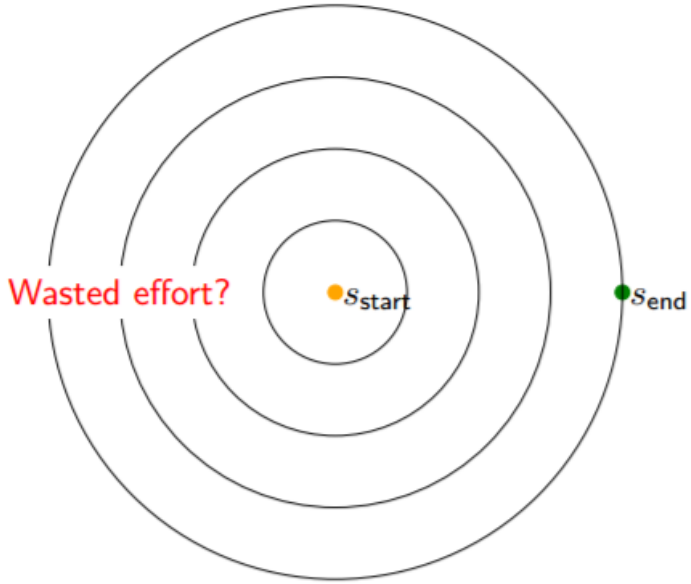

A*: Improving Dijkstra

- Dijkstra orders by optimal "*cost-to-come*"
- Faster results if order by "*cost-to-come*"*+ (approximate) "cost-to-go"*
- Take the node with lowest C(q) + h(q) from frontier set, where $h(q)$ is *a* heuristic for optimal cost-to-go (specifically, a positive *underestimate*)
- When testing to add neighbor into frontier set, use stronger

$$C(q) + C(q, q') + h(q') \leq \text{UPPER}$$

Instead of $C(q) + C(q, q') \leq \text{UPPER}$

- In this way, fewer nodes will be placed in the frontier queue
- This modification still guarantees that the algorithm will terminate with a shortest path

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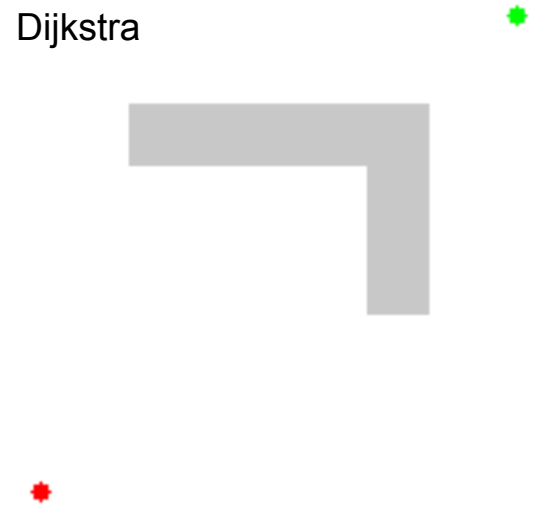

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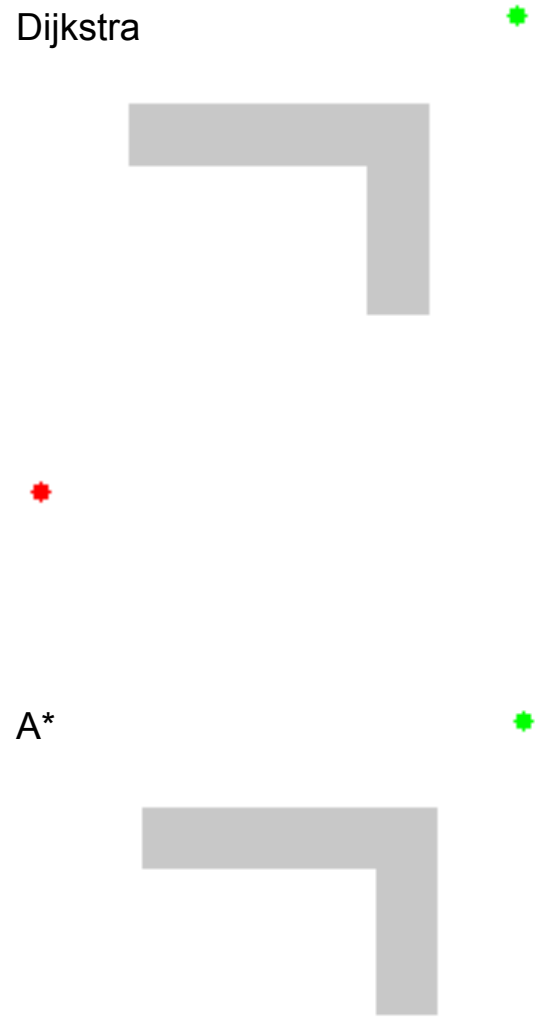

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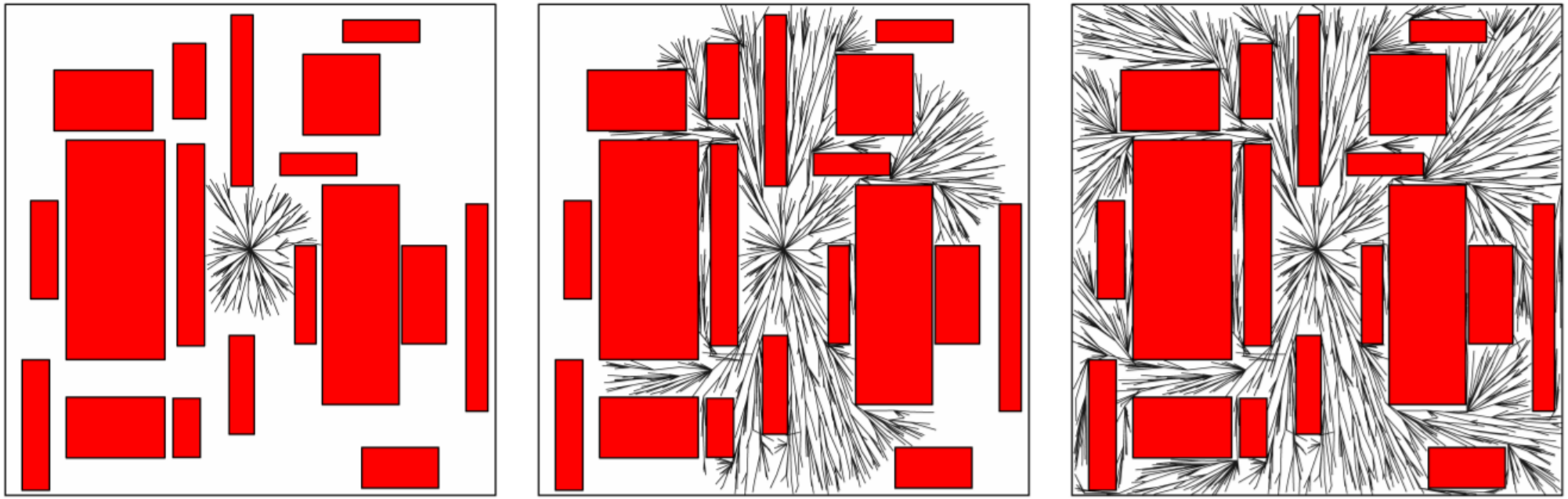
Graph search approaches: summary

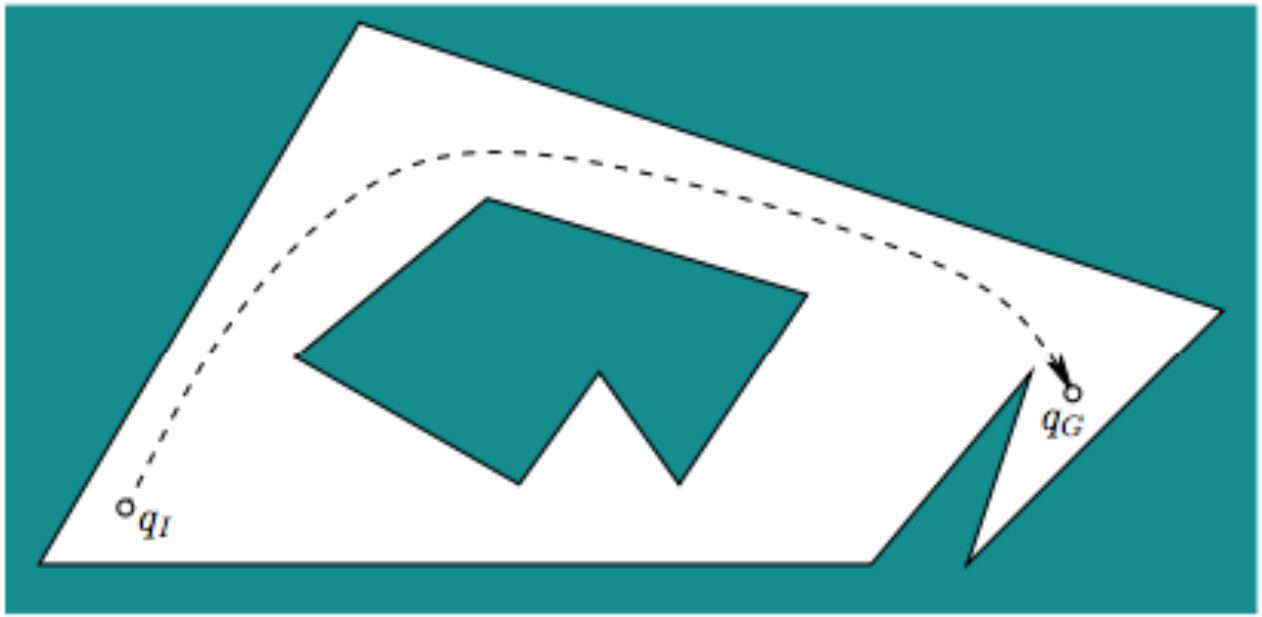
- Pros:
  - Simple and easy to use
  - Fast (for some problems)
- Cons:
  - Resolution dependent
    - Not guaranteed to find solution if grid resolution is not small enough
  - Limited to simple robots
    - C-space grid size is exponential in the number of DOFs

Next time: sampling-based planning, RRT, RRT*, PRM



Combinatorial planning

Key idea: compute a roadmap, which is a graph in which each vertex is a configuration in C_{free} and each edge is a path through C_{free} that connects a pair of vertices



Free-space roadmaps

Given a complete representation of the free space, we compute a roadmap that captures its connectivity

A roadmap should preserve:

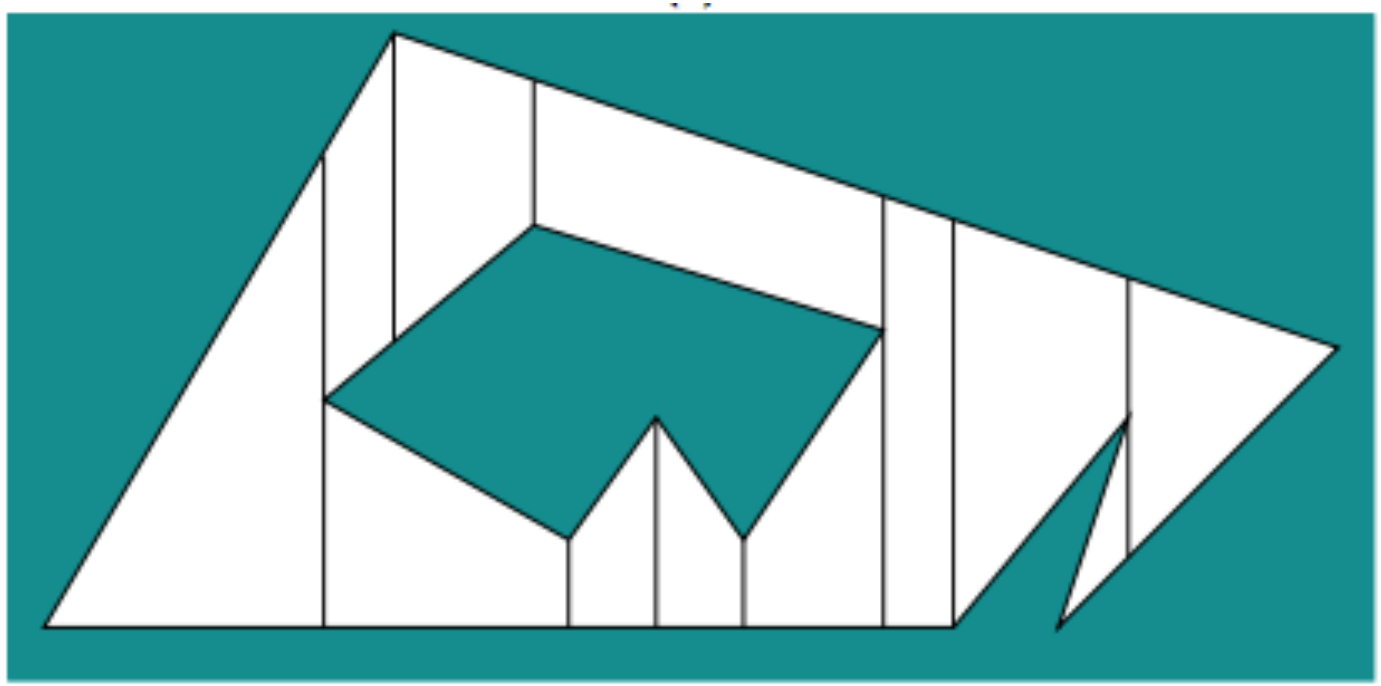
1. Accessibility: it is always possible to connect some q to the roadmap (e.g., $q_I \rightarrow s_1,\ q_G \rightarrow s_2$)
2. Connectivity: if there exists a path from q_I to q_G, there exists a path on the roadmap from s_1 to s_2

Main point: a roadmap provides a discrete representation of the continuous motion planning problem *without losing* any of the original connectivity information needed to solve it

Cell decomposition

Typical approach: cell decomposition. General requirements:

- Decomposition should be easy to compute
- Each cell should be easy to traverse (ideally convex)
- Adjacencies between cells should be straightforward to determine



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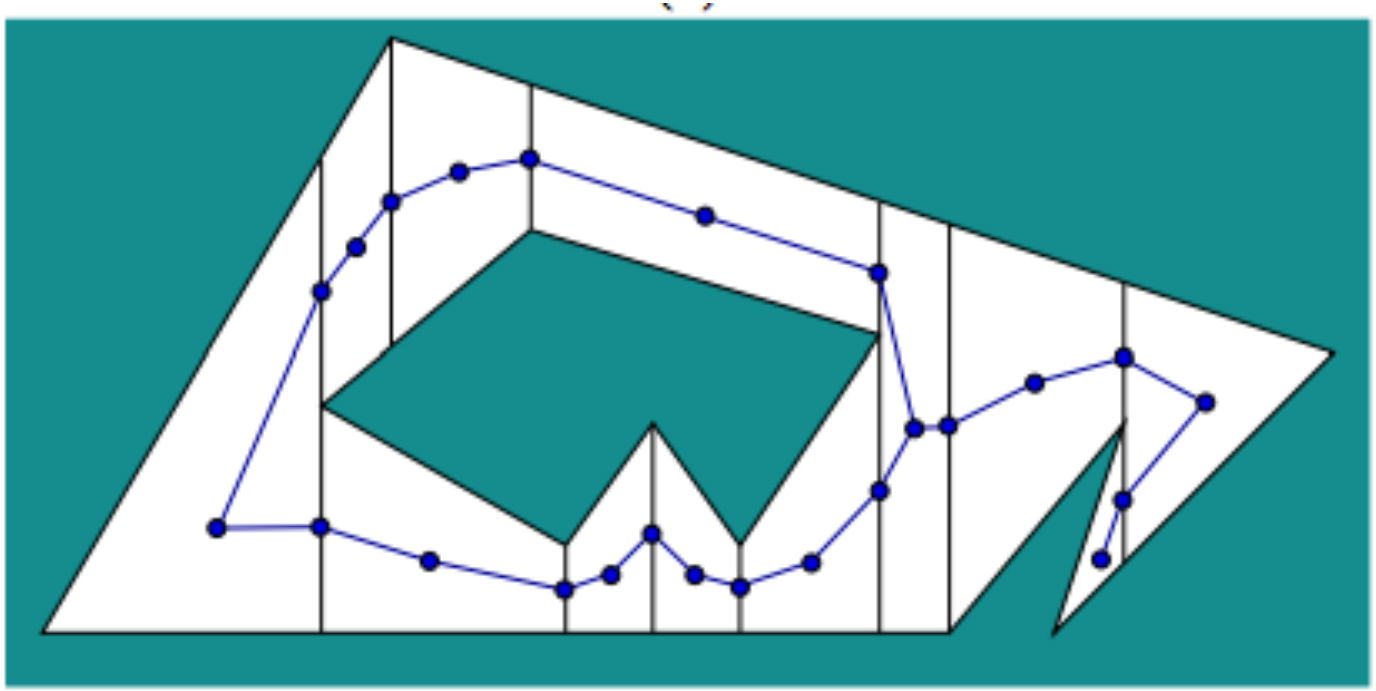

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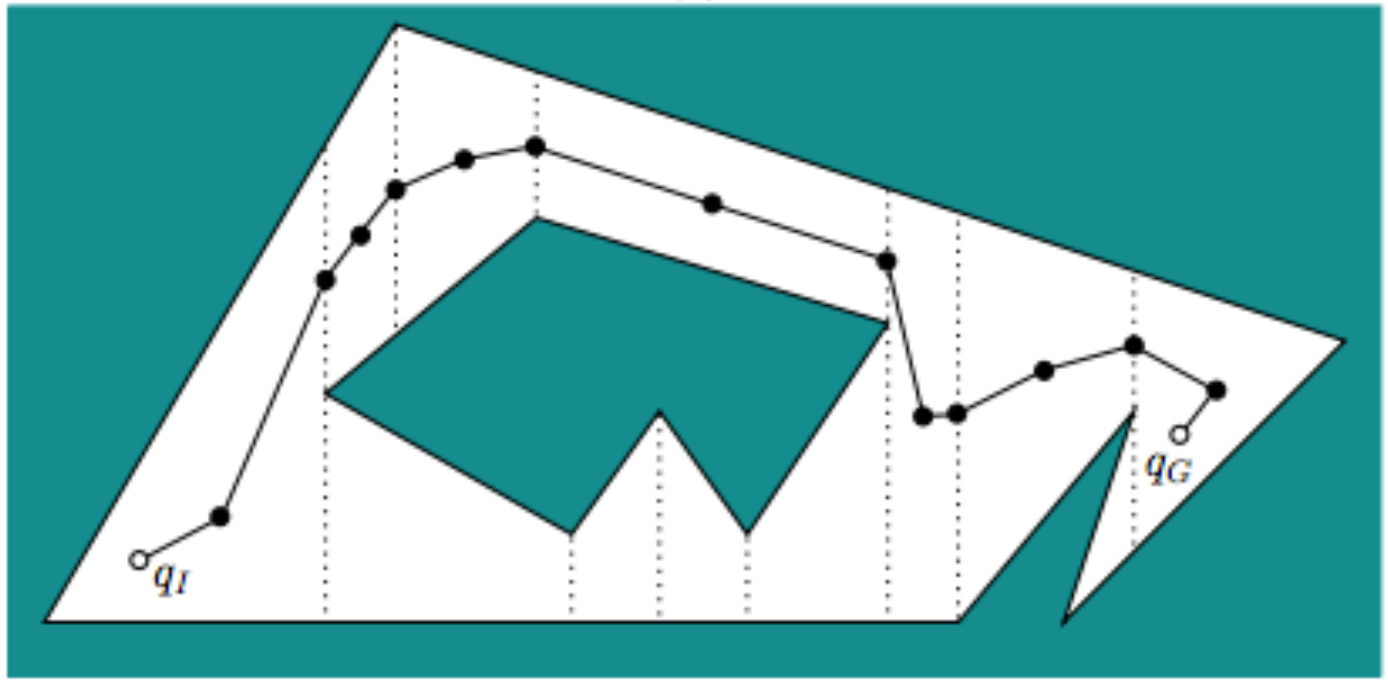

Computing a trapezoidal cell decomposition

For every vertex (corner) of the forbidden space:

- Extend a vertical ray until it hits the first edge from top and bottom
  - Compute intersection points with all edges, and take the closest ones
  - More efficient approaches exists

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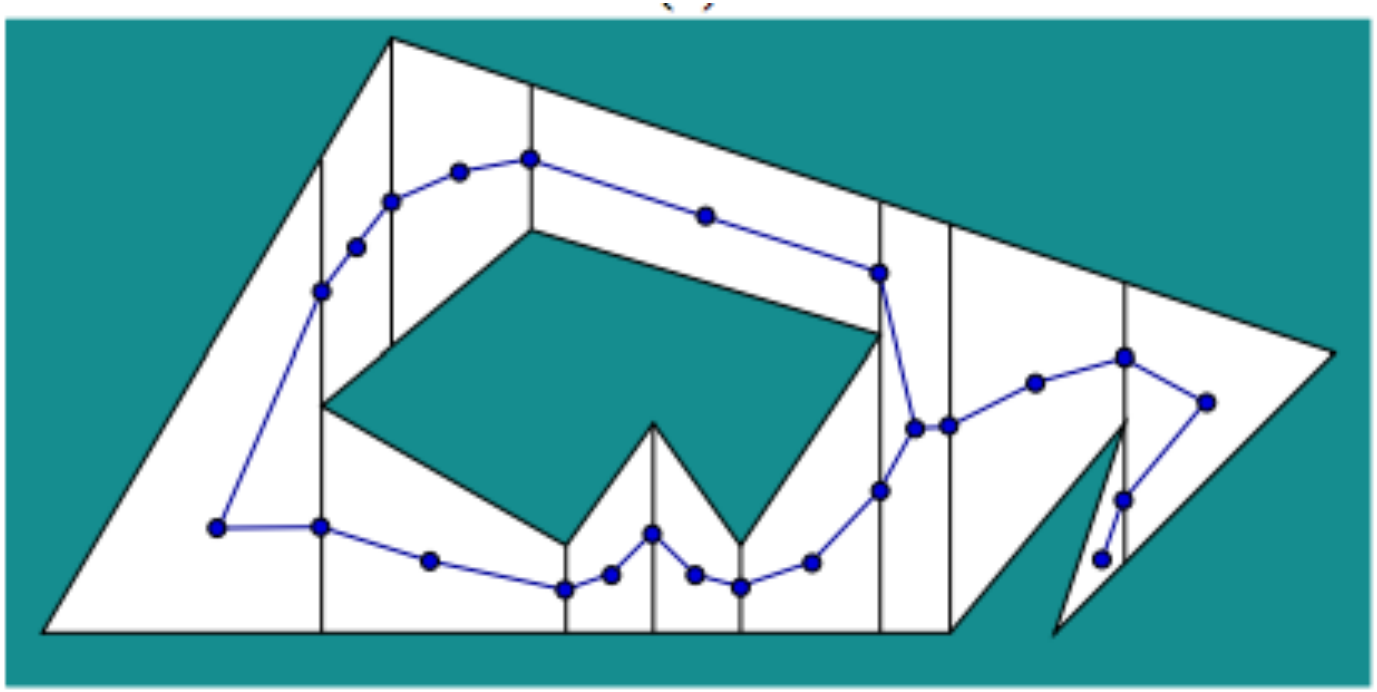
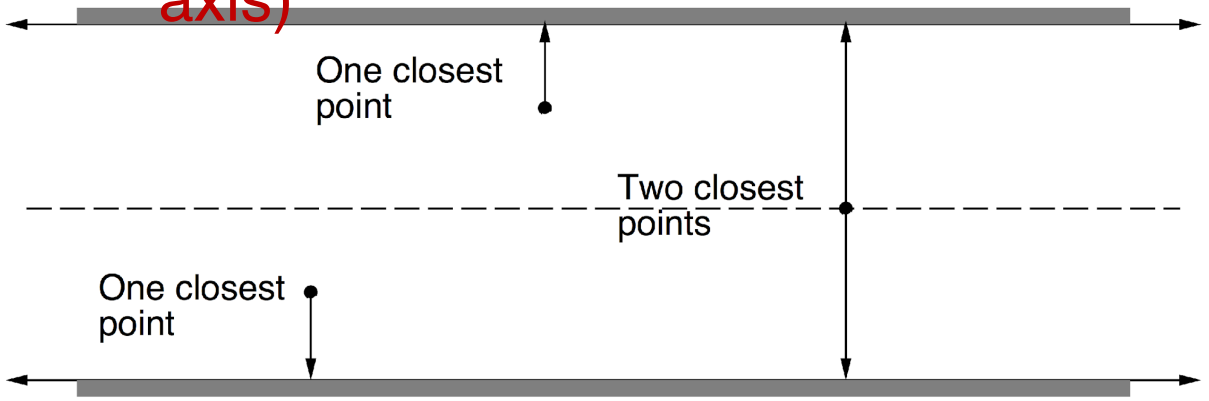
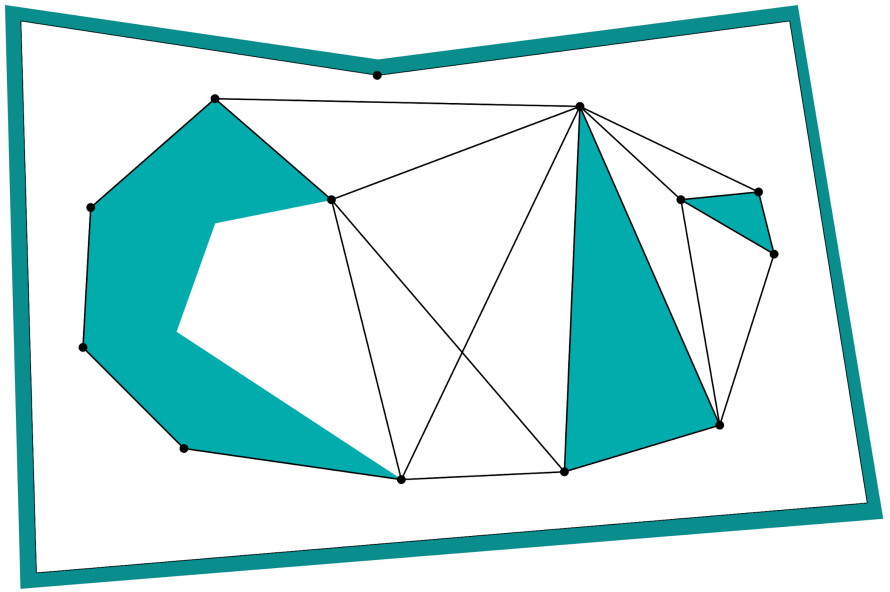


Other roadmaps

Maximum clearance (medial axis)

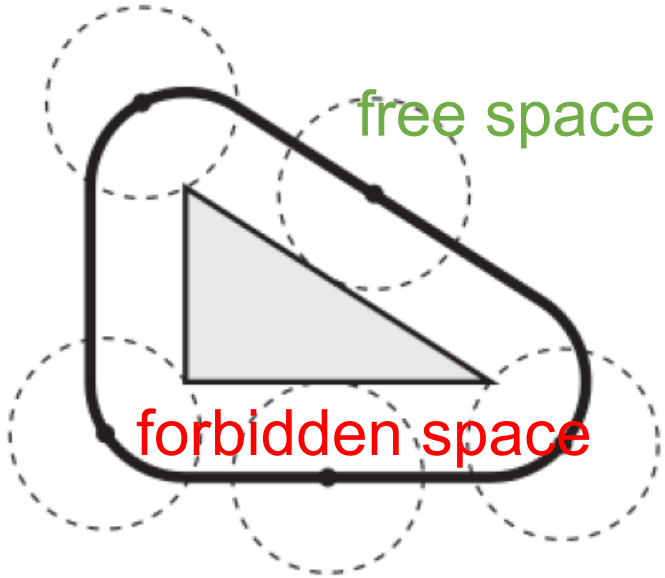


Minimum distance (visibility graph)



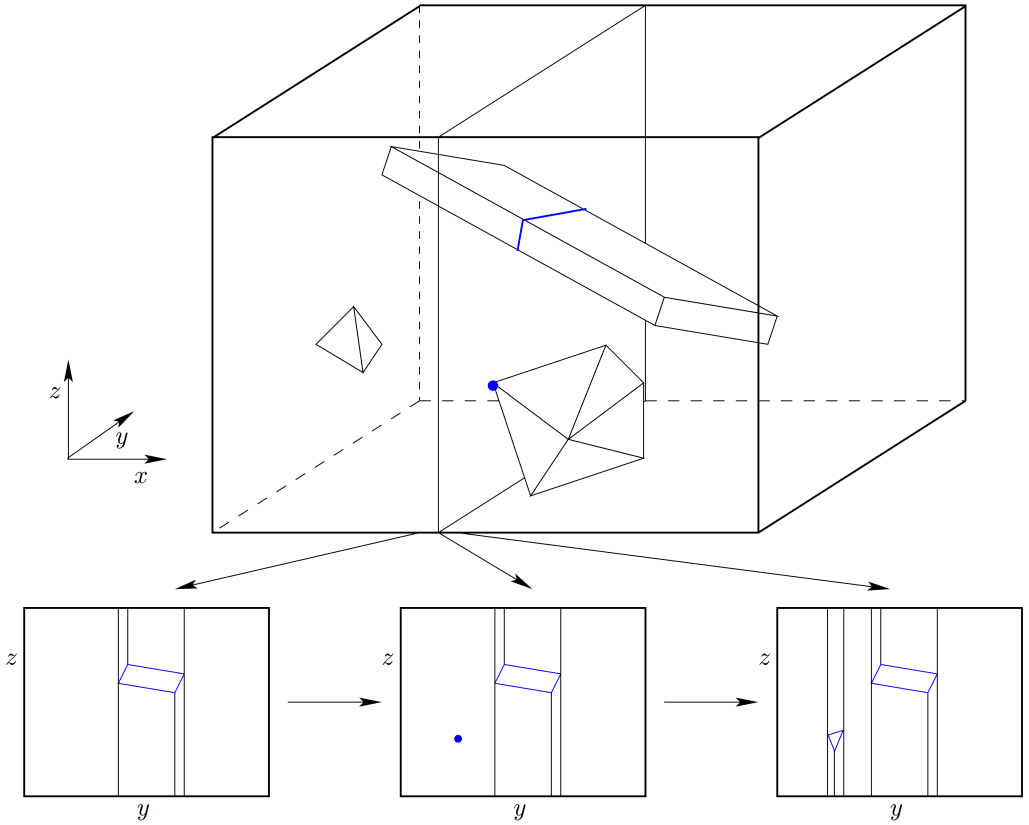
Caveat: free-space computation

- The free space is not known in advance
- We need to compute this space given the ingredients
  - Robot representation, i.e., its shape (polygon, polyhedron, …)
  - Representation of obstacles
- To achieve this, we do the following:
  - Contract the robot into a point
  - In return, inflate (or stretch) obstacles by the shape of the robots



Higher dimensions

- Extensions to higher dimensions is challenging ⇒ algebraic decomposition methods



Combinatorial planning: summary

- These approaches are complete and even optimal in some cases
  - Do not discretize or approximate the problem
- Have theoretical guarantees on the running time
  - I.e., computational complexity is known
- Usually limited to small number of DOFs
  - Computationally intractable for many problems
- Problem specific: each algorithm applies to a specific type of robot/ problem
- Difficult to implement; requires special software to reason about geometric data structures (CGAL)

Additional resources on combinatorial planning

- Visualization of C-space for polygonal robot: https://www.youtube.com/watch?v=SBFwgR4K1Gk
- Algorithmic details for Minkowski sums and trapezoidal decomposition: de Berg et al., “Computational geometry: algorithms and applications”, 2008
- Implementation in C++:
  Computational Geometry Algorithms Library

CGAL